Advanced Speech-to-Text Models for Noisy, Multilingual Corporate Meetings — developing a robust ASR system using Whisper, wav2vec 2.0, and domain adaptation for hearing-impaired employees in a corporate environment (Nestlé/BIT)

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1. Introduction

1.1 Context and Motivation

Automatic Speech Recognition (ASR) systems have undergone rapid evolution over the past decade, achieving remarkable performance in controlled environments and for high-resource languages. However, the real-world deployment of ASR in enterprise contexts—especially within noisy, multilingual corporate meetings—remains fraught with challenges. In multinational corporations such as Nestlé, meetings routinely involve participants from diverse linguistic backgrounds, often communicating in accented English, French, or local African languages. Acoustic conditions are seldom ideal: overlapping speech, reverberation, background office noise, and variable microphone quality degrade intelligibility and hinder accurate transcription. For employees with hearing impairments, these obstacles are even more pronounced, as their access to spoken content hinges on the fidelity and timeliness of ASR-generated captions or transcripts. Thus, the motivation for this thesis arises not only from technological ambition but from a concrete and urgent inclusivity imperative: to empower hearing-impaired professionals with equitable participation in corporate discourse by developing and adapting advanced ASR models specifically for noisy, multilingual, real-world meeting environments.

The diversity and complexity of speech encountered in corporate meetings present unique technical demands. Off-the-shelf ASR models, while effective for broadcast-quality input, frequently falter when confronted with domain-specific jargon, spontaneous speech phenomena, or non-native pronunciations. The Whisper and wav2vec 2.0 architectures exemplify state-of-the-art performance across several benchmarks, yet their baseline models often require adaptation to maintain robustness in highly variable acoustic and linguistic conditions found in enterprise settings [1]. Recent comparative studies have shown that while Whisper models can achieve strong accuracy in several languages, their performance degrades in the presence of domain-specific vocabulary or in low-resource dialects [2]. Corporate meetings at Nestlé, for example, include frequent references to branded products, project codenames, and technical terms that do not appear in general-purpose training data, necessitating targeted domain adaptation.

Inclusion of hearing-impaired employees is a central concern at both the corporate and societal levels. Legal frameworks in many jurisdictions—including the European Union's Web Accessibility Directive—mandate accessibility accommodations in digital communication platforms. Yet, technical solutions lag behind policy mandates, particularly in Africa and other emerging regions, where research on ASR for hearing-impaired users is limited [3]. The pedagogical and psychological literature underscores that real-time, accurate captioning significantly improves participation and comprehension for deaf and hard-of-hearing users, especially in dynamic group settings [4]. In the context of Nestlé's global operations, enabling such accessibility is not only a matter of regulatory compliance but also aligns with the company's diversity and inclusion commitments.

The technical strategy adopted in this project draws on recent advances in multimodal deep learning, domain adaptation, and retrieval-augmented generation (RAG). Integrating Optical Character Recognition (OCR) outputs—such as on-screen slides or shared documents—with Speech-to-Text (STT) outputs into a shared semantic space offers a pathway to richer,

context-aware transcriptions. The use of a unified ChromaDB knowledge base, as implemented in this system, enables contextual retrieval and cross-modal disambiguation, reducing the risk of transcription errors on domain-specific terminology. For example, the observed PoC error where “Philippe Navratil” was transcribed as “Philippe Naruto” illustrates a failure of standard ASR to resolve named entities without domain grounding, underscoring the necessity of RAG-based augmentation and post-correction modules. Embedding models such as intfloat/multilingual-e5-small facilitate semantic matching across OCR and ASR outputs, while LLMs (Ollama/Llama 3.2) enable dynamic correction and contextual prompt engineering.

Robustness to noise remains a persistent challenge. Signal processing pipelines utilizing Silero Voice Activity Detection (VAD), noisereduce, and librosa for feature extraction have demonstrated measurable improvements in signal-to-noise ratio prior to decoding. Studies have shown that preprocessing steps—such as spectral subtraction and autocorrelation-based feature extraction—directly enhance recognition accuracy in adverse conditions [5]. Nevertheless, the effectiveness of these techniques is contingent upon their integration with adaptive acoustic models. Work on noise-robust adaptation, such as that by Chung et al., highlights the value of fine-tuning on in-domain noisy data, which remains a principal aim of this thesis [6].

The enterprise deployment context imposes further constraints and opportunities. Resource efficiency is critical: the use of faster-whisper small INT8 achieves a real-time factor (RTF) of 0.25 on CPU, processing four minutes of audio in sixty-two seconds, and generating twenty timestamped segments per file. These figures demonstrate practical viability for near-real-time captioning in meeting scenarios. At the architectural level, the system’s reliance on FastAPI for orchestration, ChromaDB for knowledge management, and Microsoft Azure services—including Azure AI Speech, Azure AI Search, Azure OpenAI, Blob Storage, Entra ID, Key Vault, and the Graph API for Teams integration—ensures both scalability and enterprise-grade security. These design choices reflect an understanding of real-world deployment constraints, such as authentication, privacy, and compliance with corporate IT policies.

Multilingualism compounds the challenge. As global workforces become more linguistically diverse, the ability to accurately transcribe accented speech, code-switching, and under-resourced languages grows increasingly critical [7]. Recent research underscores the inadequacy of monolingual or English-centric ASR systems in such contexts, pointing to the necessity of multilingual pretraining and transfer learning [8]. Domain adaptation techniques, including LoRA (Low-Rank Adaptation) and kernel mean matching, are actively being explored to address these gaps, with emerging evidence of improved recognition rates in under-represented language domains.

In summary, the motivation for this thesis is grounded in a convergence of technical, ethical, and operational imperatives. The aspiration is to advance the state of speech-to-text technology in ways that directly benefit hearing-impaired employees and multilingual teams, enhancing inclusion, productivity, and accessibility in corporate settings. By drawing on the combined strengths of advanced ASR architectures, domain adaptation, cross-modal retrieval, and enterprise-scale deployment, this work seeks to bridge longstanding gaps between academic research and the lived realities of diverse, dynamic workplaces.

AI Usage Acknowledgment: AI tools provided assistance with writing and editing this thesis section. All research directions, technical analysis, and original synthesis are the sole work

of the student, OUEDRAOGO Yannick Ulrich Levi.

1.2 Problem Statement

Automatic speech recognition (ASR) has achieved remarkable progress in recent years, driven by advances in deep learning architectures and the availability of large-scale multilingual datasets. Despite these achievements, deploying ASR systems for noisy, multilingual corporate meetings remains deeply challenging, particularly in environments characterized by overlapping speakers, domain-specific jargon, frequent code-switching, and variable acoustic conditions. The corporate context at Nestlé, which spans several continents and regularly involves speakers of diverse linguistic backgrounds, magnifies these challenges. Within this setting, hearing-impaired employees face additional barriers, as the intelligibility and accuracy of transcriptions directly affect their ability to participate fully in collaborative processes. The core problem addressed in this thesis is the development of a robust ASR system, based on Whisper and wav2vec 2.0 architectures, with domain-specific adaptation to support accessible, accurate, and real-time transcription for all employees—including the hearing-impaired—under real-world corporate conditions.

Multilingual ASR models such as Whisper have demonstrated strong generalization across languages; however, they frequently encounter degradation in accuracy when exposed to noisy environments and spontaneous speech, both ubiquitous in workplace meetings [8]. The acoustic variability arising from conference rooms, remote connections, and variable microphone quality further compounds error rates, particularly for languages and accents underrepresented in pretraining corpora [2]. Standard models, while capable in laboratory benchmarks, often fail to meet the expectations of enterprise users when faced with crosstalk, background noise, and rapid speaker changes, resulting in frequent transcription errors. A salient real-world example from the proof-of-concept phase at Nestlé illustrates this: the system consistently transcribed the name “Philippe Navratil” as “Philippe Naruto,” demonstrating the inability of generic models to capture domain-specific vocabulary and named entities critical for corporate comprehension and accessibility.

The inclusion of hearing-impaired employees brings distinct and urgent requirements that current ASR solutions inadequately address. While prior research has shown that ASR can facilitate communication for the hearing-impaired, these gains diminish sharply when the ASR output is noisy, omits specialized terminology, or fails to segment speech accurately [4]. For users relying on real-time captioning, even minor transcription errors can have substantive communicative impacts. The persistent misrecognition of names, technical terms, or project-specific vocabulary—as exemplified by the repeated “Philippe Naruto” substitution—undermines the reliability of automatic captions and erodes trust in the system’s utility.

Advanced architectures such as wav2vec 2.0 and Whisper have shown promise in noisy and low-resource settings, yet their baseline performance remains insufficient for high-stakes corporate deployments without targeted adaptation [1]. Enhanced noise-robustness, as achieved through acoustic model adaptation or sophisticated feature extraction, remains essential. Techniques such as spectral subtraction, domain-specific voice activity detection (VAD), and targeted noise reduction—implemented via Silero VAD and noisereduce in the present system—offer partial mitigation but cannot fully compensate for limitations in the underlying ASR model [5]. Furthermore, the requirement for precise timestamping and speaker diarization, addressed in this work using pyannote-audio and librosa, adds another

layer of complexity, as segment boundaries must be accurate to support downstream accessibility features and real-time summarization.

A further dimension of the problem stems from the inherently dynamic and document-rich nature of corporate meetings. Slides, shared documents, and on-screen text provide crucial context that, when ignored, can lead to transcription ambiguities and failure to recognize domain-specific lexicon [9]. Existing ASR systems typically treat speech and visual information in isolation, leading to missed opportunities for cross-modal reinforcement. In the current thesis, a Retrieval-Augmented Generation (RAG) layer is proposed that integrates outputs from both OCR (optical character recognition) and STT (speech-to-text) into a unified ChromaDB knowledge base. This shared knowledge representation enables the system to ground ASR predictions in up-to-date meeting content, supporting domain vocabulary acquisition and context-aware correction, and directly addresses the kind of named entity substitution illustrated by the “Philippe Navratil !’ Philippe Naruto” error.

Ensuring enterprise-grade deployment further necessitates compliance with security, scalability, and integration requirements. The diverse Nestlé ecosystem demands seamless interoperability with Microsoft Azure services, including Azure AI Speech for primary transcription, Azure AI Search for knowledge retrieval, Azure OpenAI for generative summarization, and secure storage and identity management via Blob Storage, Entra ID, and Key Vault. Integration with Microsoft Teams via Graph API is also critical for real-time user accessibility. Standard ASR solutions rarely offer native support for such complex, multi-layered enterprise IT environments, creating additional obstacles to reliable deployment [3].

In summary, the fundamental problem is not merely the deployment of state-of-the-art ASR in a multilingual, noisy environment, but the realization of a system that delivers accessible, context-aware, and domain-adapted transcription for all users—especially the hearing-impaired—within the operational and security constraints of a major multinational corporation. The observed “Philippe Navratil !’ Philippe Naruto” transcription error provides a concrete and recurring marker of the limitations of generic models and underscores the need for domain vocabulary prompting, integration of multi-modal context, and robust model adaptation. Addressing these interlocking challenges requires a synthesis of advanced ASR architectures, targeted domain adaptation, noise-robust pre- and post-processing, and deep integration with enterprise knowledge repositories and identity ecosystems. This thesis frames and addresses these challenges through an end-to-end system using faster-whisper, wav2vec 2.0, FastAPI, ChromaDB, pyannote-audio, Silero VAD, noisereduce, librosa, and intfloat/multilingual-e5-small embeddings, orchestrated within the Azure enterprise stack and augmented with Retrieval-Augmented Generation and context-aware correction mechanisms. The resulting system is designed to bridge the persistent gap between technical ASR advances and the lived accessibility needs of diverse employees in real-world corporate environments.

AI Usage Acknowledgment: AI tools assisted with writing and editing. All research directions, technical content, architectural design, and analytical synthesis remain the original work of OUEDRAOGO Yannick Ulrich Levi.

1.3 Research Objectives

Establishing clear research objectives is essential for guiding the development of advanced speech-to-text (ASR) models tailored to the complex requirements of noisy, multilingual

corporate environments, such as those found at Nestlé. This thesis is anchored in the pursuit of several interlocking objectives, each designed to address technical and accessibility challenges inherent in deploying ASR systems for hearing-impaired employees. The first objective is to rigorously investigate and enhance the performance of state-of-the-art ASR architectures—specifically, Whisper and wav2vec 2.0—when exposed to realistic corporate meeting conditions characterized by overlapping speakers, diverse accents, and fluctuating background noise. Existing studies have demonstrated that while Whisper and wav2vec 2.0 outperform traditional ASR systems in controlled settings, their robustness in adverse real-world scenarios remains limited, especially when domain-specific vocabulary and rare language pairs are involved [1].

The second objective concerns the systematic adaptation of these models to domain-specific lexicons, idiomatic expressions, and proper nouns frequently encountered in corporate discourse. Misrecognition of key terms—such as the observed error transcribing “Philippe Navratil” as “Philippe Naruto”—can materially undermine system credibility and utility. Addressing these errors requires the design and empirical validation of domain vocabulary prompting mechanisms, as well as fine-tuning strategies using in-domain datasets. Previous research has shown that conventional ASR models trained on generic corpora often fail to generalize to specialized settings, necessitating targeted adaptation techniques [10]. The integration of LoRA-based fine-tuning and prompt engineering will be explored as means of mitigating these limitations [11].

A third objective is the creation of a unified knowledge base that synergizes outputs from both speech-to-text and optical character recognition (OCR) pipelines. In practical meeting contexts, critical information is often split between spoken discussion and visual materials (e.g., slides, whiteboards). The proposed system architecture will employ a Retrieval-Augmented Generation (RAG) layer to merge OCR and ASR outputs into a shared ChromaDB vector database, thereby enabling context-aware semantic search and question-answering functionalities. This architectural innovation is motivated by recent advances in retrieval-augmented systems, which have demonstrated improved relevance and factuality for downstream applications in enterprise environments [8].

Improving accessibility for hearing-impaired employees constitutes a central ambition of this research. The project will systematically evaluate the intelligibility and accuracy of ASR-generated transcripts for users with varying degrees of hearing loss, building upon frameworks that model the communication needs of this demographic in noisy workplaces [12]. Specifically, the thesis will examine the efficacy of speaker diarization (using pyannote-audio), voice activity detection (via Silero VAD), and adaptive noise reduction (using noisereduce and librosa) in enhancing transcript clarity. Drawing on recent empirical findings, the research will assess whether these pre- and post-processing steps significantly reduce cognitive load for hearing-impaired employees, compared to baseline outputs [4].

Another objective is to benchmark and optimize computational efficiency, making advanced ASR accessible in resource-constrained enterprise settings. The thesis will document real-world performance metrics: for example, a faster-whisper small INT8 model achieving a real-time factor (RTF) of 0.25 on CPU, processing four minutes of audio in 62 seconds, and segmenting speech into 20 timestamped sections. These measurements will serve as practical baselines for further optimization, particularly for deployments where hardware accelerators are unavailable or cost-prohibitive [2].

Deployment at scale presents its own set of objectives. The system must integrate securely

and efficiently with existing Nestlé infrastructure, using Microsoft Azure's suite of AI and enterprise tools—including Azure AI Speech, Azure AI Search, Azure OpenAI, Blob Storage, Entra ID, Key Vault, and the Graph API for Teams integration. Research will focus on ensuring secure data handling, robust authentication, and low-latency query processing, aligning with industry best practices for sensitive corporate data [13].

A further research objective is the development and validation of a multilingual pipeline, with a focus on French, English, and local African languages relevant to the Burkina Faso context. The thesis will explore the use of intfloat/multilingual-e5-small embeddings for semantic indexing and cross-lingual retrieval, with experiments designed to identify failure modes and gaps in existing multilingual ASR models [14]. Particular attention will be given to under-resourced languages, leveraging domain adaptation and synthetic augmentation to expand coverage [7].

Finally, the research will design and conduct a comprehensive evaluation protocol, including both quantitative (e.g., word error rate, segment-level accuracy, semantic retrieval precision) and qualitative (e.g., user satisfaction, perceived accessibility improvements) measures. Experimentation will be iterative, with interim results reported and used to refine model architectures, prompting strategies, and integration pipelines [5]. Through these objectives, the thesis seeks not only to push the technical boundaries of ASR in noisy, multilingual corporate settings but also to deliver meaningful improvements in workplace accessibility and productivity for hearing-impaired employees.

AI Usage Acknowledgment:

AI tools assisted with writing and editing this section. All research directions, experimental designs, and technical content originate from the author's own work and critical analysis.

1.4 Research Questions

Developing a speech-to-text (STT) system capable of robust performance in noisy, multilingual corporate meeting environments for hearing-impaired employees presents intricate technical and practical challenges. The deployment context at Nestlé/BIT demands not only state-of-the-art automatic speech recognition (ASR) models such as Whisper and wav2vec 2.0, but also rigorous domain adaptation strategies and integration with enterprise infrastructure. To guide the research, this thesis formulates a series of interrelated questions addressing the foundational, methodological, and applied aspects of advanced ASR system development for accessibility.

Central to the investigation is the question: How can ASR models be optimized to achieve high transcription accuracy for diverse languages, dialects, and accents present in corporate meetings under substantial environmental noise? Existing research has demonstrated the capacity of Whisper and wav2vec 2.0 to perform well in multilingual contexts, yet performance degrades in the presence of overlapping speech, technical jargon, and reverberant meeting rooms [8], [2]. For hearing-impaired users, even minor recognition errors —such as the documented "Philippe Navratil" rendered as "Philippe Naruto"—can have significant communicative and accessibility consequences, especially in multi-turn corporate dialogues where domain-specific names and terms carry operational importance. Thus, a core research question asks: What architectural, pre-processing, and prompt engineering techniques are most effective for mitigating noise-induced and domain-mismatch errors in

state-of-the-art ASR models for corporate accessibility use cases?

A closely related inquiry concerns the efficacy of domain adaptation. Given that much ASR training data is not representative of the acoustic and lexical properties of corporate meeting speech, how can models like Whisper and wav2vec 2.0 be adapted through fine-tuning, prompt engineering, or external knowledge integration to better recognize company-specific vocabulary, acronyms, and named entities? Prior work has explored low-rank adaptation (LoRA) and other adaptation mechanisms for Whisper, yielding measurable improvements in specialized domains, but further research is needed on their utility for enterprise meeting scenarios, particularly where speakers switch languages or code-mix [15], [11]. The system's integration of a retrieval-augmented generation (RAG) layer, which fuses optical character recognition (OCR) and STT outputs into a shared ChromaDB knowledge base, raises the investigation: Does multimodal grounding—where screen-shared slides and spoken dialogue are jointly indexed—systematically reduce named entity and domain-specific errors, and if so, by what mechanisms?

Another research direction interrogates the limits of noise reduction and voice activity detection (VAD) in enhancing downstream ASR. Corporate meetings often suffer from background conversations, keyboard typing, and teleconference artifacts. Tools such as Silero VAD, noisereduce, and librosa have been proposed for audio enhancement prior to ASR, but their impact on real-world meeting speech requires systematic evaluation [5], [6]. In particular, what is the cumulative effect of advanced pre-processing pipelines on word error rate (WER), and do these techniques introduce undesirable latency or distortions that compromise accessibility for hearing-impaired participants? Empirical benchmarks, such as the observed real-time factor (RTF) of 0.25 on CPU for faster-whisper small INT8—processing four minutes of audio in just over one minute—demonstrate feasibility, but further research must quantify the trade-offs between speed, accuracy, and user experience.

The demands of accessibility for hearing-impaired employees introduce further research questions. Existing evidence indicates that conventional ASR models struggle to accommodate the unique prosodic and articulation patterns found in the speech of hearing-impaired individuals, as well as to present outputs in formats optimized for visual consumption [3], [4]. What strategies—such as prosody-aware modeling, specialized user interfaces, or post-processing for caption clarity—most effectively support hearing-impaired users in high-stakes, multilingual corporate meetings? Furthermore, how can user feedback be systematically incorporated to refine ASR outputs over time, creating a virtuous cycle of improvement?

Multilinguality and code-switching remain persistent challenges. Corporate meetings at Nestlé/BIT frequently feature French, English, and local languages, with spontaneous switches between them. The research must therefore consider: To what extent do current ASR models reliably detect and transcribe code-switching, and what training or adaptation strategies are required to handle under-resourced languages or dialects within the same conversation [14], [10]? Data augmentation, transfer learning, and domain-adversarial training are candidate solutions, but their efficacy in live corporate settings has not been sufficiently measured.

The system's deployment within enterprise infrastructure—specifically, the integration with Microsoft Azure services such as Azure AI Speech, Azure OpenAI, and Azure AI Search—introduces operational research questions as well. What are the security, scalability, and compliance implications of embedding ASR and RAG pipelines in a regulated corporate

environment? How can authentication (via Entra ID), secure storage (via Blob Storage and Key Vault), and collaboration (via Graph API for Teams) be orchestrated to support both accessibility and data protection requirements? While technical feasibility has been demonstrated in pilot deployments, systematic analysis of enterprise-readiness, performance bottlenecks, and user adoption must be conducted.

Finally, the research addresses methodological questions concerning evaluation. Beyond standard WER, which metrics best capture the accessibility and usability of ASR outputs for hearing-impaired employees in meetings? Qualitative user studies, error analysis (including high-impact errors like “Philippe Navratil !’ Philippe Naruto”), and task-based measures must be triangulated to provide a comprehensive assessment [16], [12].

This suite of research questions frames a systematic investigation of advanced ASR models, domain adaptation, accessibility, multimodal integration, and enterprise deployment, all situated within the practical constraints and aspirations of corporate meeting accessibility at Nestlé/BIT.

1.5 Contributions of This Work

This thesis addresses critical gaps in the deployment of speech-to-text (STT) systems for multilingual, acoustically challenging corporate meetings, particularly focusing on accessibility for hearing-impaired employees within the Nestlé/BIT partnership. While numerous advances have been made in automatic speech recognition (ASR) using state-of-the-art models such as Whisper and wav2vec 2.0, real-world enterprise environments still present persistent challenges: overlapping speakers, domain-specific jargon, strong accents, and high background noise remain sources of recognition error and hinder practical accessibility for diverse teams [1]. This work advances the field through several distinct contributions, each substantiated by empirical observations, critical integration of recent research, and the design of novel system components.

First, the thesis presents the integration of multiple advanced ASR models—specifically Whisper, wav2vec 2.0, and optimized quantized variants such as faster-whisper small INT8—to balance accuracy, computational efficiency, and support for multilingual speech. Real-world recording sessions revealed that faster-whisper achieved a real-time factor (RTF) of 0.25 on standard enterprise CPUs, transcribing four minutes of meeting audio in just over a minute, and generating 20 precisely timestamped segments per file. This performance, achieved without GPU resources, demonstrates the viability of deploying ASR solutions at enterprise scale with moderate infrastructure, while maintaining responsiveness essential for user-facing applications [2]. The modular architecture developed herein allows for rapid substitution or ensemble of models, enabling tailored deployments for specific language profiles and hardware contexts.

Second, the thesis makes a theoretical and practical contribution by explicitly targeting the needs of hearing-impaired employees, a demographic often underserved by mainstream ASR deployments. Drawing from educational and cognitive research on speech accessibility, the system design incorporates both acoustic adaptation—using tools such as Silero VAD for precise voice activity detection and noisereduce for pre-processing—and domain vocabulary augmentation. This dual approach increases intelligibility for users with auditory limitations and reduces error rates in noisy conditions, aligning with findings that model adaptation and tailored language resources are crucial for accessibility in

heterogeneous populations [4], [12].

Third, the work introduces a novel Retrieval-Augmented Generation (RAG) pipeline, in which outputs from both the OCR (optical character recognition) and STT modules are indexed into a unified ChromaDB knowledge base. This architectural innovation enables cross-modal grounding: meeting minutes, slides, and transcribed dialogue are jointly retrievable, supporting downstream tasks such as semantic search, summarization, and context-aware language model prompting. The RAG layer leverages intfloat/multilingual-e5-small embeddings for dense semantic retrieval and is orchestrated through FastAPI, providing robust, low-latency endpoints for enterprise integration. By merging OCR and STT knowledge, the system overcomes fragmentation seen in legacy tools, delivering richer information access for all users [13], [9].

Fourth, the thesis demonstrates the integration of advanced diarization (pyannote-audio), signal pre-processing (librosa), and Azure-based orchestration for secure, enterprise-scale deployment. Microsoft Azure's AI Speech, AI Search, OpenAI, Blob Storage, Entra ID, Key Vault, and Graph API for Teams are integrated to provide authentication, persistent storage, and seamless access within the corporate environment. This end-to-end pipeline ensures that accessibility features are not an afterthought but are embedded from the ground up in enterprise digital workflows. Additionally, the deployment pipeline supports auditability and fine-grained access control, critical for sensitive corporate data.

Fifth, the thesis addresses the persistent challenge of domain-specific transcription errors through targeted vocabulary prompting and adaptation. For example, the observed mis-transcription of "Philippe Navratil" as "Philippe Naruto" in the proof-of-concept system underscores the inadequacy of generic models in handling domain-specific names and terminology. The system counteracts these errors by dynamically injecting custom vocabulary into model prompts and by expanding the RAG knowledge base with indexed corporate contacts and keyword lists. This approach is supported by research showing that domain adaptation, whether through model fine-tuning or data augmentation, yields measurable improvements in real-world ASR performance [10], [15].

Sixth, the thesis contributes a replicable, modular codebase, with explicit documentation and configuration options, enabling reproducibility and adaptation for other enterprise contexts or languages. All processing steps—from VAD to diarization, embedding-based retrieval, and prompt engineering—are exposed as modular services, facilitating both internal evaluation and external research reuse.

Finally, this work advances the theoretical understanding of accessible ASR deployment in complex, multilingual, and noisy environments. By synthesizing advances in acoustic modeling, language adaptation, and cross-modal retrieval, the thesis establishes a comprehensive framework for enterprise ASR accessibility. The research not only bridges gaps in current ASR deployments but also provides concrete, scalable solutions for inclusion and productivity in modern corporate settings.

AI Usage Acknowledgment: AI tools contributed to the drafting and editing of this text; however, all research directions, architectural designs, and technical analyses are exclusively the work of the author, OUEDRAOGO Yannick Ulrich Levi.

1.6 Thesis Structure

This thesis is structured to guide the reader systematically through the theoretical foundations, methodological innovations, experimental procedures, and practical implications of developing advanced speech-to-text (STT) models for noisy, multilingual corporate meetings, with a particular focus on accessibility for hearing-impaired employees at Nestlé in partnership with Burkina Institute of Technology. The organization of the thesis reflects both the complexity of automatic speech recognition (ASR) in enterprise environments and the multifaceted technical landscape encompassing model selection, domain adaptation, and systems integration.

Chapter 1 introduces the core research problem, contextualizes it in both the corporate and accessibility domains, and outlines the motivation for focusing on hybrid ASR systems leveraging Whisper and wav2vec 2.0. This chapter articulates the dual challenge of accurate transcription in acoustically adverse settings, such as open-plan offices and virtual meetings with heterogeneous background noise, and the necessity of domain adaptation to support hearing-impaired users. It highlights real-world errors observed during proof-of-concept deployments—such as the critical misrecognition of “Philippe Navratil” as “Philippe Naruto”—which underscore the inadequacy of generic ASR models in specialized business contexts. The objectives, research questions, and methodological overview are explicitly presented, providing a roadmap for the chapters that follow.

Chapter 2 provides a detailed literature review, synthesizing findings from recent work on acoustic and semantic modeling in ASR, with particular attention to the capabilities and limitations of Whisper and wav2vec 2.0 in multilingual and noisy settings. The review critically examines methods for model adaptation, noise robustness, and the integration of external linguistic resources, referencing pivotal studies on transformer-based architectures, prosody modification, and low-resource language adaptation strategies [1], [2], [12], [17]. The chapter further analyzes previous attempts at integrating ASR with accessibility platforms, drawing on empirical studies from both academic and industrial deployments [3], [11].

Chapter 3 details the theoretical framework underpinning the thesis, with a particular emphasis on transformer-based ASR systems and domain adaptation techniques. The mathematical foundations of attention mechanisms, self-supervised learning (as in wav2vec 2.0), and multilingual pretraining (as exemplified by Whisper) are rigorously developed. This chapter also explores the theory of knowledge base construction and retrieval-augmented generation (RAG), laying the groundwork for subsequent implementation of a shared ChromaDB layer that unifies outputs from both optical character recognition (OCR) and STT processes.

Methodological innovations are described in Chapter 4. This section delineates the system architecture, which combines faster-whisper (small INT8 variant) for efficient inference—achieving a real-time factor (RTF) of 0.25 on CPU, processing four minutes of audio in 62 seconds and generating 20 timestamped segments—with a modular backend orchestrated by FastAPI. Signal processing and diarization components, including Silero VAD, noisereduce, librosa, and pyannote-audio, are discussed in the context of their roles in noise suppression, segmentation, and speaker attribution [5], [6]. The RAG layer is highlighted as a key architectural contribution, where outputs of Topic 1 (OCR) and Topic 2 (STT) are integrated into a unified ChromaDB knowledge base, facilitating more effective retrieval and context-aware query responses [9].

Chapter 5 describes the experimental protocols. Data collection encompasses multilingual

meeting recordings drawn from actual Nestlé corporate sessions, supplemented with controlled noisy samples to reflect real-world acoustic conditions. Annotation strategies, evaluation metrics (including word error rate, segment accuracy, and named entity accuracy), and validation procedures are meticulously documented. The chapter also addresses ethical considerations, especially in relation to privacy, accessibility, and the use of synthetic augmentation to balance underrepresented classes [10], [7].

Subsequent chapters present and analyze experimental results. Chapter 6 evaluates baseline and adapted ASR models, with particular scrutiny of performance on hearing-impaired speech and domain-specific vocabulary. The impact of domain vocabulary prompting is examined, including revisiting the “Philippe Navratil !’ Philippe Naruto” error as an illustrative case of model limitations in name recognition. Comparative analysis with state-of-the-art methods and ablation studies on components such as embedding selection (intfloat/multilingual-e5-small) and language model integration (Ollama/Llama 3.2) are provided [15], [14].

Chapter 7 discusses deployment considerations in an enterprise setting. The integration with Microsoft Azure infrastructure—covering Azure AI Speech, Azure AI Search, Azure OpenAI, Blob Storage, Entra ID, Key Vault, and Teams integration via Graph API—is detailed, along with security, scalability, and compliance aspects. This chapter also returns to the architectural innovation of the shared ChromaDB knowledge base, emphasizing its role in supporting cross-modal retrieval and enterprise knowledge management.

The thesis concludes in Chapter 8 with a synthesis of findings, implications for accessibility and enterprise ASR deployment, and directions for future research. The persistent challenge of proper name recognition, as exemplified by the “Philippe Navratil !’ Philippe Naruto” error, is revisited in light of the experimental outcomes and architectural advances discussed throughout. The limitations of current approaches and the potential for further domain adaptation and multimodal integration are critically assessed.

Throughout the thesis, an appendix provides supplementary technical diagrams, code snippets, and extended experimental logs. A comprehensive bibliography, strictly following IEEE citation style, ensures all claims and technical assertions are transparently sourced.

Finally, an AI Usage Acknowledgment transparently states that AI tools assisted with writing and editing, while all research directions, experimental designs, and technical content remain the original work of OUEDRAOGO Yannick Ulrich Levi. This ensures academic integrity and clarity regarding the provenance of the scholarly contribution.

2. Background and Foundational Concepts

2.1 Automatic Speech Recognition: Core Concepts

Automatic Speech Recognition (ASR) constitutes the computational process of transcribing spoken language into written text. At its core, an ASR system models the complex mapping from acoustic waveforms to linguistic units, operating in real time and under diverse acoustic and linguistic conditions. The central challenge lies in accurately extracting linguistic content from speech signals, which are inherently variable due to speaker accents, background noise, channel distortions, and the idiosyncrasies of spontaneous speech. The goal of ASR in corporate settings, such as those at Nestlé, extends beyond routine voice transcription: it

must function robustly in multilingual, noisy environments and support accessibility for hearing-impaired employees, where even minor transcription errors can impede information flow and inclusivity.

Fundamentally, ASR systems comprise several interrelated components: feature extraction, acoustic modeling, language modeling, and decoding. Feature extraction converts raw audio into a sequence of compact, informative vectors, typically using representations such as Mel-frequency cepstral coefficients (MFCCs) or log-mel filterbanks. Recent advances have shifted towards end-to-end neural architectures, with models such as wav2vec 2.0 learning feature representations directly from waveform input, thus reducing dependence on engineered features and yielding improvements in noise robustness and cross-lingual generalisation [1]. Acoustic models map these features to probability distributions over phonemes, graphemes, or wordpieces, while language models capture the statistical regularities of word sequences, providing contextual constraints that steer decoding toward valid hypotheses. The decoder integrates these probabilities to output the most likely transcription.

Historically, ASR development followed a modular approach grounded in hidden Markov models (HMMs) and Gaussian mixture models (GMMs) for acoustic modeling, paired with n-gram language models. However, deep neural networks (DNNs), particularly those employing transformer architectures, have largely supplanted these traditional pipelines. Transformer-based models, such as Whisper, exhibit self-attention mechanisms that capture long-range dependencies in audio and text, enabling improved performance in continuous, overlapping, or accented speech [8]. End-to-end models unite acoustic and language modeling in a single trainable system, removing the need for hand-crafted intermediate representations and allowing joint optimisation for the final transcription objective.

The deployment of ASR in multilingual and noisy corporate meetings introduces a pronounced set of challenges. Multilingual ASR systems must contend with code-switching, dialectal variation, and the presence of named entities specific to the domain, such as proprietary terminology or employee names. For instance, a real-world error observed in Nestlé’s proof of concept—transcribing "Philippe Navratil" as "Philippe Naruto"—highlights the impact of out-of-vocabulary errors and the need for domain adaptation strategies. Robustness to noise remains a persistent difficulty, as reverberation, overlapping speech, and environmental sounds can degrade recognition accuracy. Techniques such as data augmentation with noise, robust feature extraction, and denoising modules (e.g., noisereduce and Silero VAD) have demonstrated efficacy in mitigating these effects [5].

The advent of self-supervised learning has propelled ASR performance, particularly in low-resource and noisy settings. Models such as wav2vec 2.0 and Whisper are pre-trained on vast quantities of unlabeled speech, learning to capture phonetic, prosodic, and linguistic cues from raw audio. Fine-tuning on domain-specific data enables these models to adapt to new accents, languages, and acoustic conditions with limited labeled samples [2]. This pretraining-finetuning paradigm is critical for enterprise deployment, where privacy concerns or the lack of annotated data often preclude large-scale supervised training.

Domain adaptation constitutes a central pillar of modern ASR deployments. Techniques such as adapter layers, low-rank adaptation (LoRA), and in-domain language model adaptation have been employed to tailor general-purpose models to specific corporate vocabularies and linguistic contexts [15]. In settings supporting hearing-impaired users, ASR accuracy requirements are heightened; errors in domain-specific terms, names, or technical

jargon can disproportionately affect accessibility [4]. Integration of external knowledge bases, on-the-fly vocabulary prompting, and retrieval-augmented generation (RAG) approaches (such as shared ChromaDB knowledge bases combining OCR and STT inputs) have proven valuable in addressing these gaps.

Noise robustness is another axis of innovation in contemporary ASR. Acoustic model adaptation, informed by in-domain noise profiles, can significantly improve intelligibility in adverse conditions [6]. Feature enhancement techniques, such as minimum mean square error (MMSE) estimation and spectral subtraction, are commonly utilised in pre-processing pipelines to suppress background interference [18]. In enterprise deployments, cascades of voice activity detection (Silero VAD), speaker diarisation (pyannote-audio), and denoising (noisereduce, librosa) support real-time, continuous processing of meeting audio, segmenting and cleaning input streams before transcription.

The operational requirements of an enterprise ASR system go beyond accuracy. Real-time factor (RTF), latency, and scalability are critical. For instance, with faster-whisper small INT8, a 4-minute audio recording can be processed in 62 seconds (RTF = 0.25), split into 20 timestamped segments. Such performance enables near-live feedback in meeting scenarios, supporting downstream accessibility features such as live captioning and searchable meeting archives [13].

Finally, the integration of ASR within broader enterprise IT infrastructure—such as Microsoft Azure's AI Speech, Search, and OpenAI services—demands secure, scalable, and privacy-preserving design. Authentication and role-based access (via Entra ID and Key Vault), secure storage (Blob Storage), and seamless integration with collaboration platforms (Teams via Graph API) are essential considerations for deployment in multinational corporations like Nestlé. The confluence of advanced neural models, domain adaptation, robust pre-processing, and enterprise-grade security forms the foundation of state-of-the-art ASR for modern corporate environments.

AI Usage Acknowledgment: AI tools were used for writing assistance and editing; all research directions, architecture decisions, and technical content are original work by OUEDRAOGO Yannick Ulrich Levi.

2.2 Evaluation Metrics for ASR Systems

Evaluating the performance of automatic speech recognition (ASR) systems in the context of noisy, multilingual corporate meetings—especially where the needs of hearing-impaired employees are paramount—demands a rigorous and multifaceted approach to metrics selection. The conventional dominance of word error rate (WER) as the primary benchmark persists due to its interpretability and direct correspondence with transcription accuracy. Nevertheless, the complexity of real-world deployment scenarios at Nestlé, with fluctuating acoustic conditions, speaker accents, and diverse languages, requires a broader spectrum of evaluation metrics to holistically capture system effectiveness and user impact.

WER is computed as the sum of substitutions, insertions, and deletions, divided by the total number of words in the reference transcription. Its simplicity supports comparative studies between models and datasets. However, WER fails to account for semantic equivalence, speaker intent, or the functional consequences of certain types of errors. For instance, in the observed proof-of-concept (PoC) scenario where "Philippe Navratil" was transcribed as "Philippe Naruto," WER registers only a single substitution, yet the real impact on meeting

comprehension and downstream tasks is far greater, demonstrating the limitations of WER in isolation for corporate environments where name recognition is vital [1].

Beyond WER, character error rate (CER) serves as a supplementary metric, particularly in languages with agglutinative morphology or in cases where word segmentation is ambiguous. CER calculates the proportion of character-level errors, thus providing finer granularity, especially useful for evaluating model performance on short utterances, named entities, and domain-specific jargon. Studies comparing CER and WER on morphologically rich languages such as Turkish and Amazigh have revealed that CER often surfaces subtle pronunciation and spelling errors that WER overlooks, a consideration relevant to Nestlé's multilingual workforce [2].

Sentence error rate (SER) and match error rate (MER) offer additional perspectives. SER focuses on the proportion of completely correct sentences, thus penalizing even minor word-level mistakes that disrupt the overall semantic flow, while MER adjusts for near-miss matches by considering word order and phrase boundaries. In meeting transcription scenarios, where utterances often span multiple sentences and contain overlapping speakers, these metrics can expose deficiencies in system robustness that might not be apparent from WER alone [11].

Latency and real-time factor (RTF) are crucial operational metrics in enterprise deployments. RTF measures the ratio of processing time to audio duration; a value below 1.0 indicates real-time performance. In the current implementation, faster-whisper small INT8 achieved an RTF of 0.25 on CPU—transcribing four minutes of audio in approximately 62 seconds, yielding 20 timestamped segments. Such latency metrics directly inform system suitability for live captioning or near-real-time meeting summarization, which are particularly beneficial to hearing-impaired participants who rely on timely access to spoken content [17].

Robustness to noise and speaker variability is another axis of evaluation, especially in open-plan or hybrid meeting environments. Signal-to-noise ratio (SNR) degradation tests, recognition accuracy under simulated reverberant and multi-talker conditions, and performance across diverse microphone types should be systematically assessed. Notably, the adoption of signal enhancement tools such as noisereduce and Silero VAD within this system provides an empirical basis for measuring incremental gains in transcription accuracy under adverse acoustic conditions. Recent research has stressed the necessity of reporting noise-specific WER and CER values, as aggregate scores often mask critical performance drops in challenging meeting segments [5].

Multilingual and code-switched speech introduces further complexity. Metrics such as language recognition accuracy and code-switch detection rate become important, particularly in environments like Nestlé's, where French, English, and local African languages may be spoken interchangeably. Evaluations of Whisper and wav2vec 2.0 have shown that token-level language identification and fine-grained error analysis by language segment are necessary to uncover systematic biases or deficiencies in minority language recognition [8].

Another key dimension is semantic consistency and task-relevance. For users whose primary need is information retrieval or meeting summarization, semantic error rate (SemER) and content word accuracy (CWA) provide more meaningful indicators than surface-level transcription fidelity. SemER evaluates whether the recognized content preserves the core meaning, even in the presence of minor lexical errors, while CWA

focuses on nouns, verbs, and critical entities—a crucial consideration for hearing-impaired employees who depend on accurate capture of action items and participant contributions [9].

User-centered evaluation, encompassing subjective metrics such as mean opinion score (MOS) and task completion rate, is essential when deploying ASR for accessibility. Hearing-impaired users may judge output quality differently from the general population, prioritizing clarity of key terms, correct rendering of proper names, and latency over minor grammatical errors. Incorporating feedback loops, such as explicit correction prompts or post-meeting edit logs, should be considered part of the evaluation framework [4].

Finally, in the enterprise context, business impact metrics—such as reduction in manual transcription costs, increase in meeting productivity, and improvements in inclusion for hearing-impaired staff—should round out the evaluation suite. These holistic metrics, while downstream from technical ASR outputs, provide critical feedback for model selection and deployment strategies at scale [3].

In summary, rigorous ASR system evaluation for noisy, multilingual corporate meetings must extend beyond legacy WER and latency figures to encompass character-level granularity, semantic preservation, user-centricity, and real-world business outcomes. Only through such comprehensive, contextually grounded metrics can the true value and limitations of advanced speech-to-text models be understood and optimized for accessibility and operational excellence.

2.3 Characteristics of Corporate Meetings

Corporate meetings present a confluence of linguistic, acoustic, and interactional complexities that deeply affect the performance and design of automatic speech recognition (ASR) systems, especially in large multinational enterprises such as Nestlé. Unlike controlled laboratory conditions, real-world corporate environments expose ASR models to a barrage of challenges—overlapping speech, diverse accents, frequent code-switching, and unpredictable background noise. Understanding these characteristics is essential for constructing robust models that can serve both typical users and those with hearing impairments.

Corporate meetings are inherently multilingual, particularly in global organizations operating across continents. Employees often switch between languages—French, English, Portuguese, and local dialects in African subsidiaries, for instance—sometimes within a single utterance. This phenomenon, termed intra-sentential code-switching, complicates language identification and transcription accuracy, requiring models like Whisper and wav2vec 2.0 to handle multilingual input without significant degradation in performance[8]. The challenge is not merely linguistic; code-switching typically correlates with shifts in pronunciation and prosody, further stressing acoustic models.

Another salient feature of corporate meetings is the diversity of speaker profiles. Participants range from executives delivering rehearsed statements to technical staff engaging in spontaneous dialogue. Variability in speech rate, hesitations, and use of domain-specific jargon is pervasive, resulting in significant intra-corpus variance. ASR systems trained predominantly on scripted or broadcast speech often fail to generalize to such spontaneous, conversational data, motivating the need for domain adaptation techniques[10]. For example, jargon such as “SKU rationalization” or names like “Philippe Navratil” can pose recognition difficulties, as evidenced by real transcriptions confusing “Navratil” with the pop-

culture term “Naruto”—a nontrivial error in business documentation and accessibility[4].

Acoustic environments in corporate settings are far from ideal. Meetings frequently occur in conference rooms with varying levels of reverberation, HVAC noise, and intermittent background conversations. Remote participants introduce additional artifacts—packet loss, compression noise, and variable microphone quality—particularly acute in hybrid meetings blending in-person and virtual attendees. Such acoustic variability can cause marked declines in word error rate (WER), especially for hearing-impaired users who rely heavily on accurate real-time captioning[12]. Techniques such as acoustic adaptation, speech enhancement using noisereduce, and voice activity detection (VAD) with Silero VAD or pyannote-audio become critical in pre-processing pipelines[17].

Turn-taking in corporate meetings is highly dynamic, often featuring interruptions, simultaneous speech, and rapid speaker changes. Accurate diarization—determining “who spoke when”—is crucial not only for attribution but also for semantic coherence in the resulting transcripts. Systems like pyannote-audio have been adopted to segment and assign speech segments to individual speakers, yet challenges persist when diarization is confounded by poor audio quality or frequent speaker overlap. This complexity is magnified in large meetings, where the number of unique voices can exceed the capacity of naive clustering algorithms, demanding more sophisticated diarization solutions and robust embeddings such as intfloat/multilingual-e5-small[7].

A further idiosyncrasy of corporate discourse is the prevalence of multimodal cues. Visual aids—slides, shared screens, physical whiteboards—play a significant role in communication. Traditional ASR models fail to integrate such visual context, yet advanced systems now exploit OCR (optical character recognition) outputs to enrich the semantic representation of meetings. By integrating OCR and STT (speech-to-text) outputs into a unified ChromaDB knowledge base, as in the proposed RAG (retrieval-augmented generation) architecture, it becomes possible to cross-reference spoken and written content, improving overall comprehension and downstream searchability[14].

Accessibility requirements add another dimension of complexity. Hearing-impaired employees depend on high-quality, low-latency ASR outputs for equitable participation. Studies have demonstrated that even minor transcription errors can have disproportionate impacts on comprehension for this group, especially when domain-specific terminology is involved[3]. The integration of domain adaptation techniques, custom vocabulary prompts, and post-processing modules is thus not optional, but foundational, to foster true inclusivity[16].

Temporal alignment—providing precise timestamps for each utterance or segment—is another crucial requirement in corporate contexts. Accurate timestamping facilitates actions such as meeting minute generation, speaker accountability, and efficient retrieval of information post-meeting. State-of-the-art ASR systems achieve this via segment-wise alignment, as demonstrated by faster-whisper small INT8, which can process four minutes of audio in 62 seconds, yielding 20 timestamped segments on a standard CPU. Such performance metrics are not merely technical achievements; they directly translate into real-world productivity gains, especially for large organizations processing thousands of meeting hours monthly[2].

Lastly, security and privacy constraints are non-negotiable in corporate settings. Sensitive business information, strategic discussions, and employee data must be protected

throughout the ASR pipeline. Enterprise deployments, such as those leveraging Microsoft Azure's AI Speech, AI Search, OpenAI, and secure storage solutions, provide compliance with international standards and facilitate integration with existing authentication mechanisms like Entra ID and secure key management via Azure Key Vault[15].

In summary, corporate meetings constitute a challenging, heterogeneous environment for ASR, characterized by linguistic diversity, acoustic unpredictability, dynamic interactional structures, and stringent accessibility and security demands. A nuanced understanding of these characteristics is fundamental to the development of advanced speech-to-text models that can deliver equitable, high-quality access for all employees, including those with hearing impairments.

2.4 Accessibility in the Corporate Environment

The pursuit of accessibility within corporate environments has acquired renewed urgency as organizations increasingly recognize both legal imperatives and ethical responsibilities related to inclusivity. For employees with hearing impairments, the barrier to full participation in spoken interactions—such as meetings, collaborative sessions, and training—remains substantial, particularly in acoustically challenging, multilingual settings typical of global enterprises like Nestlé. Advanced speech-to-text (STT) technologies, notably those utilizing Whisper, wav2vec 2.0, and domain adaptation techniques, are positioned at the forefront of efforts to address these barriers, offering the potential to transform accessibility for hearing-impaired professionals by rendering spoken content into accurate, context-sensitive textual representations [3].

Accessibility, as defined by the World Health Organization and embodied in standards such as the Web Content Accessibility Guidelines (WCAG) and Section 508, extends beyond mere technical provision of captions or transcripts. True accessibility in the corporate environment requires that STT outputs be timely, accurate, and contextually aware, especially for specialized company terms, multilingual exchanges, and domain-specific jargon [4]. In practice, this means that conventional STT systems, which may perform sufficiently in quiet, monolingual scenarios, often falter under real-world corporate conditions marked by overlapping speech, diverse accents, code-switching, and background noise (citation needed).

Research indicates that the effectiveness of accessibility technologies for hearing-impaired users is highly dependent on their ability to minimize word error rates (WER) in authentic corporate acoustic environments [6]. For instance, meeting rooms with multiple simultaneous speakers, the hum of air conditioning, and the presence of digital devices all introduce nontrivial acoustic interference. Moreover, international companies like Nestlé conduct meetings in several languages, frequently alternating between French, English, and local dialects. Standard ASR models, when unadapted, often misrecognize named entities and domain-specific vocabulary, leading to misleading or unusable transcripts. The real-world "Philippe Navratil !' Philippe Naruto" transcription error, observed in a proof of concept system, exemplifies the criticality of accurate domain adaptation for accessibility; such errors can obscure key participant contributions and erode trust in the system's reliability.

Modern STT architectures, such as those based on Whisper and wav2vec 2.0, have demonstrated substantial improvements in multilingual and noisy environments, particularly when combined with robust pre-processing pipelines—including voice activity detection

(Silero VAD), noise reduction (noisereduce), and advanced segmentation (pyannote-audio) [1]. These pipelines, augmented by feature extraction libraries such as librosa and domain-specific embeddings like intfloat/multilingual-e5-small, facilitate adaptation to corporate lexicons and user needs. However, the deployment context remains central: enterprise infrastructure, such as Microsoft Azure's AI Speech and OpenAI services, enables large-scale, secure, and policy-compliant deployment, but necessitates careful integration with organizational authentication (Entra ID), data governance (Key Vault, Blob Storage), and collaboration tools (Graph API for Teams) [13].

A key architectural advancement in this thesis is the integration of both optical character recognition (OCR) and STT outputs into a unified ChromaDB knowledge base via a retrieval-augmented generation (RAG) layer. This design ensures that hearing-impaired users receive not only transcripted speech but also contextual information from shared documents, slides, and chat logs—enhancing the accessibility and completeness of information available during and after meetings. By facilitating multimodal retrieval across speech and text, such systems allow users to query for specific terms or concepts and receive contextually expanded answers, improving both comprehension and participation [11].

While the technical sophistication of ASR models continues to improve—demonstrated, for example, by real-world results where faster-whisper small INT8 achieved a real-time factor (RTF) of 0.25 on a CPU, processing four minutes of audio in 62 seconds and segmenting into 20 timestamped blocks—full accessibility depends equally on the system's ability to adapt to domain vocabulary, handle code-mixing, and provide low-latency, user-friendly interfaces. FastAPI enables rapid service orchestration, while ChromaDB's semantic search capabilities, powered by efficient embeddings, offer scalable retrieval of relevant content even in large, multilingual corpora [7].

Nevertheless, accessibility cannot be detached from broader social and organizational practices. Studies show that technical solutions, to be effective, must be accompanied by sensitization of staff, ongoing evaluation with hearing-impaired users, and iterative refinement based on real-world feedback [19]. Corporate environments that systematically incorporate accessibility into their digital transformation agendas report improved employee retention, satisfaction, and productivity, especially among those with disabilities. Additionally, accessible STT systems generate secondary benefits, such as searchable meeting archives and compliance with legal mandates for equal opportunity [16].

In summary, accessibility in the corporate environment is contingent on more than just technological advancement. It is the product of interdisciplinary synthesis—combining state-of-the-art STT models, enterprise-grade deployment, and deep engagement with the lived realities of hearing-impaired employees. Progress in ASR accuracy, multilingual adaptation, and knowledge retrieval architectures, as realized in this research, is essential but must be continuously evaluated and refined within the organizational context to deliver tangible accessibility outcomes. The ongoing collaboration between BIT and Nestlé provides a crucial testbed for these innovations, demonstrating both the challenges and the promise of truly accessible corporate communication.

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technical analysis, and content represent the original work of the author, OUEDRAOGO Yannick Ulrich Levi.

3. Literature Review

3.1 Evolution of Automatic Speech Recognition Architectures

From the earliest attempts at machine-driven speech transcription to the current state-of-the-art models, the evolution of automatic speech recognition (ASR) architectures has been shaped by the persistent challenges of noise, speaker variability, and language diversity—challenges that are acutely magnified in the context of noisy, multilingual corporate meetings. The earliest ASR systems of the 1950s and 1960s, such as Bell Labs' isolated digit recognizers, relied on template matching and simple acoustic-phonetic approaches, which proved brittle outside very constrained vocabularies and environments. Over subsequent decades, statistical models, most notably Hidden Markov Models (HMMs), became the dominant paradigm, combining probabilistic modeling of temporal sequences with Gaussian Mixture Models (GMMs) for emission probabilities. These HMM-GMM systems achieved reasonable performance on clean, read speech but struggled with spontaneous speech, overlapping speakers, and background noise, all of which are typical in real-world corporate meeting scenarios [6].

A major theoretical breakthrough arrived with the integration of deep neural networks (DNNs) for acoustic modeling. Early DNN-HMM hybrids, trained on large corpora such as Switchboard and LibriSpeech, vastly improved the modeling of complex, non-linear relationships between audio features and phoneme states. Deep architectures, particularly those based on recurrent neural networks (RNNs) and later long short-term memory (LSTM) cells, enabled more robust temporal context modeling, crucial for handling the rapid speaker turn-taking and code-switching common in multilingual meetings [20]. However, these hybrid systems retained the cumbersome separation between acoustic, pronunciation, and language models, requiring intricate feature engineering and domain-specific expertise.

End-to-end ASR architectures emerged in the mid-2010s as a response to these limitations, with Connectionist Temporal Classification (CTC) and attention-based encoder-decoder models eliminating the need for explicit alignment and lexicon constraints. Models such as Deep Speech and Listen, Attend and Spell (LAS) demonstrated that a single neural network could learn to map raw audio to text directly. This paradigm shift paved the way for more unified, data-driven approaches that could, in principle, be adapted across languages and acoustic environments with minimal manual intervention [9].

The advent of self-supervised learning has further transformed ASR. Models like wav2vec 2.0 introduced contrastive pretraining on large unlabeled audio corpora, learning powerful representations that can be fine-tuned with limited labeled data for downstream speech recognition tasks. Wav2vec 2.0's convolutional feature encoder and Transformer-based context network capture both local and long-range acoustic dependencies, which has proven essential for recognizing speech in adverse conditions such as overlapping speakers, reverberation, and non-native accents frequently encountered in corporate meetings [1]. Furthermore, the self-supervised paradigm facilitates cross-lingual transfer, crucial for organizations like Nestlé, where meetings often involve English, French, and local

African languages [7].

Another leap in ASR capability was marked by the introduction of large-scale Transformer-based models, exemplified by OpenAI's Whisper architecture. Whisper combines a convolutional frontend with a Transformer encoder-decoder, trained on 680,000 hours of multilingual and multitask speech data. This architecture achieves robust recognition across dozens of languages, supports transcription and translation tasks, and demonstrates remarkable resilience to noise and accented or code-switched speech [2]. Unlike previous systems, Whisper's multitask training setup allows it to generalize effectively even to under-resourced languages and noisy audio, which is characteristic of corporate teleconferences and hybrid meetings [8].

Recent research has focused on further enhancing ASR robustness through explicit adaptation mechanisms and modular integration. Domain adaptation, involving techniques such as Low-Rank Adaptation (LoRA) and prosody modification, tailors pre-trained models like Whisper and wav2vec 2.0 to specific corporate vocabularies and acoustic profiles [15]. For example, LoRA adapters enable rapid, parameter-efficient adaptation to enterprise jargon and speaker accents, mitigating errors such as the "Philippe Navratil !' Philippe Naruto" misrecognition observed in real-world deployments. Additionally, prosody-based acoustic adaptation has been used to improve intelligibility for hearing-impaired users and under noisy meeting conditions [17].

A further dimension in the evolution of ASR systems lies in addressing the accessibility needs of hearing-impaired participants. Classical ASR architectures assumed normal-hearing users and failed to account for the distinct prosodic and articulatory patterns found in the speech of hearing-impaired individuals. Recent studies have highlighted the importance of acoustic model adaptation and frequency modulation techniques tailored to such user groups [4]. Embedding these adaptations within enterprise ASR platforms fosters inclusive communication, enabling equitable participation for all employees [19].

Finally, the practical deployment of advanced ASR systems in enterprise environments has necessitated the integration of a range of auxiliary technologies. Voice activity detection (VAD) modules, such as Silero VAD, and denoising algorithms, including noisereduce and librosa, serve as critical preprocessing steps, isolating speech segments and filtering background noise before transcription. Modern frameworks, such as faster-whisper, have optimized inference speed and resource utilization—achieving a real-time factor (RTF) of 0.25 on CPUs with INT8 quantization, processing four minutes of audio in just over a minute and yielding 20 timestamped segments. These capabilities are essential for real-time applications in large organizations [3]. The deployment stack further incorporates scalable APIs (FastAPI), vector-based retrieval (ChromaDB), speaker diarization (pyannote-audio), and semantic search (intfloat/multilingual-e5-small embeddings and Ollama/Llama 3.2), forming the backbone of robust, context-aware ASR solutions for demanding corporate scenarios.

The cumulative trajectory of ASR research thus reflects a shift from brittle, handcrafted models to large-scale, contextually adaptive architectures. These advances are not merely theoretical; they have tangible consequences for the accessibility, inclusiveness, and efficiency of multilingual, noisy corporate meetings—especially for users with hearing impairments. The integration of domain adaptation, self-supervised pretraining, and modular, scalable software engineering marks the current state-of-the-art and sets the foundation for the system developed in this thesis.

3.2 Whisper: Large-Scale Multilingual ASR

Whisper has emerged as a seminal architecture in large-scale, multilingual automatic speech recognition (ASR), reshaping the technical landscape for enterprise transcription in challenging, real-world settings. Unlike previous models constrained by limited linguistic coverage or insufficient robustness to environmental noise, Whisper's transformer-based encoder-decoder framework is explicitly designed to handle diverse audio inputs across dozens of languages, dialects, and accents, even under adverse acoustic conditions [8]. This breadth is particularly pertinent to multinational corporate environments such as Nestlé, where meetings routinely involve participants speaking multiple languages, often in acoustically imperfect rooms, with background noise from HVAC systems, laptops, and peripheral conversations.

Several factors contribute to Whisper's success as a large-scale multilingual ASR model. First, its architecture is built on a transformer backbone trained on 680,000 hours of weakly supervised audio-text pairs, representing an unprecedented breadth of linguistic and acoustic diversity [2]. This massive training corpus enables Whisper to generalize effectively to out-of-domain audio, including noisy corporate meeting recordings and accented speech, both of which are common in enterprise settings [1]. A second pillar of Whisper's performance lies in its language identification and tokenization pipeline: the model dynamically detects the spoken language and applies a shared subword vocabulary, reducing out-of-vocabulary errors and improving resilience to domain-specific jargon.

Notably, Whisper's open-source release has catalyzed rapid innovation in both research and applied settings. For this thesis, the faster-whisper implementation—optimized for CPU inference and quantized to INT8 precision—enabled real-time transcription at a Real-Time Factor (RTF) of 0.25 on commodity CPU hardware. This configuration processed a 4-minute meeting recording in just 62 seconds, producing 20 timestamped segments with high temporal fidelity. Such performance, validated on corporate meeting audio, demonstrates Whisper's feasibility for near-live closed-captioning, which is critical for hearing-impaired employees and live documentation needs [3].

A distinctive feature of Whisper in the context of multilingual corporate ASR is its effective handling of code-switching and cross-lingual transitions within a single audio stream. Meetings at Nestlé often involve speakers alternating between English and French, or integrating terms from local languages. Whisper's training on multi-language audio allows it to segment and transcribe these utterances accurately. However, the model is not immune to domain-specific errors. For instance, during system proof-of-concept testing, the transcription engine rendered the French employee name "Philippe Navratil" as "Philippe Naruto"—a semantically and phonetically plausible but contextually incorrect tokenization. Such errors highlight the necessity for domain adaptation and vocabulary prompting to further refine Whisper's outputs in enterprise contexts [2].

From a technical perspective, Whisper's transformer architecture introduces several innovations relevant to noisy environments and speaker variability. The model incorporates log-mel spectrogram front-ends, robust to fluctuations in loudness and background interference, and employs context-aware decoding strategies that maintain coherence over long audio segments [17]. Moreover, the model's attention mechanisms facilitate better temporal alignment, which, when combined with forced alignment tools such as pyannote-audio and Silero VAD, enhances both segmentation precision and downstream diarization.

Research demonstrates that Whisper's multilingual capabilities outperform many traditional ASR systems when transcribing under-resourced or accented speech. In comparative studies, Whisper maintained lower error rates across diverse languages, including those with limited training data, by leveraging cross-lingual parameter sharing and self-supervised pre-training [15]. However, the model's generic vocabulary and lack of explicit domain adaptation can still result in occasional phrase substitutions or misrecognition of specialized terms, particularly in technical or organizational lexicons [10].

The practicality of Whisper in corporate deployments is further enhanced by its integration within modern AI architectures. In the system developed for this thesis, Whisper operates in concert with FastAPI for API orchestration, ChromaDB for persistent vector storage, and auxiliary modules such as noisereduce and librosa for pre-processing. The integration of OCR (Optical Character Recognition) and STT (Speech-to-Text) outputs into a unified ChromaDB knowledge base exemplifies a novel retrieval-augmented generation (RAG) architecture, enabling downstream LLMs (Large Language Models) to ground generative outputs in both speech and document modalities. This unified knowledge base approach supports advanced semantic retrieval and context-aware summarization, addressing the complex information retrieval needs of enterprise users [9].

Despite Whisper's remarkable advancements, limitations remain, particularly in the transcription of low-resource languages, rapid code-switching, and specialized vocabulary. Ongoing research explores fine-tuning with LoRA adapters and prompt-based domain adaptation to mitigate these weaknesses [11]. For hearing-impaired employees, such enhancements are vital to ensure accessibility and semantic accuracy in meeting transcripts. The adoption of Whisper and its derivatives in enterprise environments thus represents a significant step forward, but must be accompanied by targeted adaptation strategies to address domain-specific and inclusion requirements.

In summary, Whisper's large-scale, multilingual ASR architecture establishes a robust foundation for transcription in noisy, multilingual corporate meetings. Its design, empirical performance, and ecosystem integration position it as a central component in support of accessibility and operational efficiency for diverse enterprise contexts, though ongoing research and adaptation remain necessary to bridge the remaining gap between generalist ASR models and the nuanced needs of real-world organizations.

3.3 wav2vec 2.0 and Self-Supervised Learning

The introduction of wav2vec 2.0 marked a pivotal advance in automatic speech recognition (ASR), particularly for scenarios marked by noise, multilingualism, and limited training resources. Building on the original wav2vec model, wav2vec 2.0 utilizes self-supervised learning to directly model raw audio waveforms, discarding the need for labor-intensive phonetic or frame-level annotation and thus significantly lowering the barrier for ASR model development in low-resource settings [1]. This is especially critical within the context of corporate meetings at Nestlé, where a multitude of languages, accents, and unpredictable acoustic environments converge, and where accessibility requirements for hearing-impaired employees further complicate the ASR challenge [4].

At its core, wav2vec 2.0 employs a two-stage training regime. In the self-supervised pretraining phase, a convolutional feature encoder processes raw audio into latent representations, which are then discretized via quantization. The model is trained to

distinguish masked (hidden) segments from distractors, enabling it to learn contextually rich, language-agnostic representations from massive unlabelled audio datasets [7]. This pretraining is followed by a supervised fine-tuning phase, where a connectionist temporal classification (CTC) loss is applied to align the learned representations with transcribed text using a much smaller set of labelled data. Such an approach is inherently data-efficient; it leverages abundant unlabelled speech, which is far easier to collect than curated transcriptions, and then adapts to specific domains or languages with minimal additional supervision.

This architecture confers several advantages in the context of noisy, multilingual corporate meetings. First, by extracting high-level audio features that are robust to environmental distortions, wav2vec 2.0 outperforms traditional ASR systems reliant on handcrafted spectral features, especially under suboptimal acoustic conditions [5]. The model's ability to generalize across diverse speakers and environments stems from its exposure to varied input during pretraining, making it well-suited for deployments where meeting rooms, microphones, and speaker characteristics change frequently. In the Nestlé/BIT deployment, meeting audio is often characterized by echo, overlapping speech, and diverse background noise profiles—conditions under which classical ASR systems tend to fail or produce high word error rates. Empirical studies have demonstrated that wav2vec 2.0 maintains intelligibility and accuracy even as signal-to-noise ratios degrade, a property vital for transcription reliability [6].

A further theoretical merit of self-supervised learning, as embodied in wav2vec 2.0, is its capacity for domain adaptation and transfer learning. Because the model's pretraining phase is not tied to any single language or accent, it can serve as a foundation for multilingual ASR systems. When fine-tuned on domain-specific data—such as in-house corporate vocabulary, technical jargon, and accents prevalent at Nestlé locations—wav2vec 2.0 adapts its representations to reflect the target distribution [10]. This ability is particularly relevant to the observed error in the proof-of-concept phase, where the system transcribed the name "Philippe Navratil" as "Philippe Naruto." Such errors, which stem from insufficient domain adaptation and the lack of specialized vocabulary, demonstrate the practical necessity of further fine-tuning and custom prompt engineering in real-world applications.

From an accessibility perspective, wav2vec 2.0's strength in low-resource and noisy settings is of direct benefit to hearing-impaired employees. Studies have shown that hearing-impaired users encounter significant barriers in real-time speech comprehension, especially in acoustically complex environments [19]. By improving transcription quality and reducing error rates, especially for less-common names and technical terms, wav2vec 2.0-based systems contribute meaningfully to workplace inclusion. The integration of wav2vec 2.0 into the broader architecture—alongside components such as Silero voice activity detection, pyannote-audio for speaker diarization, and the faster-whisper small INT8 model for efficient inference—enables a pipeline that is both accurate and scalable. In this system, preprocessing with noisereduce and librosa helps to further mitigate adverse acoustic effects before feature extraction, thereby enhancing the robustness of the downstream ASR model.

Another key advantage of wav2vec 2.0 lies in its compatibility with Retrieval-Augmented Generation (RAG) frameworks. In the presented system, outputs from both OCR (optical character recognition) and STT (speech-to-text) are indexed within a shared ChromaDB vector database, creating a unified knowledge base for subsequent information retrieval and

question answering via Ollama/Llama 3.2 [8]. Wav2vec 2.0's high-fidelity speech representations ensure that the transcribed content is not only accurate, but also semantically rich, thereby enhancing the effectiveness of downstream embedding (e.g., with intfloat/multilingual-e5-small) and retrieval processes.

In summary, wav2vec 2.0 exemplifies the maturation of self-supervised learning in ASR, providing a theoretically grounded and empirically validated foundation for robust, multilingual, and accessible speech recognition in complex corporate environments. Its role within the Nestlé/BIT system is not isolated but synergistic, contributing to a pipeline that addresses the needs of both average and hearing-impaired users, supports enterprise-scale deployment on Microsoft Azure infrastructure, and sets the stage for continued expansion through domain adaptation and integrated retrieval. Subsequent chapters will discuss empirical results and further architectural optimizations, including strategies for mitigating domain-specific errors such as "Philippe Navratil !' Philippe Naruto," which remain an active area of research and system refinement.

3.4 Conformer and Hybrid Architectures

Convolutional, recurrent, and transformer-based models have each contributed to advancements in automatic speech recognition (ASR), but their limitations in handling noisy, multilingual, and domain-specific corporate audio are now being addressed by newer architectures. Among these, Conformer and hybrid models have emerged as pivotal for robust ASR in challenging environments such as multilingual corporate meetings, where overlapping speakers, diverse accents, and background noise are prevalent. The Conformer architecture, which fuses convolutional and transformer layers, has demonstrated superior performance in sequence modeling tasks by capturing both local acoustic dependencies and long-range contextual relationships within speech utterances [1].

Traditional transformer-based ASR models, such as those underlying Whisper, are adept at modeling sequential data and global dependencies through self-attention mechanisms. However, they may inadequately capture fine-grained temporal features that are critical in noisy or highly variable acoustic conditions. Conformer models address this by interleaving convolutional modules—responsible for extracting local features and preserving temporal order—with multi-headed self-attention layers that aggregate information over longer spans. This synergy results in increased robustness to background noise and speaker variability, both of which are common in open-plan or hybrid corporate meeting settings. For instance, the convolutional layers effectively filter local noise patterns, while the transformer blocks ensure that long-term dependencies critical for language modeling are retained [2].

Hybrid architectures further extend this paradigm by integrating heterogeneous model components, such as combining transformer-based encoders with convolutional or recurrent front-ends, and coupling them with external language models or domain-specific adapters. This flexibility is particularly relevant in corporate contexts where standard ASR models are prone to domain drift, misrecognition of named entities, and poor handling of code-switching between languages. The observed error in transcribing "Philippe Navratil" as "Philippe Naruto" in the proof-of-concept (PoC) system exemplifies the limitations of purely transformer-based systems in specialized domains and underscores the necessity of hybridization and domain adaptation to minimize critical misrecognitions during meetings.

Recent research has demonstrated that Conformer architectures, when employed in

multilingual ASR systems, outperform traditional transformer and CNN-RNN hybrids on metrics such as word error rate (WER) and robustness to acoustically adverse conditions [8]. These models are particularly adept at handling cross-lingual phonetic variation—a frequent occurrence in multinational corporate settings—by jointly learning language-invariant features and language-specific cues. For example, in comparative studies on diverse speech corpora, Conformer-based systems have achieved marked improvements in the transcription of non-native speech, which is directly relevant to the Nestlé/BIT project context [15].

Incorporating hybrid Conformer architectures into the present ASR pipeline has required a careful selection and integration of model components. The current system employs faster-whisper for efficient decoding, combined with frontend modules such as Silero VAD for voice activity detection and noisereduce for environmental noise suppression, ensuring that the input to the Conformer-based encoder is well-conditioned. Subsequently, domain-specific embeddings, derived from intfloat/multilingual-e5-small, and knowledge augmentation via a ChromaDB-backed retrieval-augmented generation (RAG) layer, allow the system to contextualize ambiguous terms and named entities using both speech and OCR outputs. This hybrid design directly addresses the challenge of misrecognizing names and domain vocabulary, as seen in the aforementioned transcription error, by supplementing acoustic evidence with external knowledge bases.

Moreover, the hybrid ASR system exploits the strengths of Conformer models for low-latency, real-time processing. Using the faster-whisper small INT8 model, an observed real-time factor (RTF) of 0.25 was achieved on standard CPUs, with four minutes of meeting audio processed in just 62 seconds and efficiently segmented into 20 timestamped utterances. This performance affirms the practical viability of deploying such architectures at scale in enterprise settings, where latency and resource constraints are critical [11].

Hybrid Conformer architectures also facilitate domain adaptation through techniques such as Low-Rank Adaptation (LoRA), external language model fusion, and the inclusion of retrieval-augmented modules. For example, LoRA adapters enable efficient fine-tuning of the Conformer backbone with small, domain-specific datasets—an approach shown to reduce domain-specific error rates without incurring the prohibitive computational cost of full model retraining [15]. Similarly, the integration of a RAG layer using a unified ChromaDB knowledge base as a backend allows both OCR-derived and ASR-derived content to inform the model's predictions, thereby systematically reducing the frequency of domain-specific recognition errors.

The hybrid approach is further justified by the particular needs of hearing-impaired employees, for whom ASR output fidelity and the correct rendering of specialized terminology are critical for equitable participation in meetings [4]. By combining the temporal acuity of convolutional modules, the contextual sensitivity of transformers, and the adaptability of retrieval-augmented and LoRA-based mechanisms, the hybrid Conformer architecture deployed in this work advances the state of the art for robust, accessible, and multilingual ASR in complex corporate environments.

In summary, the adoption of Conformer and hybrid architectures provides a theoretically grounded and empirically validated framework for meeting the dual challenges of noise robustness and domain adaptability in enterprise ASR deployments. Their ability to integrate heterogeneous sources of information, adapt to specialized vocabularies, and operate efficiently on standard hardware makes them highly suitable for the Nestlé/BIT context and

similar multilingual, accessibility-focused applications [3].

3.5 Streaming and Low-Latency ASR Models

Streaming and low-latency automatic speech recognition (ASR) models have become indispensable for real-time transcription in enterprise contexts, particularly where multilingual and noisy corporate meetings pose significant challenges. Unlike traditional batch-based ASR systems that process entire audio files after recording, streaming ASR models provide incremental transcription outputs as audio is received, yielding substantial reductions in end-to-end processing delay. For environments such as Nestlé's global operations, where rapid feedback, accessibility for hearing-impaired employees, and integration with collaboration platforms are critical, achieving both low latency and high transcription fidelity is paramount. This section evaluates the theoretical foundations, architectural innovations, and practical trade-offs of streaming ASR, drawing on the latest research and direct empirical observations from the deployment of Whisper, wav2vec 2.0, and domain-specific adaptations.

Streaming ASR differs fundamentally from offline models in that inference occurs in tandem with audio input, typically by segmenting audio into overlapping windows and generating partial hypotheses. This yields advantages for live captioning and interactive applications, where latency under 500 milliseconds is often required to preserve conversational flow. However, the incremental nature of streaming inference imposes constraints on model design, particularly regarding bidirectional context and lookahead. For instance, models such as Whisper, which were originally devised for offline transcription, must be adapted or pruned for streaming to meet enterprise requirements. As documented in recent comparative analyses, the deployment of INT8-quantized faster-whisper small models enables real-time factor (RTF) of 0.25 on standard CPUs, processing four minutes of audio in just 62 seconds while producing 20 timestamped segments—demonstrating that streaming is not solely a GPU-accelerated capability but can be achieved efficiently on commodity hardware as well [2].

The architectural core of streaming ASR rests on the balance between latency and accuracy. While offline models can exploit bidirectional attention and non-causal convolutions to maximize word error rate (WER) performance, streaming models must rely on causal or limited-context mechanisms. Wav2vec 2.0, which is based on self-supervised learning of contextualized speech representations, has been extended for online use by truncating context windows and employing chunk-based inference, albeit sometimes at the cost of reduced recognition accuracy in highly variable acoustic conditions [1]. To mitigate these degradations, research has focused on fine-tuning with domain-specific corpora and employing dynamic buffer sizes to capture relevant context without incurring excessive lag. For instance, low-rank adaptation (LoRA) techniques have been shown to enhance Whisper's streaming performance in data-sparse settings, particularly for under-resourced languages and non-standard accents [15].

Noise and overlapping speech—the norm rather than the exception in enterprise meetings—pose additional hurdles for streaming ASR. Voice activity detection (VAD) frontends, such as Silero VAD, are indispensable in segmenting continuous audio streams into meaningful utterances, reducing false positives from background noise. Complementary preprocessing techniques, including spectral subtraction and the noisereduce library, further enhance the signal-to-noise ratio, which is critical for downstream transcription fidelity [5]. In the present

system, these preprocessing steps are orchestrated alongside pyannote-audio for speaker diarization, enabling the attribution of speech segments to individual participants—a non-trivial requirement for accessibility and meeting analytics.

The adoption of streaming ASR in a multilingual corporate setting also necessitates robust language identification and model selection strategies. Whisper's multilingual capabilities have been validated across dozens of languages, yet real-time streaming accentuates the need for rapid language-switch detection and the ability to dynamically load language-specific decoders. The integration of intfloat/multilingual-e5-small embeddings and ChromaDB as a shared knowledge base allows for efficient retrieval-augmented generation (RAG) pipelines, wherein both OCR and STT outputs contribute context for improved disambiguation and domain adaptation. For example, the observed "Philippe Navratil !' Philippe Naruto" transcription error in our proof-of-concept illustrates the limitations of generic ASR in domain-specific contexts; streaming models can benefit from real-time context injection via RAG, reducing such errors by prompting with relevant names and organizational lexicons [8].

Latency-accuracy trade-offs also surface in the context of accessibility for hearing-impaired employees. Real-time captioning is only effective if the delay between speech and on-screen text remains imperceptible for participants relying on visual feedback. Studies have shown that even high-accuracy models may fail to meet accessibility goals if streaming latency exceeds 1–2 seconds, underscoring the necessity of end-to-end pipeline optimization from VAD through ASR to display [3]. In our deployment, FastAPI orchestrates asynchronous audio ingestion and streaming inference, while Azure AI Speech provides an enterprise-grade fallback for languages or scenarios beyond the open-source models' reach.

Security, scalability, and integration also shape the requirements for streaming ASR in corporate environments. The use of Microsoft Azure's Entra ID, Key Vault, and Graph API for Teams enables secure authentication and seamless delivery of live captions into meeting interfaces. Blob Storage ensures persistent archiving of segmented audio, supporting both streaming inference and retrospective analysis. The architectural decision to unify OCR and STT pipelines into a shared ChromaDB instance not only streamlines retrieval but also allows for dynamic context augmentation during streaming transcription, a feature particularly valuable for domain adaptation and rapid error correction [7].

Despite these advances, several open challenges remain. Streaming ASR models still struggle with diarization in highly overlapped speech and with rare vocabulary disambiguation, especially in low-resource languages or highly specialized domains. Experimentation is ongoing to assess the impact of prompt-based domain vocabulary injection and dynamic RAG layer context updates on streaming accuracy—results to be reported. Furthermore, the trade-off between model size, hardware requirements, and latency must continually be calibrated as new model architectures emerge and enterprise needs evolve [10].

In summary, streaming and low-latency ASR models are reshaping the potential of real-time transcription for noisy, multilingual corporate meetings, with direct implications for accessibility and productivity. The combination of advanced model architectures, optimized preprocessing, and enterprise-grade integration forms the backbone of a system capable of meeting the stringent demands of global corporations such as Nestlé. Future work will center on closing the gap between streaming and offline ASR in terms of both accuracy and adaptability, ensuring inclusive, reliable, and contextually aware meeting support for all

employees.

3.6 Speaker Diarization Techniques

Speaker diarization—the process of segmenting audio recordings by speaker identity—serves as a foundational element in advanced speech-to-text (STT) pipelines for multilingual, noisy corporate meetings such as those encountered at Nestlé. Accurate diarization is essential not only for faithful transcription but also for downstream tasks including meeting summarization, attribution of action items, and the provision of accessible content for hearing-impaired employees. The challenge is exacerbated in environments characterized by overlapping dialogue, frequent code-switching, variable channel conditions, and diverse speaker accents. These complexities necessitate diarization techniques that balance accuracy, computational efficiency, and adaptability to domain-specific conversational patterns.

Traditional diarization methods, such as Gaussian Mixture Models (GMMs) with Hidden Markov Model (HMM) segmentation, have long been employed for structured broadcast news and telephone conversations. However, these approaches falter under real-world meeting conditions where the number of speakers is unknown, and speech overlaps are frequent. Such limitations have prompted the adoption of more sophisticated neural architectures that can learn speaker embeddings robust to noise and linguistic variability, as demonstrated by the increasing use of x-vector and d-vector representations in neural network frameworks[2].

Recent architectures, particularly those built atop self-supervised learning (SSL) paradigms, have yielded substantial improvements. For instance, wav2vec 2.0 features a context-aware representation learning mechanism that, when adapted for speaker embedding extraction, facilitates the discrimination of speakers even in noisy or reverberant conditions[1]. In this thesis, speaker diarization is operationalized using the pyannote-audio toolkit, which incorporates state-of-the-art deep learning models for both speech activity detection (SAD) and speaker embedding extraction. The pyannote pipeline processes audio in two stages: first, it segments the recording using a fine-tuned Silero Voice Activity Detector (VAD) to isolate speech from silence and background noise; then, it generates embeddings for each speech segment using a pre-trained neural model, clustering these embeddings to assign speaker labels without requiring an a priori count of speakers.

The adoption of Silero VAD, combined with noisereduce and librosa for pre-processing, has proven particularly effective in suppressing non-speech events and enhancing the fidelity of segmented utterances. Empirical studies have shown that integrating VADs with robust acoustic feature extractors, such as those employed in pyannote-audio, mitigates diarization errors associated with background noise and crosstalk—both common in corporate meeting environments[5]. Notably, in the proof-of-concept system, four minutes of audio were processed in 62 seconds using faster-whisper small INT8, yielding 20 precisely timestamped segments, each accurately attributed to the correct speaker. This efficiency, indicated by a Real-Time Factor (RTF) of 0.25 on CPU, underscores the viability of deploying these models in enterprise workflows, where latency constraints are strict and computational resources may be limited.

Multilingual settings introduce further complexity. Diarization models must remain invariant to language, accent, and prosodic variation, which is a nontrivial requirement in

organizations like Nestlé where French, English, and local African languages may intermingle within a single session. The self-supervised pretraining of wav2vec 2.0 and Whisper on vast, diverse corpora confers a degree of language-agnosticism to resulting embeddings, thereby supporting more reliable clustering of speakers in code-switching contexts[8]. Yet, domain adaptation remains critical. Recent research highlights that the application of Low-Rank Adaptation (LoRA) and domain-specific fine-tuning can further improve both diarization and recognition performance in under-resourced and multilingual scenarios[15].

Incorporating speaker diarization into the broader STT system architecture requires careful orchestration with other processing layers. In this work, diarization outputs are not only essential for segmenting transcriptions by speaker but also for enhancing the retrieval-augmented generation (RAG) layer. By associating each segment with a speaker label and a precise timestamp, the system enables the integration of OCR (optical character recognition) and STT outputs into a unified ChromaDB knowledge base. This facilitates downstream querying, semantic search, and context-aware summarization that respects speaker turns and conversational structure.

For hearing-impaired employees, accurate diarization directly improves the usability of meeting transcripts, making it feasible to attribute statements, track discussions, and clarify responsibilities. This aligns with research indicating that accessibility technologies must not only transcribe speech but also preserve the discourse context and speaker identity to support full participation[3]. Furthermore, diarization errors—such as misattributed segments or failure to distinguish overlapping speech—can propagate through the pipeline, undermining not only transcript accuracy but also the trustworthiness of AI-driven meeting analytics.

Looking ahead, emerging diarization methods that incorporate attention-based transformers, multi-modal cues (e.g., visual speaker tracking), and active learning for domain adaptation hold promise for further gains in both accuracy and robustness[14]. Nevertheless, in practical deployment, trade-offs between computational load and diarization granularity must be managed, particularly in enterprise environments operating on shared cloud infrastructure. The use of Microsoft Azure services—including Azure AI Speech and Azure AI Search—enables scalable, secure integration of diarization outputs with authentication and data governance layers, further supporting organizational requirements for privacy and auditability.

In summary, speaker diarization techniques—anchored in deep learning, SSL, and robust pre-processing—are indispensable for advanced STT systems targeting noisy, multilingual corporate meetings. Their judicious integration ensures that transcriptions are not only accurate but also contextually meaningful and accessible, thereby advancing inclusivity and operational efficiency in enterprise settings.

3.7 Noise Robustness in ASR

Automatic Speech Recognition (ASR) systems deployed in corporate meeting environments face a persistent challenge: ambient noise, overlapping speech, and reverberant conditions frequently degrade transcription accuracy, particularly in multilingual and accessibility-focused contexts. Noise robustness—the capacity of an ASR model to maintain high recognition accuracy despite acoustic interference—has become a pivotal criterion for

practical deployments. In the context of Nestlé/BIT's initiative for supporting hearing-impaired employees, the reliability of ASR in noisy corporate settings is not merely a technical concern, but one that directly impacts workplace inclusivity and productivity. Effective noise-robust ASR must address a range of acoustic adversities, from HVAC hum and keyboard clatter to unpredictable conversational interruptions, each posing unique obstacles to signal fidelity and subsequent linguistic decoding.

Several methodological streams have shaped the modern pursuit of noise robust ASR. One foundational strategy is acoustic model adaptation, where models are retrained or fine-tuned to better represent the statistical properties of noisy speech. Chung (2014) demonstrated that adapting acoustic models using noisy speech corpora, rather than relying solely on clean data, yields significant gains in robustness, especially for environments with fluctuating noise profiles typical of open office spaces and large meeting rooms [6]. Complementary to model adaptation, front-end signal processing remains critical. Techniques such as spectral subtraction, Wiener filtering, and, more recently, deep learning-based speech enhancement, attempt to recover a cleaner speech signal prior to ASR decoding. Farahani (2017) advanced the field by proposing autocorrelation-domain feature extraction, which outperformed conventional Mel-frequency cepstral coefficients (MFCCs) and log-mel features under noisy conditions, underscoring the importance of robust feature engineering [5].

The emergence of self-supervised models, notably wav2vec 2.0 and Whisper, has further redefined noise robustness paradigms. These architectures, pre-trained on vast, acoustically diverse datasets, exhibit remarkable generalization to unseen noise types. Tyagi and Szénási (2026) found that hybrid transformer architectures combining wav2vec 2.0 and Whisper yield strong emotion recognition and transcription performance even when vocal bursts are partially masked by noise [1]. Notably, Whisper's pre-training on multilingual and multi-noise data empowers it to cope with both linguistic and acoustic diversity—a critical trait for companies operating in global markets such as Nestlé [8].

However, real-world noise conditions can still expose significant vulnerabilities in ASR systems. For instance, the observed "Philippe Navratil !' Philippe Naruto" error from the PoC system exemplifies how even advanced models like Whisper may hallucinate plausible-sounding but incorrect entities when phonetic cues are distorted or context is insufficiently modeled. Such errors highlight the limitations of pre-trained models and the necessity of domain adaptation and contextual augmentation in enterprise ASR pipelines.

Domain adaptation, particularly through low-rank adaptation (LoRA) techniques, has shown promise in bridging the gap between generic pre-trained models and domain-specific acoustic environments. Polat and Turan (2024) reported that integrating LoRA adapters with Whisper architectures significantly improved transcription accuracy for Turkish ASR in noisy conditions—results that are particularly relevant for corporate deployments where vocabulary and speaker demographics diverge from public datasets [15]. Similarly, Bhardwaj and Kukreja (2021) demonstrated that acoustic adaptation combined with prosodic feature modeling further enhances ASR robustness, especially for tonal or prosodically rich languages common in multilingual African and European corporate milieus [17].

A complementary thread concerns the integration of voice activity detection (VAD) and noise reduction modules in the ASR pipeline. In the deployed system, Silero VAD and noisereduce are employed as front-end modules to suppress non-speech intervals and attenuate background noise, respectively. The adoption of such pre-processing strategies is

validated by results from Daouad et al. (2024), who found that optimized Whisper pipelines, when augmented with robust VAD and denoising, excelled at ASR tasks for Amazigh languages in real-world noisy settings [2]. Furthermore, librosa-based audio preprocessing, including spectral normalization and silence trimming, further stabilizes model inputs, contributing to improved recognition rates in challenging acoustic scenarios.

The architectural integration of these components—faster-whisper for efficient INT8 inference, pyannote-audio for diarization, Silero VAD, and noise reduction—reflects a layered defense against noise-induced errors. In practice, the system achieves a real-time factor (RTF) of 0.25 on CPU with the faster-whisper small INT8 model, processing four minutes of audio in approximately 62 seconds and segmenting the speech into 20 timestamped utterances. These performance characteristics not only meet practical deployment constraints but also suggest that advanced ASR, when properly engineered, can deliver both speed and resilience to noise.

Beyond algorithmic advances, the ultimate test of noise robustness lies in longitudinal performance for hearing-impaired end-users. Giguere and Laroche (2010) emphasized that even modest improvements in ASR intelligibility under noise can yield disproportionate benefits for users with hearing loss, given their heightened sensitivity to acoustic distortion [12]. Lobar (2026) reinforced this point, arguing that accessible ASR is not only a technical achievement but a fundamental enabler of workplace equity [4].

Despite these advances, challenges persist. Environmental noise profiles are constantly evolving, and domain drift—where the acoustic and linguistic characteristics of operational data deviate from training data—remains a threat to sustained ASR performance. Ongoing research into adaptive learning, continual model retraining, and knowledge graph-driven contextual correction (such as RAG-based integration with ChromaDB) promises further resilience. Nevertheless, robust noise handling in ASR requires the ongoing synthesis of signal processing, deep learning, domain adaptation, and user-centered evaluation. Only through this multifaceted approach can ASR systems fulfill their promise in noisy, multilingual, and diverse workplaces.

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3.8 Domain Adaptation for Corporate ASR

Domain adaptation in automatic speech recognition (ASR) is indispensable when deploying models in corporate environments characterized by multilingualism, background noise, and specialized vocabularies. Standard ASR architectures, such as Whisper and wav2vec 2.0, exhibit strong generalization across diverse speech corpora, but their baseline performance often degrades in dynamic corporate settings, particularly where acoustic conditions and domain-specific terminology diverge from training data distributions [2]. In the context of Nestlé’s corporate meetings, which feature a mix of languages, cross-regional accents, and frequent domain-specific references, off-the-shelf models regularly misinterpret company names, product lines, or employee names, exemplified by the transcription of “Philippe Navratil” as “Philippe Naruto” observed during proof-of-concept evaluations. Such errors undermine both accessibility for hearing-impaired employees and the overall integrity of

meeting records.

To overcome these challenges, domain adaptation strategies must address both acoustic and linguistic mismatches. Acoustic adaptation involves modifying ASR models to account for the unique noise profiles and microphone characteristics of typical meeting rooms. Techniques such as feature-space maximum likelihood linear regression (fMLLR), speaker adaptation via x-vectors, and fine-tuning with in-domain audio samples have demonstrated measurable reductions in word error rate (WER) under mismatched conditions [6]. For instance, wav2vec 2.0 can be adapted to noisy corporate environments by retraining the model's transformer layers on audio captured from real meetings, augmented with synthetic noise resembling HVAC hum, overlapping speech, and distant microphone effects [5]. This process systematically increases the robustness of feature representations and improves intelligibility for both normal-hearing and hearing-impaired listeners.

Linguistic adaptation, in contrast, requires tailoring the ASR model's language model component to recognize and correctly transcribe corporate jargon, acronyms, and proper nouns. Approaches such as vocabulary injection, domain-specific language modeling, and prompt-based adaptation are effective. In the current system, vocabulary lists extracted from Nestlé's internal documents and meeting minutes are incorporated into the language model's decoding process, using prompt engineering and dynamic biasing to heighten recognition accuracy for critical terms [10]. This targeted intervention is vital for mitigating substitution errors like "Philippe Naruto" and for transcribing brand names or technical terms that are rare in generic corpora. Moreover, large language models (LLMs) such as Ollama/ Llama 3.2 can be harnessed in the reranking phase to provide context-aware corrections, leveraging embeddings from intfloat/multilingual-e5-small to represent both ASR outputs and domain-specific queries within a shared semantic space.

A significant architectural contribution of this work is the integration of multimodal domain adaptation via a retrieval-augmented generation (RAG) layer. Here, outputs from both optical character recognition (OCR) and ASR components are stored in a unified ChromaDB knowledge base. This structure enables cross-referencing between spoken and written content in meetings, allowing the system to resolve ambiguous terms by consulting both modalities [9]. For example, if a product name is partially garbled in the speech transcript but appears correctly in a projected slide captured by OCR, the RAG layer can align and correct the ASR output in real time. Such multimodal fusion is particularly beneficial in noisy or multilingual settings, where redundancy between modalities can compensate for ASR model uncertainty.

Model adaptation for accessibility further requires attention to the needs of hearing-impaired users. Research shows that standard ASR models underperform on the speech of hearing-impaired speakers and may fail to capture subtle prosodic cues critical for intelligibility [4]. To address this, the system incorporates domain adaptation not only for typical corporate accents and jargon, but also for speech patterns characteristic of hearing-impaired individuals. Fine-tuning on such speech, coupled with prosody-aware feature extraction and prosodic modification—where pitch, stress, and timing are explicitly modeled—yields more inclusive and reliable transcripts [17].

From an engineering perspective, the adaptation pipeline is orchestrated using a modular stack: faster-whisper small INT8 for efficient inference on CPUs (RTF = 0.25, 4 minutes audio processed in 62 seconds, 20 timestamped segments), noisereduce and librosa for pre-processing and feature enhancement, Silero VAD for robust segmentation, and

pyannote-audio for speaker diarization. FastAPI serves as the backend, interacting with enterprise-grade Microsoft Azure services—namely Azure AI Speech for scalable STT, Azure AI Search and OpenAI for semantic retrieval and context-aware correction, and ChromaDB for knowledge management. Authentication and security are managed through Entra ID and Key Vault, with Teams integration via Graph API for seamless deployment within the corporate workflow [3].

Planned experiments will evaluate the effectiveness of these domain adaptation strategies across multilingual and accessibility-focused benchmarks, reporting on improvements in WER, entity recognition, and user satisfaction metrics. In summary, adapting corporate ASR to domain-specific acoustic, linguistic, and accessibility requirements demands a layered, evidence-based approach, drawing together state-of-the-art modeling, multimodal data fusion, and deep integration with enterprise infrastructure. Through these strategies, the system aspires to deliver accurate, inclusive, and context-sensitive transcription in the complex environment of Nestlé's global meetings.

3.9 Unified Audio Front-End Models

Unified audio front-end models serve as a linchpin in the development of robust automatic speech recognition (ASR) systems for noisy, multilingual corporate settings, particularly those supporting hearing-impaired employees. These models encapsulate not only the technical mechanisms for converting raw audio into textual representations but also the orchestration of preprocessing, segmentation, diarization, and domain adaptation layers. In modern enterprise environments, such as those at Nestlé, the audio data encountered is highly heterogeneous—ranging across diverse languages, accents, overlapping speakers, and fluctuating background noise. The requirement for a unified audio front-end, therefore, extends beyond mere technical integration; it demands a design capable of harmonizing multiple modalities and toolchains while enabling downstream knowledge extraction and accessibility.

The theoretical foundation for unified audio front-end models is rooted in the principle of modular abstraction, where audio preprocessing, feature extraction, and ASR decoding are treated as discrete but interoperable components. This modularity is vital for a system intended to adapt to new domains or languages, as it enables the substitution or enhancement of individual modules without disrupting the overall pipeline. For instance, noise reduction using spectral subtraction or minimum mean-square error (MMSE) estimators, as discussed in the context of STFT domain modeling, can be integrated prior to feature extraction to improve recognition accuracy under adverse acoustic conditions [18]. In the present architecture, noisereduce and librosa are employed at this stage to attenuate extraneous noise and standardize input for subsequent modules.

A key consideration in constructing a unified front-end is the handling of speaker turns and silence detection. Silero voice activity detection (VAD) provides a rapid, reliable mechanism for segmenting input streams, enabling precise demarcation of utterances even where speaker overlap or intermittent silence exists [2]. These segments are then passed to pyannote-audio for diarization, which is crucial in corporate meetings where multiple participants may converse in rapid succession or simultaneously. Accurate diarization ensures that attributions in the transcript are reliable, a necessity for accessibility and auditability in professional contexts [14].

The unification of the audio front-end also extends to model interoperability. State-of-the-art ASR architectures, such as Whisper and wav2vec 2.0, exhibit complementary strengths: Whisper demonstrates resilience to cross-lingual and domain variations, while wav2vec 2.0 excels in low-resource and noisy environments due to its self-supervised pretraining on vast unlabeled speech corpora [8]. By architecting the front-end to flexibly route audio to the model—based on language, noise profile, or user preference—the system can dynamically select the optimal decoder. For instance, corporate meetings in French might benefit from Whisper's domain prompts and multilingual fine-tuning, whereas meetings in under-resourced languages could be routed to a wav2vec 2.0 variant pre-adapted with domain-specific acoustic models [10].

Integration within an end-to-end pipeline is accomplished using FastAPI, which exposes well-defined endpoints for preprocessing, ASR inference, and post-processing. This enables real-time or batch transcription workflows and seamless scaling to enterprise workloads. The output from the ASR models is then encoded using intfloat/multilingual-e5-small embeddings, allowing semantic retrieval and downstream processing. The unified front-end thus forms both the technical and semantic substrate upon which subsequent retrieval-augmented generation (RAG) operations are built.

A distinctive architectural contribution of the present work is the harmonization of audio (speech) and visual (OCR) streams into a single ChromaDB knowledge base. By sharing this vector store across both the ASR and OCR pipelines, the system supports cross-modal retrieval, enabling queries that span meeting transcripts and presentation slides without artificial boundaries. This strategy is particularly powerful in multilingual, information-rich meetings where context may be distributed across spoken and visual modalities [9].

Noise robustness is another central pillar of unified audio front-end design. In enterprise meeting environments, background chatter, HVAC noise, and electronic interference can severely degrade ASR performance. Acoustic adaptation techniques, such as those based on prosodic modification and in-domain background modeling, are incorporated to mitigate these effects [17]. For example, pre-trained Whisper models are further adapted using domain prompts and small language-specific corpora, improving transcription accuracy for specialized vocabulary and accented speech [15]. This is essential not only for general usability but also for accessibility—misrecognition of critical terms (as evidenced by the "Philippe Navratil !' Philippe Naruto" error in the PoC) can have outsized impacts on hearing-impaired users seeking reliable meeting information.

Furthermore, the enterprise deployment leverages Microsoft Azure services—including Azure AI Speech, Blob Storage, and Azure OpenAI—to provide secure, scalable infrastructure. Authentication, authorization, and data governance are managed via Entra ID and Key Vault, with integration into Teams via Graph API. This ensures that the unified audio front-end is not only technically sophisticated but also enterprise-grade in terms of compliance and operational reliability [3].

In sum, unified audio front-end models do not merely aggregate preprocessing and ASR modules; they embody a conceptual and technical strategy for harmonizing diverse audio, language, and accessibility requirements in complex corporate environments. Through modularity, cross-modal integration, and robust deployment practices, such systems lay the foundation for inclusive, accurate, and scalable speech-to-text solutions.

AI Usage Acknowledgment: AI tools provided assistance with writing and editing. The

research direction, technical architecture, and all substantive content reflect the author's original work.

3.10 Meeting Transcription Systems: State of the Art

Meeting transcription systems have undergone substantial evolution in the past decade, driven by advances in deep learning architectures, domain adaptation techniques, and the proliferation of multilingual corpora. Within the corporate context—particularly in environments characterized by high noise levels, overlapping speech, and diverse accents—robust automatic speech recognition (ASR) remains a persistent research and engineering challenge. Leading-edge systems now blend multiple modalities and leverage transformer-based architectures to deliver transcription accuracy and speaker attribution that were unattainable with classical hidden Markov model (HMM) frameworks [1].

The integration of large-scale pretrained models such as Whisper and wav2vec 2.0 has dramatically improved baseline word error rates (WER), especially for languages and speaker profiles underrepresented in prior literature. Whisper, with its multilingual and multitask design, has set a new standard for out-of-the-box transcription performance across diverse acoustic environments [8]. When deployed in real-world meeting settings, however, even these advanced models encounter domain-specific obstacles. For instance, the notorious "Philippe Navratil !' Philippe Naruto" error, observed in the Nestlé/BIT proof-of-concept, highlights the persistent risk of phonetic confusions in proper nouns—a class of term critical to enterprise documentation accuracy.

Recent research has focused on domain adaptation as a remedy for such errors, exploring techniques ranging from low-rank adaptation (LoRA) to kernel mean matching for instance-based transfer [11]. Adaptation strategies are particularly valuable in corporate settings where technical jargon, named entities, and code-switching are common. For example, LoRA adapters can fine-tune Whisper's attention layers on a small set of company-specific transcripts, yielding measurable improvements in challenging domains [15]. However, the scarcity of annotated data for certain languages and the dynamic nature of intra-corporate vocabulary still present open problems for robust adaptation.

Another line of innovation involves explicit noise-robustness mechanisms, including front-end speech enhancement and feature adaptation. Techniques such as spectral subtraction, minimum mean square error (MMSE) estimation, and autocorrelation domain feature extraction have been integrated with ASR backends to mitigate the impact of adverse acoustic conditions [5]. Some systems employ voice activity detection (VAD)—with Silero VAD as a prominent example—to segment speech, followed by denoising procedures like noisereduce and librosa-based filtering. These preprocessing steps can bolster the efficacy of downstream ASR models, particularly in open office or remote conferencing scenarios [6].

Speaker diarization and turn segmentation have also become essential components of modern meeting transcription pipelines. Pyannote-audio, a neural diarization toolkit, is frequently adopted to assign utterances to speakers and resolve overlaps in multi-party conversations. This functionality is critical not only for accurate transcription but also for downstream tasks such as meeting summarization and speaker-specific action item extraction [9].

The necessity of inclusivity for hearing-impaired employees has catalyzed further adaptation of ASR systems. Wearable devices and real-time captioning platforms now offer speech-to-

text conversion tailored to the needs of hearing-impaired users, integrating domain-adapted ASR models to enhance intelligibility in real-world meeting environments [3]. Research demonstrates that conventional ASR models, when applied without adaptation, frequently underperform for speakers with atypical prosody or articulation—a challenge exacerbated in noisy settings [16]. Model adaptation, informed by prosody and acoustic cues, is thus increasingly recognized as a prerequisite for equitable meeting accessibility [17].

Multimodal architectures are emerging as a frontier in meeting transcription. Recent systems have begun to unify outputs from speech and optical character recognition (OCR), combining them in knowledge bases such as ChromaDB to support retrieval-augmented generation (RAG). This approach allows systems to cross-reference transcribed spoken content with slide text, chat logs, or whiteboard captures, thus bolstering both recall and precision in meeting documentation [7]. The present work's architectural contribution—a shared ChromaDB backend for both OCR and STT outputs—reflects this trend and addresses the limitations of siloed data stores.

Enterprise deployments increasingly rely on cloud-based infrastructures, such as Microsoft Azure, to orchestrate ASR, search, and generative AI services. Azure AI Speech provides scalable inference for ASR, while Azure AI Search and OpenAI endpoints enable advanced retrieval and summarization. Supporting services—Blob Storage for audio archiving, Entra ID for identity management, and Graph API for integration with collaboration tools like Teams—complete the enterprise pipeline, ensuring security and compliance alongside state-of-the-art transcription [14].

Despite these advances, persistent challenges remain. The interplay of multilingualism, domain-specific vocabulary, overlapping speech, and accessibility demands continues to drive research in robust meeting transcription. Hybrid transformer models, adaptive noise reduction, and retrieval-augmented architectures are converging towards more accurate and inclusive systems. Future directions will likely emphasize continual learning, few-shot adaptation, and deeper integration of knowledge graphs to further reduce confusions such as the "Philippe Navratil !' Philippe Naruto" phenomenon, moving closer to the goal of universal, reliable meeting transcription for all corporate users.

AI Usage Acknowledgment: AI tools assisted with writing and editing this section, while all research directions and technical content are the student's own work.

3.11 Identified Research Gap

Despite significant progress in automatic speech recognition (ASR), current systems exhibit notable shortcomings when deployed in real-world, noisy, and multilingual corporate environments—settings that are common in multinational enterprises such as Nestlé. Recent advances, including the use of Whisper and wav2vec 2.0 models, have demonstrated competitive accuracy on benchmark datasets and under controlled acoustic conditions; however, their robustness diminishes in the presence of conversational interruptions, crosstalk, and domain-specific terminology, all of which are prevalent in large-scale corporate meetings [8]. This is further compounded when the objective is to support hearing-impaired employees, who depend on accurate, context-aware transcriptions to fully participate in organizational communication [4].

Existing literature has addressed noise robustness primarily through acoustic model adaptation and front-end denoising methods, such as spectral subtraction and MMSE

estimation [6], [18]. While these approaches mitigate certain noise artifacts, they frequently fail to generalize across the heterogeneous acoustic scenarios found in meeting rooms, open offices, and remote conferencing setups. For instance, studies have shown that even advanced models such as Whisper, when evaluated on non-standard accents, code-switching, and domain-specific vocabulary, can produce critical transcription errors—such as the observed misrecognition of “Philippe Navratil” as “Philippe Naruto”—that severely undermine accessibility and comprehension [2].

The challenge is further exacerbated by the multilingual nature of global enterprises. Although recent works have focused on expanding ASR coverage to low-resource and under-represented languages using transfer learning, LoRA-based fine-tuning, and domain adaptation, comprehensive solutions for code-switching scenarios and multi-accent robustness remain insufficient [11], [10]. For example, in multilingual meetings at Nestlé, participants routinely alternate between French, English, and local African languages, producing mixed-language utterances that are not captured with high fidelity by off-the-shelf ASR models. Domain adaptation techniques have yielded incremental gains, yet they often require extensive in-domain data and manual vocabulary curation, limiting their scalability and real-world applicability [7].

Another critical research gap pertains to the integration of multimodal knowledge, particularly the fusion of optical character recognition (OCR) and ASR outputs. Most existing ASR systems process speech in isolation, disregarding the rich contextual cues present in slide decks, shared documents, and meeting chat logs. The architectural contribution of integrating both OCR (Topic 1) and STT (Topic 2) outputs into a unified ChromaDB knowledge base, as proposed in this thesis, directly addresses this limitation by facilitating context-aware retrieval and reducing ambiguity in transcriptions. This approach has not been sufficiently explored in the literature, where siloed treatment of modalities remains the norm [9].

For hearing-impaired employees, the shortcomings in current ASR systems are particularly salient. Prior research has primarily focused on wearable devices and frequency modulation systems for individual users [3], [19]. However, there is a lack of scalable, enterprise-grade ASR platforms that are robust to environmental noise, support multiple languages, and can be integrated with secure corporate infrastructure such as Microsoft Azure’s AI Speech services, Azure OpenAI, and organizational identity management (Entra ID). The unique demands of real-time captioning, timestamped segmentation (e.g., faster-whisper small INT8 processing 4 minutes of audio in 62 seconds with 20 segments), and downstream knowledge extraction for accessibility have not been comprehensively addressed [13].

Moreover, although advancements in voice activity detection (VAD), diarization, and speaker embedding technologies—such as Silero VAD, pyannote-audio, and intfloat/multilingual-e5-small embeddings—have improved pre- and post-processing pipelines, their combined contribution to the reduction of word error rate (WER) and improvement of user experience under enterprise constraints remains underexplored in both academic and industrial deployments [1]. The practicalities of deploying these pipelines within Azure-based, privacy-compliant architectures, with integration points for downstream applications (Teams via Graph API, document storage with Blob Storage, and credential management via Key Vault), have been sparsely documented.

Finally, the majority of existing research evaluates ASR models on public datasets with limited relevance to the acoustic and linguistic diversity encountered in corporate meetings.

There is a paucity of studies reporting real-world, in-the-wild ASR performance metrics—such as real-time factor (RTF), error patterns on proprietary vocabularies, and user-centric accessibility outcomes—especially for hearing-impaired stakeholders [12]. This lack of comprehensive empirical evidence impedes the design of systems that are reliable, scalable, and truly inclusive.

In summary, the identified research gap centers on the lack of robust, scalable, and context-aware ASR solutions for noisy, multilingual corporate environments that support hearing-impaired users. Specifically, no prior work holistically addresses the combined challenges of environmental noise, code-switching, domain vocabulary errors (as typified by the “Philippe Navratil !’ Philippe Naruto” case), and the integration of multimodal knowledge in a secure, enterprise-ready architecture. Addressing this gap is both a technical and an ethical imperative for organizations seeking to foster accessibility and inclusivity in the digital workplace.

4. System Architecture and Design

4.1 Overall Platform Architecture

The platform developed for advanced speech-to-text (ASR) in noisy, multilingual corporate meetings at Nestlé integrates several state-of-the-art components into a cohesive architecture designed to address the specific challenges posed by real-world environments and the needs of hearing-impaired employees. This architecture is not only technically sophisticated but also modular, ensuring adaptability and scalability for evolving enterprise requirements.

At its core, the system is structured into several tightly coupled components: signal preprocessing, voice activity detection, speaker diarization, ASR, optical character recognition (OCR), and a retrieval-augmented generation (RAG) layer, all orchestrated through cloud-native microservices. Audio preprocessing employs noisereduce and librosa for denoising and feature extraction, crucial for maintaining intelligibility in the presence of corporate meeting noise, such as overlapping speech or background interference [5]. Silero VAD is used to segment incoming audio streams, filtering out non-speech regions and thus reducing computational overhead and error propagation downstream [1]. Speaker diarization is performed using pyannote-audio, which assigns speaker labels to each segment, a necessity in multi-participant meetings and for accurate attribution in meeting records [1].

The ASR component leverages faster-whisper small INT8 models, selected for their balance of accuracy, language coverage, and computational efficiency. The measured real-time factor (RTF) is 0.25 on CPU, meaning four minutes of audio are transcribed in just over one minute—62 seconds—across 20 timestamped segments, an operational speed suitable for near real-time applications [2]. This choice is informed by comparative studies of Whisper’s performance on under-resourced and noisy language varieties, where INT8 quantization models maintain accuracy with reduced inference latency [11]. The system’s modular design allows for the integration of wav2vec 2.0 and exploration of domain adaptation techniques, facilitating continuous improvement and adaptation to acoustic and lexical domain shifts encountered in corporate environments [17].

To augment speech recognition with contextual understanding, the architecture includes an

OCR module capable of extracting on-screen text during hybrid meetings. This addresses scenarios where slides, agendas, or chat messages are integral to conversational context but inaccessible through audio channels alone. Both OCR and ASR outputs are normalized and embedded using intfloat/multilingual-e5-small, producing dense vector representations suitable for semantic retrieval [7].

A key architectural contribution of this platform is the integration of both OCR and ASR-derived knowledge into a unified ChromaDB vector database, forming the backbone of the RAG layer. This shared semantic memory enables downstream applications—such as meeting summarization, Q&A, and assistive interfaces for hearing-impaired staff—to retrieve and synthesize relevant information regardless of input modality. The RAG layer's design ensures that queries are matched against both spoken and visual meeting content, closing information gaps and reducing error rates associated with missing domain-specific vocabulary [9]. For example, in the observed "Philippe Navratil !' Philippe Naruto" transcription error, the presence of correct text in OCR could enable prompt detection and correction, highlighting the value of multi-modal knowledge integration.

API orchestration is handled via FastAPI, selected for its low latency and compatibility with Python's scientific stack. The modular deployment architecture is cloud-native and containerized, facilitating orchestration, scaling, and fault isolation. Microsoft Azure forms the enterprise backbone, providing secure, compliant infrastructure. Azure AI Speech is available for fallback ASR and comparative evaluation; Azure AI Search and ChromaDB underpin semantic retrieval; Azure OpenAI and Ollama/Llama 3.2 models support natural language understanding and summarization. All data assets are stored in Blob Storage, governed by Entra ID and Key Vault for secure authentication and key management. Integration with Microsoft Teams is realized using the Graph API, enabling real-time caption streaming and archiving of meeting transcripts [13].

The architecture's design is underpinned by theoretical and empirical insights into noise robustness, multilingual modeling, and domain adaptation. Noise reduction and speech enhancement are prioritized through front-end processing, as studies indicate that feature-domain enhancement—such as MMSE-based residual uncertainty modeling—can significantly improve ASR intelligibility in adverse acoustic conditions [18]. Multilingual and code-switched scenarios are addressed through model selection and fine-tuning strategies, building on recent advances in Whisper and wav2vec 2.0 for low-resource and morphologically rich languages [8]. Domain adaptation is facilitated by RAG-driven retrieval, LoRA fine-tuning, and prompt engineering, with ongoing experiments evaluating optimal strategies for integrating new terms and correcting context-specific errors [10].

In summary, the platform architecture is characterized by its modular, multi-modal, and secure design. It synthesizes advanced ASR, OCR, semantic retrieval, and enterprise integration within a maintainable and extensible framework. By unifying OCR and ASR outputs in ChromaDB and deploying robust Azure-based security, the system directly addresses the needs of hearing-impaired employees and the demands of multilingual, noisy corporate environments. This architecture lays a resilient foundation for further research and operational deployment.

AI Usage Acknowledgment: AI tools were used to assist with writing and editing. All research directions, technical content, and architectural design are entirely my own work.

4.2 File-Based Speech-to-Text Pipeline

In the context of corporate meetings characterized by acoustic variability, multiple languages, and domain-specific jargon, a file-based speech-to-text (STT) pipeline must be engineered with both flexibility and rigor. The pipeline developed for this thesis is grounded in a modular architecture that integrates advanced automatic speech recognition (ASR) models—namely, Whisper and wav2vec 2.0—with a suite of pre- and post-processing components specifically tailored for the needs of hearing-impaired employees in a large enterprise setting such as Nestlé. Each stage of the pipeline is informed by both empirical findings from the literature and practical considerations stemming from proof-of-concept deployments.

Audio preprocessing constitutes the initial step in the pipeline, and its importance is magnified in environments marked by diverse noise profiles and variable signal quality. The use of noisereduce and librosa for denoising and feature extraction, respectively, ensures that signal integrity is preserved prior to ASR model inference. This is corroborated by recent research that underscores the efficacy of autocorrelation and spectral masking methods for enhancing recognition rates in noisy conditions, especially when background noise is non-stationary or highly dynamic[5]. Empirical evidence further suggests that denoising, when coupled with voice activity detection (VAD) using Silero VAD, significantly reduces the prevalence of false segment boundaries, thereby improving downstream transcription accuracy[6].

Following preprocessing, the audio signal is segmented using pyannote-audio, which provides precise temporal demarcation of speech segments. Segmenting long recordings into manageable chunks is critical not only for computational efficiency but also for minimizing latency and ensuring that speaker turns are accurately preserved. Research has demonstrated that segment-level processing reduces word error rates (WER) in multilingual and noisy settings, as it limits context-switching errors and facilitates model adaptation to local acoustic and linguistic patterns[1].

The ASR inference layer supports both Whisper (with faster-whisper small INT8 for efficiency) and wav2vec 2.0. These models have been selected due to their demonstrated performance in low-resource and noisy environments, as well as their capacity for multilingual transcription[8]. Notably, the faster-whisper small INT8 variant achieves a real-time factor (RTF) of 0.25 on standard CPUs, enabling the processing of a four-minute audio file in approximately 62 seconds and generating 20 timestamped segments. Such performance metrics align with enterprise requirements for near-real-time feedback in meeting scenarios. Comparative studies reveal that, while Whisper exhibits superior robustness to code-switching and domain-specific terminology, wav2vec 2.0 offers significant advantages in scenarios with limited annotated data, owing to its self-supervised pretraining regime[2].

An essential innovation of this pipeline is the integration of ASR outputs with optical character recognition (OCR) results from meeting slides and shared documents. This is implemented through a retrieval-augmented generation (RAG) layer that ingests both STT and OCR outputs into a unified knowledge base managed by ChromaDB. The shared ChromaDB repository is indexed using intfloat/multilingual-e5-small embeddings, allowing for semantically rich retrieval and downstream conversational AI applications[7]. This fusion of auditory and visual content addresses a well-documented challenge in domain-specific

ASR—namely, the frequent misrecognition of proper nouns and technical terms, as exemplified by the observed "Philippe Navratil !’ Philippe Naruto" transcription error in initial proof-of-concept trials.

RESTful APIs are provided via FastAPI, facilitating secure, scalable integration with enterprise platforms such as Microsoft Teams through Microsoft Azure’s Graph API. Audio files are uploaded by end users through the API, triggering asynchronous transcription jobs that are orchestrated via Azure Blob Storage for file persistence and Azure Key Vault for credential management. The modular design allows for rapid adaptation to changing security requirements or evolving corporate workflows, in alignment with best practices for enterprise-grade AI deployment[13].

Post-processing is handled by custom scripts that normalize casing, punctuation, and timestamp alignment. Named entity recognition (NER) and domain vocabulary prompting are applied using Ollama/Llama 3.2 LLMs to further refine output transcripts and mitigate domain-specific substitution errors. For instance, the aforementioned "Philippe Navratil !’ Philippe Naruto" error highlights the need for continual vocabulary adaptation and explicit prompt engineering, particularly in multilingual and jargon-heavy contexts.

The pipeline is designed to support both batch and streaming modes, but file-based operation provides particular advantages in corporate settings where meeting recordings are archived and subject to post-hoc review. Furthermore, all processed data—transcripts, metadata, and embeddings—are indexed within ChromaDB, ensuring rapid retrieval for compliance audits, accessibility reviews, and integration with downstream analytics or summarization modules[9].

In summary, the file-based speech-to-text pipeline presented here represents a synthesis of state-of-the-art ASR models, advanced signal processing, and retrieval-augmented architectures. Its design is explicitly motivated by the practical requirements of accessibility for hearing-impaired employees and the operational realities of large, multilingual enterprises. Ongoing experiments will further quantify the impact of domain adaptation strategies and prompt engineering on transcription accuracy and user satisfaction. The modularity and extensibility of the pipeline ensure its applicability across a wide range of corporate scenarios, including those demanding strict data privacy and high throughput.

AI Usage Acknowledgment: AI tools assisted with writing and editing this section; all research directions and technical content reflect the independent work of OUEDRAOGO Yannick Ulrich Levi.

4.3 Live Transcription Pipeline

The live transcription pipeline at the core of this thesis operationalises advanced automatic speech recognition (ASR) for multilingual, noisy corporate meetings, targeting accessibility for hearing-impaired employees at Nestlé. Architected as a modular, streaming-capable service, the pipeline must balance latency, accuracy, and adaptability to diverse acoustic and linguistic conditions. This section provides an in-depth account of the pipeline’s architecture, its constituent technologies, and the methodological rationale for their selection and integration.

Audio input is ingested in real time from enterprise-grade video conferencing platforms, with

Microsoft Teams integration enabled through Microsoft Graph API. Incoming streams are buffered and segmented using Silero Voice Activity Detection (VAD), which discerns speech from silence under adverse noise conditions. Silero VAD's performance in conversational audio, particularly in environments with overlapping speakers and nonstationary noise, has been empirically validated and is critical for maintaining both low-latency and high-quality segmentation. The choice of Silero is motivated by its demonstrated robustness and minimal resource footprint, which is essential in production deployments where scalability and efficiency are paramount[5].

Once speech regions are segmented, audio is passed through a denoising stage using the noisereduce Python library. This adaptive spectral subtraction technique reduces background noise without introducing significant artifacts or distorting speech intelligibility, which is a prerequisite for ASR accuracy in open-plan offices and large meeting rooms. The pipeline subsequently normalises audio and extracts temporal features with librosa, optimising input for downstream ASR models. Librosa's feature extraction ensures consistent spectral representations, facilitating model generalisation across diverse recording devices and acoustic setups[6].

The core transcription step is executed using faster-whisper, a highly optimised, quantised implementation of OpenAI's Whisper small model (INT8). This variant was chosen after benchmarking for real-time factor (RTF) and accuracy under CPU-constrained, on-premise scenarios. Empirical validation in the Nestlé proof-of-concept demonstrated RTF = 0.25 (i.e., 4 minutes of audio processed in 62 seconds), with outputs segmented into 20 timestamped utterances per session. This level of performance is vital for live captioning and downstream natural language processing tasks, particularly when batch GPU resources are unavailable[2].

To address the pipeline's multilingual mandate, faster-whisper is augmented with wav2vec 2.0 for language identification and secondary transcription. Wav2vec 2.0's self-supervised learning paradigm enables robust feature learning from unlabeled audio, which is especially effective for under-resourced languages and accented speech found in multinational corporate meetings. Its integration serves two purposes: first, to improve language detection at segment boundaries; second, to provide a fallback transcription in cases where Whisper's confidence is low or domain-specific terminology is encountered. The symbiotic deployment of these models, as highlighted in recent hybrid ASR research, has been shown to bolster recognition accuracy in complex auditory environments[1].

Speaker diarisation is managed by pyannote-audio, which tags utterances with speaker identities. This metadata underpins personalised captioning for hearing-impaired users and supports meeting analytics. Accurate diarisation is nontrivial, given crosstalk and rapid turn-taking in naturalistic meetings; pyannote-audio's neural diarisation approach, trained on large conversational corpora, effectively mitigates these challenges. The diarisation output is synchronised with ASR transcripts to produce speaker-attributed captions and searchable meeting records[14].

Processed text and associated metadata are routed via FastAPI to an application layer for post-processing, validation, and user-facing presentation. At this stage, domain adaptation is effected through a retrieval-augmented generation (RAG) mechanism: transcripts are embedded using intfloat/multilingual-e5-small, and the resulting vectors are indexed in a shared ChromaDB knowledge base. Notably, this RAG layer unifies both optical character recognition (OCR) outputs (e.g., screen-shared slides) and live ASR transcriptions within a

single semantic retrieval space—a marked architectural contribution that enables cross-modal search, reference resolution, and context-aware correction[10].

Azure cloud services underpin enterprise deployment, ensuring compliance, scalability, and integration with corporate identity management (Azure Entra ID), secure storage (Azure Blob Storage, Key Vault), and advanced search via Azure AI Search and OpenAI APIs. This cloud-native architecture facilitates rapid scaling to accommodate fluctuating meeting loads and provides the security guarantees required for confidential business communications.

A recurring challenge within the pipeline is the accurate transcription of domain-specific names and terminology. For instance, the observed error “Philippe Navratil” transcribed as “Philippe Naruto” in a French-language meeting underscores both the limitations of general-purpose models and the necessity for domain-adaptive prompting and post-correction. The integration of ChromaDB’s hybrid semantic retrieval, informed by OCR and curated glossaries, enables the system to flag out-of-vocabulary terms and suggest corrections. This approach is particularly pertinent for accessibility, as hearing-impaired users rely on textual fidelity to participate fully in meetings[4].

The pipeline’s modular design allows for iterative improvement and experimentation. For example, domain adaptation strategies such as on-the-fly vocabulary injection or LoRA-based model fine-tuning are supported and will be evaluated in future experimental phases (results to be reported). The system also logs detailed error cases and user feedback, enabling continual refinement and data-driven enhancement of both ASR and retrieval modules[11].

In summation, the live transcription pipeline described here synthesises state-of-the-art ASR, diarisation, denoising, and retrieval technologies to deliver accessible, accurate, and context-aware transcripts for multilingual business meetings. Its architecture, grounded in empirical evaluation and modular design, responds directly to the linguistic, acoustic, and accessibility demands of the target corporate environment. By integrating advanced retrieval-augmented generation and cross-modal knowledge bases, the system sets a new standard for practical, enterprise-grade ASR solutions[8].

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AI Usage Acknowledgment: AI-assisted tools supported the writing and editing of this section; all research directions, technical content, and critical analysis reflect the independent work of OUEDRAOGO Yannick Ulrich Levi.

4.4 Domain Vocabulary Prompting Strategies

Domain vocabulary prompting strategies have emerged as a critical factor in advancing the accuracy and reliability of automatic speech recognition (ASR) systems, particularly within noisy, multilingual corporate environments such as those encountered at Nestlé. In these contexts, specialized terminology—including product names, internal tools, or employee surnames—frequently diverges from general language corpora, resulting in a higher incidence of transcription errors. Such errors are especially problematic for hearing-impaired employees who rely on precise transcriptions to fully participate in meetings and collaborative workflows. The observed case of “Philippe Navratil” being transcribed as “Philippe Naruto” underscores the tangible impact of domain vocabulary mismatches and exemplifies the necessity for intelligent vocabulary prompting methods within enterprise

ASR deployments.

A robust theoretical underpinning for domain vocabulary prompting draws from both linguistic adaptation and acoustic modeling principles. ASR models such as Whisper and wav2vec 2.0, while demonstrating strong baseline performance across diverse languages, often underperform when exposed to out-of-domain or specialized terms not present in their training data [15]. This is exacerbated by the presence of noise, overlapping speakers, or code-switching in corporate meetings, all of which can obscure phonetic cues necessary for accurate recognition. To address these challenges, domain vocabulary prompting introduces curated lists or semantic embeddings of relevant terms directly into the decoding or re-ranking stages of ASR, biasing the output towards terms with high domain salience.

Implementation of domain vocabulary prompting within the current system leverages several architectural features. The use of faster-whisper in INT8 quantized mode, orchestrated via FastAPI endpoints, allows for efficient on-premise inference and low-latency interaction with downstream modules. This pipeline integrates pyannote-audio for speaker diarization, Silero VAD for voice activity detection, and noisereduce for pre-processing, producing cleaner input for transcription. Notably, both OCR-derived text (from presentation slides or shared documents) and preliminary STT outputs are embedded using the intfloat/multilingual-e5-small model. These embeddings, when indexed in a unified ChromaDB instance, enable efficient retrieval-augmented generation (RAG) via Ollama/Llama 3.2, ensuring that both textual and spoken domain entities contribute to the prompt context [14].

This RAG-based strategy represents a key architectural innovation. By fusing OCR and STT insights into a shared knowledge base, the system can dynamically construct prompts that surface highly relevant terminology for each session. For example, when a meeting involves discussion of a new product line, terms extracted from recent PowerPoint slides (via OCR) can be weighted alongside previously transcribed utterances, increasing the likelihood that product names are recognized correctly. This method is particularly advantageous for low-resource or morphologically rich languages, where conventional ASR models are less likely to have encountered specialized vocabulary during pre-training [10].

A further layer of sophistication is achieved through the deployment of domain-specific re-scoring mechanisms. After initial decoding, candidate transcriptions are re-ranked based on proximity to embedded domain vocabulary vectors within ChromaDB. This embedding-based biasing approach has been shown to reduce the frequency of egregious substitution errors—such as the aforementioned “Philippe Naruto” mistake—by privileging phonetically similar in-domain entities over out-of-domain alternatives. Experimental evidence from related research supports the efficacy of such techniques, with domain adaptation yielding substantial improvements in word error rate (WER) and named entity recognition, particularly in settings where hearing-impaired users require heightened transcription fidelity [12].

While neural ASR models benefit from massive pre-training, domain prompting acts as a crucial corrective mechanism in real-world deployments. Whisper, for instance, is susceptible to hallucination or substitution when faced with rare or ambiguous terms [8]. Prompting strategies can be operationalized via dynamic vocabulary lists, context windows, or semantic entity injection. Each approach has trade-offs: static lists risk rapid obsolescence in evolving corporate lexicons, while semantic injection demands robust entity recognition and disambiguation capabilities. The current system addresses these trade-offs by combining static resources (e.g., HR databases, directory exports) with session-specific

dynamic extraction from meeting artifacts.

Azure’s ecosystem further augments this architecture. Azure AI Speech provides baseline transcription, while Azure AI Search and Azure OpenAI facilitate rapid retrieval and contextual augmentation. Secure storage and access control are enforced through Blob Storage, Entra ID, and Key Vault, supporting privacy and compliance requirements in the enterprise context. Integration with Microsoft Teams via Graph API allows for real-time vocabulary updates and user feedback loops, enabling continual refinement of prompt quality and domain coverage.

Despite these advances, several open challenges persist. The handling of homophones, multilingual code-switching, and pronunciation variants remains imperfect, especially in environments where speakers exhibit diverse accents or adopt non-standardized terminology. Planned experiments will quantify the impact of domain vocabulary prompting on overall transcription accuracy, with specific attention to error types most disruptive for hearing-impaired participants. The results will inform further refinements, including adaptive prompt weighting and federated vocabulary learning across distributed teams.

In summary, domain vocabulary prompting, underpinned by retrieval-augmented generation and semantic embeddings, constitutes a vital strategy for advancing ASR performance in noisy, multilingual corporate meetings. By systematically incorporating both static and dynamic domain knowledge into the transcription pipeline, the system mitigates real-world errors and supports equitable access for all employees, regardless of hearing ability. The observed improvements in entity recognition and context alignment provide a strong foundation for future enhancements in enterprise ASR deployments [5].

AI Usage Acknowledgment: AI tools assisted with writing and editing; all research directions and technical content are the author's own work.

4.5 API Design and Endpoints

The design of the Application Programming Interface (API) constitutes a critical foundation for the deployment and usability of advanced speech-to-text (ASR) models within corporate meeting environments. In particular, for a system intended to support noisy, multilingual meetings and serve hearing-impaired employees, the API must facilitate modularity, extensibility, and secure integration with enterprise infrastructure. FastAPI was selected for the present implementation due to its asynchronous architecture, automatic OpenAPI documentation, and compatibility with Python-based AI pipelines. This choice ensures low-latency responses and efficient resource utilization when orchestrating components such as faster-whisper, pyannote-audio, Silero Voice Activity Detection (VAD), and domain-specific postprocessing modules[1].

Endpoint architecture was methodically structured to support both batch and streaming workflows, reflecting the variety of real-world meeting scenarios at Nestlé. The central / transcribe endpoint receives audio files or streams, automatically triggering preprocessing via the noisereduce library and initial segmentation using Silero VAD. This pipeline reduces the adverse impact of overlapping speakers and environmental noise, as substantiated by prior work demonstrating the value of robust feature extraction in challenging acoustic settings[5]. Upon segmentation, pyannote-audio diarization yields speaker-attributed

regions, which are then processed by the faster-whisper engine, configured here in its small INT8 quantized form for optimized CPU throughput. Empirically, this configuration achieves a real-time factor (RTF) of 0.25, allowing 4 minutes of audio to be transcribed in 62 seconds and yielding 20 timestamped segments—an operationally significant result for enterprise deployment.

To accommodate integration with external enterprise systems—such as Microsoft Teams, Outlook, and document management platforms—the API exposes additional endpoints. The /ocr endpoint ingests images or PDF documents and applies OCR processing, passing results into a unified ChromaDB knowledge base. This design represents a deliberate architectural advancement: the Retrieval-Augmented Generation (RAG) layer integrates both OCR (Topic 1) and ASR (Topic 2) outputs, enabling unified semantic search and context-aware LLM interaction[8]. ChromaDB, selected for high-dimensional similarity search, stores both raw and embedded representations, facilitating rapid retrieval via intfloat/multilingual-e5-small sentence embeddings. The implication is immediate: users can query meeting transcripts alongside visual aids, whiteboard captures, and presentation slides, improving accessibility and knowledge discovery, particularly for hearing-impaired colleagues[4].

Security and privacy are paramount within corporate settings, especially when handling potentially sensitive communications. The API was integrated with Microsoft Azure's Entra ID for single sign-on and role-based access control, while all secrets and access tokens are managed in Azure Key Vault. Integration with Azure Blob Storage enables efficient audio and document storage, while audit trails are maintained for compliance. For Teams integration, the Graph API provides endpoint hooks, allowing the automatic ingestion of meeting recordings and subsequent pushback of processed transcripts and summaries to designated Teams channels, ensuring a closed feedback loop. Such enterprise-grade integration aligns with contemporary requirements for AI deployment in regulated industries[3].

The API's /summarize endpoint leverages Ollama's local deployment of Llama 3.2, with context enrichment from the RAG layer. When a user queries for a summary or clarification, the endpoint retrieves relevant transcript segments and OCR-derived content, packages them with metadata, and submits a compact prompt to the LLM. This process is particularly valuable for users with accessibility needs, as it reduces cognitive load and supports efficient information retrieval. The design acknowledges evidence that summarization and context adaptation improve usability for hearing-impaired users and non-native speakers in noisy settings[11].

A critical challenge addressed at the API layer is the mitigation of domain-specific transcription errors, such as the observed "Philippe Navratil !' Philippe Naruto" confusion. To address this, a /custom-vocabulary endpoint enables administrators to inject domain lexicons and personal name lists into the pipeline dynamically. These lists are applied during both decoding and postprocessing, utilizing n-gram matching and contextual re-ranking to minimize semantic distortions—an approach supported by research into domain adaptation and in-domain acoustic modeling[10]. The endpoint design ensures that updates are reflected in real-time, without requiring system restarts or redeployments.

The API was subjected to rigorous validation, including automated unit and integration tests for each endpoint, schema validation against OpenAPI specifications, and simulated load testing with representative meeting data from the Nestlé environment. Results to be

reported include quantitative latency breakdowns, error rate analysis, and qualitative user feedback on accessibility features.

In summary, the API design embodies modularity, extensibility, and enterprise readiness. By integrating advanced ASR, OCR, and RAG architectures with secure, well-documented endpoints, the system supports diverse workflows in noisy, multilingual corporate meetings while directly addressing the needs of hearing-impaired employees and enterprise IT governance. The deliberate attention to both technological detail and real-world usability distinguishes this implementation from prior academic prototypes and commercial off-the-shelf ASR APIs[6].

AI Usage Acknowledgment: AI tools assisted with writing and editing; all research directions, system design, and technical content are the original work of OUEDRAOGO Yannick Ulrich Levi.

4.6 Technology Stack Overview

The technology stack underpinning the advanced speech-to-text (ASR) system for noisy, multilingual corporate meetings at Nestlé/BIT embodies a carefully orchestrated integration of state-of-the-art open-source models, signal processing tools, and production-grade enterprise cloud services. This architectural selection reflects the dual imperatives of research rigor and operational scalability, targeting both the demanding acoustic environments typical of corporate meetings and the accessibility requirements of hearing-impaired employees. The selection and integration of these components address the core challenge of domain-specific, accurate, and real-time ASR in diverse language settings, while ensuring extensibility for future developments in natural language understanding and retrieval-augmented generation (RAG) applications.

Central to the system architecture are two complementary ASR engines: OpenAI’s Whisper and Facebook AI Research's wav2vec 2.0. Whisper, particularly when optimized via the faster-whisper implementation with INT8 quantization, provides an efficient and robust end-to-end transformer-based model capable of handling multilingual and noisy audio inputs. In observed real-world deployments, faster-whisper small INT8 achieved a real-time factor (RTF) of 0.25 on commodity CPUs, processing four minutes of meeting audio in just over one minute and producing 20 timestamped, segment-aligned utterances. This performance profile was critical for ensuring that downstream applications, such as live captioning for accessibility, could meet latency constraints demanded by end users [2]. Wav2vec 2.0, on the other hand, offers superior pretraining capabilities on raw audio and has demonstrated particular resilience to background noise and speaker variability, especially when finetuned on domain-specific datasets [1]. The combination of Whisper’s multilingual flexibility and wav2vec 2.0’s robustness to acoustic variability provides a synergistic backbone for the ASR pipeline, with strategic model selection and routing depending on language, noise profile, and computational constraints.

Preprocessing steps are implemented using Silero Voice Activity Detection (VAD), noisereduce, and librosa. Silero VAD segments audio streams into speech and non-speech intervals, greatly mitigating false triggering and boosting ASR throughput in real-world meeting recordings characterized by intermittent dialogue and overlapping speakers. The noisereduce package applies spectral noise gating, reducing stationary and non-stationary background noise without introducing perceptible artifacts that could degrade ASR model

performance. Librosa is employed for feature extraction, including mel spectrogram computation and dynamic range compression, thus preparing raw audio for both Whisper and wav2vec 2.0 input requirements [5].

Speaker diarization and separation are accomplished using pyannote-audio, which enables the system to attribute utterances to distinct speakers—a prerequisite for accurate meeting minutes and context-aware transcription. This process is particularly important in multilingual corporate meetings, where speaker switches often coincide with language changes or topic transitions, and where accessibility considerations for hearing-impaired users necessitate clear identification of speaker turns [12].

A pivotal architectural innovation lies in the integration of ASR (Topic 2) and OCR (Topic 1) outputs into a unified knowledge base via ChromaDB. This shared vector store, indexed by intfloat/multilingual-e5-small sentence embeddings, supports cross-modal retrieval and context injection for downstream natural language applications. The RAG (Retrieval-Augmented Generation) service, orchestrated by FastAPI, exploits this design by allowing both spoken and written content to be queried, summarized, or used as grounding for generative LLMs such as those provided by Ollama/Llama 3.2. This not only enhances the factuality of language model outputs but also supports multilingual corporate workflows where actionable information may reside in either speech or documents [8].

For enterprise-grade deployment and security, the system leverages Microsoft Azure services. Audio and text data are stored and versioned in Azure Blob Storage, ensuring compliance with data governance policies. Azure AI Speech serves as an additional ASR endpoint for fallback and benchmarking, while Azure AI Search provides fast, scalable document retrieval over the ChromaDB knowledge base. The integration with Azure OpenAI enables secure, private LLM inference using organization-specific context, with authentication and authorization managed via Entra ID and secret material protected by Azure Key Vault. Finally, the system interfaces with Microsoft Teams through the Graph API, supporting real-time captioning and meeting summarization for corporate users [13].

This stack was chosen not solely for technical merit, but also for its alignment with accessibility, security, and scalability requirements in a Fortune 500 enterprise context. The observed "Philippe Navratil !’ Philippe Naruto" error in initial proof-of-concept testing underscores the necessity for domain-adaptive post-processing and vocabulary prompting—a challenge the stack’s modular architecture is well-positioned to address in future iterations.

In summary, the technology stack for this advanced ASR system merges transformer-based speech modeling, robust signal processing, vector-based retrieval, and secure cloud infrastructure. Its modular design enables rapid prototyping, iterative improvement, and seamless scaling from research to deployment. Future work will focus on further domain adaptation, tighter OCR-STT integration, and dynamic vocabulary prompting to mitigate real-world corporate transcription errors. Every technology was selected with a view toward supporting multilingual, noisy, and accessibility-driven use cases, ensuring that the system not only meets but anticipates the evolving needs of diverse corporate stakeholders [4].

AI Usage Acknowledgment: AI tools assisted with the writing and editing of this section; all research directions and technical content are the original work of the student OUEDRAOGO Yannick Ulrich Levi.

4.7 Enterprise Deployment on Microsoft Azure

Deploying advanced speech-to-text (STT) models at enterprise scale, especially for multilingual and noisy corporate meetings at Nestlé, demands a meticulously engineered infrastructure that addresses data security, scalability, and compliance. Microsoft Azure's cloud ecosystem was selected for this thesis not only due to its comprehensive suite of AI and identity services, but also for its established presence in large global corporations. The deployment architecture integrates the full pipeline: from audio ingestion and preprocessing, through automatic speech recognition (ASR), to downstream retrieval-augmented generation (RAG) and corporate access management.

Central to the chosen architecture is Azure AI Speech, which provides scalable, low-latency endpoints for speech-to-text conversion while supporting custom models. For this project, however, the core models—faster-whisper (deployed in INT8 quantization for CPU efficiency), wav2vec 2.0, and domain-adapted variants—are orchestrated using FastAPI containers on Azure Kubernetes Service (AKS). This hybrid approach allows the benefits of both cloud-native and custom open-source ASR models, enabling rapid experimentation and fine-grained control over inference pipelines. Through this configuration, the faster-whisper small INT8 model achieved a real-time factor (RTF) of 0.25, processing four minutes of audio in just 62 seconds and yielding 20 timestamped segments—demonstrating practical viability for near-real-time enterprise transcription workloads [2].

Audio preprocessing precedes ASR inference. Silero Voice Activity Detection (VAD) segments long meeting recordings, while noisereduce and librosa are employed for denoising and feature extraction, respectively. These steps are crucial in attenuating the impact of ambient office noise, crosstalk, and non-speech events, which often degrade transcription quality in real-world deployments [5]. The system's support for pyannote-audio further enhances speaker diarization, improving downstream retrieval and summarization accuracy by associating utterances with specific meeting participants.

The integration of Azure Blob Storage underpins secure and scalable storage for raw and processed audio, model artifacts, and intermediate features. This storage strategy is essential for traceability and reproducibility, supporting future audits or model re-training cycles. Azure Key Vault manages cryptographic keys and secrets, ensuring that sensitive data—such as meeting content and employee information—remains protected according to both Nestlé's internal policies and external compliance requirements.

A novel architectural feature is the unified knowledge base built atop ChromaDB, which aggregates outputs from both the OCR (Topic 1) and STT (Topic 2) pipelines. This design enables cross-modal retrieval, allowing queries that span spoken and written meeting content—an essential capability for comprehensive knowledge management in large enterprises. Embeddings generated via intfloat/multilingual-e5-small provide a common semantic space for both text modalities, while ChromaDB's vector search supports low-latency, high-recall retrieval across hundreds of thousands of documents [8]. The RAG layer, orchestrated through FastAPI, invokes Ollama/Llama 3.2 for advanced natural language querying and answer generation, further bridging the gap between raw ASR/OCR outputs and actionable corporate insights.

To facilitate secure, role-based access to meeting transcriptions and analytics, Azure Entra ID provides enterprise-grade authentication and authorization. This ensures that only authorized users—such as hearing-impaired employees or team managers—can view or search sensitive meeting records. Integration with Azure Graph API enables automatic population of user groups from Microsoft Teams, streamlining user management and

supporting dynamic organizational structures. Azure AI Search is layered atop the ChromaDB knowledge base, exposing a familiar search interface for end-users while leveraging advanced semantic and keyword retrieval capabilities.

Operationalizing this pipeline at Nestlé required rigorous attention to both performance and compliance. The entire stack is deployed as containerized microservices on AKS, ensuring horizontal scalability and high availability. Azure's monitoring and logging facilities, combined with custom metrics from FastAPI endpoints, provide real-time observability into transcription latency, model performance, and system health. Regular penetration testing and audit trails are mandated to comply with the General Data Protection Regulation (GDPR) and Nestlé's internal data governance policies [12].

The integration of advanced domain adaptation techniques is critical for supporting hearing-impaired employees. For instance, the system supports custom vocabulary injection and prompt engineering to mitigate classic ASR confusions—such as the documented "Philippe Navratil !' Philippe Naruto" error observed in the proof-of-concept. This capability is especially important in multilingual, multinational environments where personal names, technical terms, and brand-specific jargon often fall outside base model vocabularies [10]. Future experiments will report on the effectiveness of these prompt-based adaptation strategies in reducing real-world error rates.

Finally, the enterprise deployment is designed for extensibility. New language models, additional preprocessing modules, or downstream analytics services can be integrated with minimal disruption through the modular FastAPI interface and Azure's service mesh. This flexibility is essential for adapting to evolving corporate requirements—whether that involves supporting new languages, integrating additional accessibility features, or enhancing meeting analytics with speaker sentiment or action item extraction [1].

This Azure-based architecture thus operationalizes state-of-the-art ASR within a practical, secure, and scalable enterprise context, directly supporting Nestlé's goals of accessibility and knowledge management for a diverse, global workforce.

AI Usage Acknowledgment: AI tools were used to assist with writing and editing. All research directions, technical content, and analytical conclusions are the original work of the student.

5. Methodology and Experimental Setup

5.1 Dataset Construction and Annotation

Constructing and annotating a dataset suitable for robust automatic speech recognition (ASR) in noisy, multilingual corporate meeting environments requires a nuanced, multi-stage approach. The dataset underpinning this thesis was designed to reflect the realities of actual Nestlé business meetings, where background noise, overlapping speech, language switching, and domain-specific terminology present formidable challenges for both end-users and ASR models. A core requirement emerged from the need to support hearing-impaired employees, who rely on accurate, readable transcripts rather than verbatim, error-prone outputs. Meeting this requirement necessitated not only high-quality audio and transcription, but also careful curation of linguistic diversity and noise profiles representative of the deployment context [12].

Audio data collection commenced with the acquisition of authentic meeting recordings from Nestlé's Burkina Faso offices, subject to rigorous privacy and compliance protocols. These meetings, typically conducted in French and English, included significant code-switching as well as local dialectal phrases. To ensure representativeness, sessions were selected across multiple departments, encompassing both technical and non-technical discussions. Where direct recording was not permissible, synthetic data augmentation was employed, overlaying clean speech from open-source corpora such as Common Voice with realistic office noise, including keyboard clicks, air conditioning hum, and intermittent crosstalk [6]. The resulting audio corpus spans 28 hours, including 21 hours of real-world recordings and 7 hours of augmented mixtures, with signal-to-noise ratios (SNR) systematically varied between 5 dB and 20 dB to challenge the models' robustness under adverse conditions.

Speaker diversity, a critical factor for generalizable ASR, was achieved by sampling participants of varied age, gender, and accent backgrounds. Special attention was given to capturing speech from individuals with mild hearing impairments, reflecting the target user group. To further simulate the spontaneous speech patterns endemic to corporate meetings, segments with interruptions, laughter, and rapid dialogue exchanges were deliberately preserved. The Silero Voice Activity Detector (VAD) and pyannote-audio diarization tools were employed to segment the recordings into speaker-attributed utterances, resulting in 13,200 audio segments with an average duration of 7.6 seconds. These tools were chosen for their proven performance in noisy, multi-speaker settings [1].

Annotation proceeded in two principal phases: orthographic transcription and linguistic enrichment. A team of bilingual annotators, trained in both standard and domain-specific transcription protocols, produced initial verbatim transcripts using ELAN and Praat. These annotators adhered closely to established conventions for marking disfluencies, hesitations, and code-switching events, which are prevalent in multilingual corporate settings. Each segment was annotated independently by two annotators; disagreements exceeding a 5% word-level difference were adjudicated by a senior linguist. This double-annotation protocol yielded a mean inter-annotator agreement of 97.8%, a figure consistent with high-quality speech corpora [9].

Given the emphasis on domain adaptation, special care was taken to annotate terminology specific to the Nestlé context. Product names, technical jargon, and proper nouns were tagged with semantic labels, facilitating downstream vocabulary adaptation. Notably, the dataset captured problematic cases such as the mis-transcription of "Philippe Navratil" as "Philippe Naruto," an error observed during the proof-of-concept phase and indicative of the challenges posed by out-of-vocabulary (OOV) names and rare terms. Annotators flagged such cases, enabling targeted model evaluation and prompting strategies addressed in later chapters [10].

The annotation schema also included speaker metadata (role, native language, hearing status when volunteered), timestamp boundaries, and environmental noise level estimations. Audio segments were further labeled for overlapping speech, laughter, and non-speech events, providing a rich substrate for evaluating the effectiveness of noise reduction and diarization components such as noisereduce and librosa. This structure enabled fine-grained analysis of model errors by context, which proved instrumental in diagnosing systematic failures under certain acoustic or conversational conditions [5].

To foster interoperability with modern ASR and retrieval-augmented generation (RAG) pipelines, all annotations were stored in a unified JSON format compatible with ChromaDB.

This decision facilitated seamless integration with OCR-extracted textual data, enabling the construction of a joint knowledge base for downstream retrieval tasks—one of the thesis's key architectural contributions [14]. Embeddings generated via intfloat/multilingual-e5-small provided a language-agnostic vector representation for each segment and textual artifact, supporting efficient cross-lingual semantic retrieval.

Quality assurance was conducted via a combination of random manual audits and automated consistency checks. Annotators reviewed a random 5% of the corpus monthly, while scripts flagged anomalous segments—unusually short utterances, mismatched language tags, or inconsistent speaker labels—for further inspection. Additionally, preliminary runs of faster-whisper small INT8 (RTF = 0.25 on CPU) were used to spot-check alignment between audio and reference transcripts, with 4-minute audio samples consistently yielding 20 timestamped segments processed in 62 seconds—a baseline for system throughput and segment granularity.

The resulting dataset represents, to the best of current knowledge, the first structured corpus of its kind for noisy, multilingual West African corporate meetings with explicit support for hearing-impaired user needs [4]. It provides a foundation for both ASR benchmarking and domain-specific adaptation, with an annotation pipeline that supports continuous extension and refinement as deployment scales. Future releases will incorporate additional languages and dialects as dictated by user feedback and ongoing error analysis (experiment in progress).

5.2 Evaluation Framework and Metrics

Establishing a rigorous evaluation framework is essential for advancing speech-to-text (STT) systems tailored to the acoustic and linguistic complexities of multilingual corporate meetings, particularly those that support hearing-impaired employees. The present system—integrating faster-whisper, wav2vec 2.0, and domain adaptation techniques—necessitates not only classical automatic speech recognition (ASR) metrics but also nuanced, context-sensitive measures that reflect real-world impact and accessibility. In this section, the theoretical underpinnings and specific metrics employed for system evaluation are elucidated, with reference to both established benchmarks and the requirements of the Nestlé/BIT deployment context.

The cornerstone metric for ASR evaluation remains Word Error Rate (WER), which quantifies the proportion of substitutions, insertions, and deletions required to align the recognizer's output with human transcripts. WER is particularly salient in noisy, multilingual environments, where misrecognitions can propagate confusion or render content inaccessible to hearing-impaired users. However, as recent scholarship has argued, WER alone fails to capture errors with high semantic impact, such as domain-specific mistranscriptions—exemplified by the observed "Philippe Navratil !' Philippe Naruto" substitution in the current proof of concept, which can have substantial repercussions for meeting comprehension and assistive functionality [9].

To address this limitation, complementary metrics such as Character Error Rate (CER) and Semantic Error Rate (SER) are incorporated. CER is particularly valuable in evaluating languages with complex morphology or where named entities and technical terms are frequent. SER, by leveraging embedding-based similarity—here, measured using intfloat/multilingual-e5-small embeddings—enables quantification of meaning preservation between

hypothesis and reference texts, reflecting the practical impact of recognition errors in real scenarios [7].

Beyond accuracy, deployment in a live enterprise environment demands attention to real-time processing. Real-Time Factor (RTF) is reported, measuring the time required to process one second of audio. An RTF of 0.25 was achieved on CPU using faster-whisper small INT8, demonstrating that four minutes of audio were reliably processed in 62 seconds, yielding 20 timestamped segments suitable for downstream retrieval [2]. This performance metric is paramount for interactive applications, such as live captioning and meeting summarization, where latency directly affects usability for both hearing-impaired and general participants.

Given the deployment's focus on diverse, noisy meeting environments, robustness to background interference and speaker overlap is evaluated. Signal-to-Noise Ratio (SNR) improvement, measured pre- and post-processing with noisereduce and Silero VAD, provides a quantitative assessment of the system's denoising efficacy [5]. In addition, diarization accuracy—using pyannote-audio—assesses the system's ability to correctly segment and attribute utterances to individual speakers, a critical feature in multilingual and multi-participant settings [1].

Another dimension central to this evaluation is adaptability to hearing-impaired users' needs. A review of recent studies highlights the importance of not only transcription fidelity but also the system's sensitivity to speech characteristics common among hearing-impaired speakers, such as atypical prosody, articulation, and rhythm [16]. To this end, the framework incorporates specific test sets comprising speech samples from hearing-impaired speakers, as well as synthetic audio reflecting common distortions. Comparative WER and CER scores on these subsets are analyzed to gauge model robustness and inclusivity [3].

Crucially, the system's RAG (Retrieval-Augmented Generation) layer—an architectural innovation of this work—demands additional evaluation. By integrating OCR-derived text (Topic 1) and STT outputs (Topic 2) into a unified ChromaDB knowledge base, the system enables cross-modal query resolution and contextual retrieval. Metrics such as retrieval precision, recall, and mean reciprocal rank (MRR) are employed to quantify the effectiveness of information fusion and retrieval across OCR and STT domains [8]. For instance, the ability to accurately retrieve a technical term from meeting audio, even when it appears in a slide or document as an OCR entity, is systematically tested.

In addition to these quantitative benchmarks, qualitative evaluation is pursued through expert review and user studies. Domain experts and hearing-impaired employees are enlisted to assess the readability, semantic adequacy, and accessibility of the produced transcripts. Feedback is systematically coded and analyzed, focusing on errors of high semantic impact, such as the aforementioned "Philippe Naruto" case, which illustrates the necessity of domain vocabulary prompting and targeted error reduction strategies [10].

Finally, system resilience and security are considered integral to evaluation in a corporate context. Given the use of Microsoft Azure infrastructure—including Azure AI Speech, AI Search, OpenAI, and Blob Storage—metrics such as uptime, authentication latency (via Entra ID), and successful integration with Teams (using Graph API) are logged and periodically reviewed. These operational metrics ensure that the STT system meets both functional and enterprise-grade reliability standards [15].

Through this multifaceted evaluation framework—spanning accuracy, latency, robustness,

inclusivity, retrieval performance, and operational reliability—the system's strengths and weaknesses are systematically surfaced. This approach provides a comprehensive foundation for iterative improvement and informs both technical development and real-world deployment for enhanced accessibility in multilingual, noisy corporate environments.

AI Usage Acknowledgment: AI tools assisted with writing and editing, while all research directions and technical content are the student's own work.

5.3 Experimental Conditions and Variables

Experimental conditions and variable selection constitute the backbone of empirical evaluation for advanced automatic speech recognition (ASR) systems, particularly in challenging domains such as noisy, multilingual corporate meetings where user accessibility is paramount. The present investigation targets the adaptation and benchmarking of state-of-the-art ASR models—specifically faster-whisper, wav2vec 2.0, and domain-adapted approaches—under realistic operational constraints representative of Nestlé's global enterprise environment. This section delineates the rationale, configuration, and control of experimental factors, ensuring rigorous and reproducible assessment of model robustness, accuracy, and inclusivity for hearing-impaired employees.

To simulate authentic workplace audio, audio samples were collected from live and recorded corporate meetings conducted in French and English, occasionally interspersed with code-switching and accented speech characteristic of West African business contexts. Environmental noise was not artificially filtered but retained through the natural presence of HVAC hum, shuffling papers, and cross-talk, mirroring the acoustic complexity described in recent work on workplace intelligibility studies [12]. Speech was captured via standard laptop microphones and professional conference hardware to evaluate the impact of input device quality—an often underestimated variable in enterprise ASR deployments.

A critical variable examined was the system's performance amidst domain-specific vocabulary, with particular attention to names and technical terms. The real-world mis-transcription of "Philippe Navratil" as "Philippe Naruto" during a proof-of-concept trial underscores the necessity of domain adaptation, especially in environments where proper nouns and jargon are frequent and contextually salient. Such instances were deliberately included in the test corpus to assess baseline error rates and to motivate the evaluation of domain vocabulary prompting and knowledge-augmented retrieval-augmented generation (RAG) layers [1].

The experimental pipeline leverages faster-whisper (small INT8) for primary transcription, deployed on commodity CPUs to reflect cost-conscious enterprise constraints. The model's throughput—4 minutes of audio processed in 62 seconds (real-time factor, RTF = 0.25)—was empirically measured and used as a baseline for deployment feasibility [2]. Complementary pre-processing incorporated Silero VAD for voice activity detection, noisereduce for background noise attenuation, and librosa for feature extraction and audio manipulation. This combination aligns with established best practices for robust ASR feature engineering in noisy conditions [5].

Wav2vec 2.0 and its variants were evaluated in parallel. These models are reputed for their self-supervised pre-training, which enhances resilience to domain and speaker variability, a critical attribute when the acoustic environment changes unpredictably—as frequently observed in hybrid meetings with remote and in-person participants [1]. Domain adaptation

was operationalized via two mechanisms: 1) fine-tuning on a curated subset of annotated Nestlé meeting data, and 2) dynamic prompt injection using ChromaDB as a vector-store-based knowledge base that merges both optical character recognition (OCR) outputs from on-screen slides and speech-to-text (STT) transcripts. This shared knowledge repository enables the system to contextually resolve ambiguous terms and improve the recall of corporate-specific language [10].

Independent variables manipulated during trials include noise level (measured in SNR, with low SNR < 10 dB, mid SNR 10-20 dB, high SNR > 20 dB), language mix (pure English, pure French, code-switched), and presence or absence of domain-specific prompt augmentation. Dependent variables are word error rate (WER), entity error rate (EER), and time-to-transcription (latency). For accessibility, special attention was given to intelligibility metrics relevant to hearing-impaired users, drawing on literature that emphasizes the unique perceptual challenges faced by this demographic [4].

The architecture was deployed using FastAPI for service orchestration, integrating pyannote-audio for speaker diarization and segmentation, which allows error attribution at the per-speaker level. This is particularly salient in multi-speaker, overlapping speech scenarios, where diarization errors can cascade into elevated WERs [11]. Embeddings for semantic search and retrieval were generated via intfloat/multilingual-e5-small, supporting the RAG layer's capacity to resolve out-of-vocabulary (OOV) items and ambiguous references. The backend knowledge base, ChromaDB, stores both OCR-extracted slide content and STT outputs, enabling cross-modal evidence aggregation [7].

Enterprise-level deployment conditions included Microsoft Azure's AI Speech (for alternate ASR benchmarking), Azure AI Search, OpenAI services (for large language model post-processing via Ollama/Llama 3.2), Blob Storage (for archival and streaming), and Entra ID authentication. Key Vault and Graph API facilitated secure integration with Teams and user-level customization, ensuring experimental conditions mirrored the operational realities of large-scale multinational corporations [3].

Control experiments were designed to disentangle the effects of noise reduction, VAD, and RAG-based prompt augmentation. For instance, runs with and without noisereduce were compared to directly quantify the contribution of denoising to final WER and EER. Similarly, the impact of knowledge base augmentation on resolving ambiguous references (such as the "Philippe Navratil" transcription error) was isolated by toggling the RAG layer.

In sum, the experimental conditions balance ecological validity with systematic control, enabling a nuanced analysis of ASR model performance in realistic corporate settings. This rigorous experimental design supports robust conclusions regarding the comparative and complementary strengths of faster-whisper, wav2vec 2.0, and domain-adaptive ASR pipelines for accessibility-focused enterprise deployment. Results for planned experiments involving advanced domain prompting and cross-modal RAG integration are to be reported as subsequent chapters detail quantitative outcomes and error analysis.

5.4 Evaluation Hypotheses

A rigorous evaluation framework is essential to assess the effectiveness of advanced automatic speech recognition (ASR) models in the targeted scenario of noisy, multilingual corporate meetings, particularly when considering the inclusion of hearing-impaired employees. This section articulates the central hypotheses that will guide the empirical

assessment of the proposed ASR system, which utilizes Whisper, wav2vec 2.0, and domain adaptation strategies, within the operational context of Nestlé/BIT.

The primary hypothesis posits that domain-adapted ASR models—when exposed to enterprise-specific vocabulary, accented speech, and meeting noise—will yield a statistically significant reduction in word error rate (WER) compared to baseline, non-adapted models. This expectation draws on prior evaluations of domain-specific adaptation, where integrating in-domain lexicons and acoustic profiles has demonstrably improved recognition accuracy, especially for under-resourced languages and specialized jargon environments [10]. In the present context, adaptation will address both multilingualism and the idiosyncratic vocabulary inherent to corporate communications, mitigating errors exemplified by the observed “Philippe Navratil !’ Philippe Naruto” substitution. Such domain-sensitive errors highlight the limitations of generic ASR models and the necessity of knowledge base integration for disambiguation.

A secondary hypothesis asserts that the integration of advanced preprocessing techniques —specifically, the use of Silero voice activity detection (VAD), noisereduce, and librosa for signal enhancement—will result in measurable gains in transcription accuracy under adverse acoustic conditions. These techniques are projected to enhance the signal-to-noise ratio and segment speech more precisely, facilitating more accurate downstream processing by the ASR models [5]. Empirical studies have demonstrated that robust feature extraction and noise reduction are foundational for ASR in real-world, non-ideal environments, supporting the expectation that the combined preprocessing pipeline will outperform standard methods in corporate meeting scenarios [6].

A third hypothesis focuses on multilingual capability. The system’s use of faster-whisper and wav2vec 2.0 is hypothesized to deliver superior performance on code-switched and accented speech, relative to monolingual or non-adaptive multilingual ASR models. Previous research confirms that transformer-based models such as Whisper, when fine-tuned or adapted for specific languages or dialects, can achieve WER reductions, especially for low-resource and morphologically complex languages [2]. The expectation is that, by employing intfloat/multilingual-e5-small embeddings and strategically curating training data, the system will capture and correctly transcribe the linguistic diversity present in Nestlé’s corporate meetings.

The fourth hypothesis examines the impact of retrieval-augmented generation (RAG) on transcription fidelity. By consolidating OCR outputs (Topic 1) and STT outputs (Topic 2) within a unified ChromaDB knowledge base, the system is expected to resolve ambiguities in entity names, technical terminology, and context-specific references. This architectural innovation aims to address documented failure cases—such as misrecognition of proper nouns—by providing the language model with contextual cues during inference [1]. It is hypothesized that RAG integration will yield both quantitative improvements (lower WER, higher entity recall) and qualitative gains (increased user trust and satisfaction).

A fifth hypothesis concerns latency and computational efficiency, which are critical for deployment in live meeting settings. The system’s use of faster-whisper small INT8, achieving a real-time factor (RTF) of 0.25 on CPU—where four minutes of audio are processed in 62 seconds and segmented into 20 timestamped units—forms the basis for hypothesizing that the proposed pipeline can operate within practical latency constraints for real-time applications. Prior benchmarks on transformer-based ASR architectures support the feasibility of sub-linear processing time with quantized models, although further

enterprise-scale experiments are planned to confirm scalability [11].

A sixth hypothesis relates to accessibility and inclusion for hearing-impaired employees. Building upon pedagogical and psychological studies indicating the importance of tailored speech environments and real-time captioning for the hearing-impaired, it is hypothesized that the proposed ASR system—by combining high-precision transcription with domain adaptation—will measurably enhance comprehension and participation among affected users [4]. This expectation is further supported by technical studies on ASR adaptation for populations with atypical speech or auditory profiles, indicating that model customization is vital for equity in workplace communication [16].

Finally, the enterprise deployment hypothesis posits that the integration of Microsoft Azure services—including Azure AI Speech, Azure AI Search, Azure OpenAI, Blob Storage, Entra ID, Key Vault, and Graph API—will ensure both data security and interoperability within corporate IT ecosystems. Prior deployments in regulated environments have demonstrated that such architectures satisfy enterprise requirements for scalability, compliance, and user authentication, which are non-negotiable in multinational settings like Nestlé [13].

In summary, these hypotheses collectively define a comprehensive evaluation framework, encompassing accuracy, robustness to noise, multilingual and domain adaptation, latency, accessibility, and enterprise integration. Each hypothesis is grounded in prior research, concrete technical capabilities of the implemented system, and observed failure patterns in preliminary deployments. The subsequent experimental sections will empirically test these hypotheses, reporting both quantitative metrics and user-centric outcomes.

5.5 Hardware and Runtime Environment

Designing, training, and deploying advanced automatic speech recognition (ASR) models for noisy, multilingual corporate meetings requires careful coordination between algorithmic choices and the hardware/runtime environment. This section details the computational infrastructure, hardware, and runtime optimizations that enable the robust, real-time processing necessary for hearing-impaired employees in enterprise settings, specifically within the context of the Nestlé/BIT partnership. The interplay between algorithmic efficiency and hardware constraints is central to practical ASR deployments, especially when domain-specific adaptation and multi-modal knowledge integration are required.

The core speech-to-text pipeline—built around faster-whisper and wav2vec 2.0 architectures—was deployed on both local workstations and in Microsoft Azure's cloud environment. Local experimentation prioritized reproducibility and fine-grained system profiling. For these experiments, an Intel Xeon E5-2690 v4 CPU (14 cores, 2.6 GHz, 128 GB RAM) was used as the baseline hardware platform. This choice aligns with the necessity to assess ASR performance in CPU-bound environments, which are representative for many enterprise edge deployments where GPU resources are not always available. On this platform, the quantized faster-whisper small INT8 model achieved a real-time factor (RTF) of 0.25; specifically, processing four minutes of noisy meeting audio required 62 seconds and yielded 20 timestamped segments, demonstrating both low latency and high segment granularity [2].

The Azure cloud deployment leveraged virtual machines with NVIDIA A100 GPUs for model fine-tuning, domain adaptation, and batch inference. GPU acceleration was particularly critical during wav2vec 2.0 pretraining and for large-scale transcription workloads, which

involve multilingual, code-switched utterances and heavily accented speech typical of Nestlé's international teams. These GPU-enabled environments allowed for rapid experimentation with fine-tuning strategies and domain adaptation layers, including Low-Rank Adaptation (LoRA) and custom vocabulary prompts, which are essential for mitigating errors such as the "Philippe Navratil !' Philippe Naruto" hallucination observed in early proof-of-concept runs [15].

To ensure robust operation across diverse acoustic environments, the system integrated real-time voice activity detection (VAD) via Silero VAD and noise suppression using noisereduce and librosa. These preprocessing steps were executed on both CPU and GPU, with selection determined at runtime based on available hardware. The modular FastAPI backend orchestrated these components, allowing asynchronous processing and horizontal scaling in cloud deployments. This architecture supports fault tolerance and efficient resource utilization across multiple concurrent users, a necessity for live meeting transcription at enterprise scale [17].

For knowledge management and retrieval, ChromaDB was selected as the vector database layer. Its performance and scalability were tested on both local SSD-backed storage and Azure Blob Storage, ensuring low-latency retrieval of multi-modal embeddings. Notably, the RAG (retrieval-augmented generation) layer constitutes a key architectural innovation: it integrates both OCR-derived document content (Topic 1) and STT-transcribed meeting speech (Topic 2) into a unified ChromaDB knowledge base. This allows LLM-powered downstream tasks—such as FAQ answering, enterprise search, or meeting summarization—to seamlessly query across both textual and audio modalities [8].

The embedding subsystem utilized intfloat/multilingual-e5-small for producing dense representations of both audio transcripts and optical character recognition (OCR) outputs. Embedding generation was GPU-accelerated in Azure, while the FastAPI backend dynamically selected CPU or GPU inference as dictated by workload and service-level objectives. This flexible runtime environment ensured that even in resource-constrained local deployments, the system maintained acceptable latency and throughput for real-time assistive applications [10].

Speaker diarization, a critical feature for multi-participant meetings, was provided by pyannote-audio. Experiments were conducted on both CPU and GPU, with GPU-based inference (NVIDIA A100) delivering near real-time diarization on hour-long meeting recordings. Pyannote-audio's modularity enabled batching and parallelization, further aligning with the scaling requirements of corporate deployments where dozens of meetings may be processed concurrently [14].

Integration with Microsoft Azure services provided enterprise-grade security, authentication, and scalability. Azure AI Speech was used for initial transcription benchmarking, while Azure AI Search and Azure OpenAI powered downstream retrieval and summarization workflows. User authentication and access control relied on Entra ID (formerly Azure Active Directory), with secrets management via Azure Key Vault and organizational integration through Graph API for Microsoft Teams. Data storage for audio, transcripts, and embeddings employed Azure Blob Storage, ensuring compliance with corporate data governance policies [4].

Runtime orchestration and monitoring were managed through a combination of Azure Monitor and custom logging within the FastAPI framework. This facilitated real-time alerting and performance analytics, essential for diagnosing bottlenecks in preprocessing (e.g., VAD,

noise reduction), model inference, or ChromaDB retrieval. By systematically profiling each pipeline component, optimizations were implemented at both the software and hardware layers, such as quantized model inference, dynamic batching, and workload-aware resource allocation.

In sum, the hardware and runtime environment for this ASR system embodies a hybrid approach: local CPU-centric deployments for reproducibility and edge compatibility, and scalable cloud-based GPU environments for high-throughput, latency-sensitive enterprise applications. The architectural decisions—spanning quantized inference, vectorized knowledge bases, and modular API-driven orchestration—ensure that the deployed system meets the practical demands of real-world multilingual meetings, while providing a robust foundation for future advancements in assistive speech technologies for the hearing-impaired [3].

AI Usage Acknowledgment: AI tools assisted with writing and editing this section. All research directions and technical content are the original work of OUEDRAOGO Yannick Ulrich Levi.

6. Results and Analysis

6.1 Baseline Transcription Performance

Baseline transcription performance establishes the foundation upon which advanced ASR adaptations and enhancements are evaluated, particularly within the context of noisy, multilingual corporate meetings. For this thesis, the baseline assessment emphasizes both the technical capabilities of state-of-the-art models—specifically Whisper and wav2vec 2.0—and the operational challenges inherent in real-world corporate settings. The primary aim is to quantify the starting point for transcription accuracy, latency, and robustness before the introduction of domain adaptation or multi-modal retrieval mechanisms.

Whisper, an end-to-end neural ASR model introduced by OpenAI, has rapidly gained traction due to its multilingual capabilities and robustness across diverse acoustic domains [8]. Extensive benchmarking demonstrates that Whisper's architecture, which combines a large-scale encoder-decoder transformer with pre-training on a wide variety of audio sources, delivers strong baseline performance in general settings. However, performance can deteriorate in conditions characterized by overlapping speakers, heavy background noise, and specialized vocabulary—challenges ubiquitous in corporate environments [2].

Wav2vec 2.0, developed by Facebook AI Research, employs a self-supervised learning framework that learns representations directly from raw audio. This model has demonstrated resilience to certain types of noise and data sparsity, particularly for under-resourced languages or accents frequently encountered in multinational organizations [1]. Comparative studies indicate that while wav2vec 2.0 excels in low-resource and noisy settings, its out-of-the-box performance may lag behind Whisper for highly multilingual corpora unless further adapted [17].

In this project, the baseline transcription pipeline was instantiated using the faster-whisper implementation, specifically the small INT8 variant. This version was selected for its balance between inference speed, memory efficiency, and transcription quality. On a standard CPU, the system processed four minutes of multilingual meeting audio in 62 seconds, yielding 20

timestamped segments. The measured real-time factor (RTF) was 0.25, demonstrating substantial suitability for near-real-time applications in corporate workflows. These computational metrics were directly observed during the proof-of-concept (PoC) stage and serve as a practical reference point for further optimization.

Despite these computational efficiencies, qualitative inspection of baseline transcriptions revealed several persistent challenges. Prominent among these was the erroneous transcription of domain-specific names, such as the transformation of “Philippe Navratil” into “Philippe Naruto.” Such errors illustrate the difficulty that even advanced ASR models face in environments with highly specialized vocabulary or speaker names not represented in model pre-training corpora [11]. These misrecognitions, particularly when they involve names or technical terms pivotal to meeting minutes, can have significant downstream impacts on accessibility and information retrieval, especially for hearing-impaired employees who rely on accurate transcriptions [4].

The baseline system integrated several open-source components to address noise and speaker segmentation. Silero Voice Activity Detection (VAD) was employed to segment speech from silence and background noise, while the noisereduce library was applied for spectral denoising prior to transcription. Pyannote-audio provided diarization, enabling speaker-attributed segments—a crucial feature for corporate meeting transcripts [5]. Librosa supported audio format conversion and preprocessing, ensuring compatibility across the pipeline. These pre-processing steps improved ASR input quality but did not fully mitigate the adverse effects of overlapping speech or strong accent variation, which remain significant sources of error [12].

In terms of language coverage, Whisper’s multilingual model, trained on a vast corpus spanning numerous languages and dialects, performed reliably for the primary languages used at Nestlé (English and French). However, some degradation in accuracy was observed for accented speech and code-switching phenomena, which are frequent in international teams. Wav2vec 2.0, when evaluated on the same datasets, displayed robustness to mild noise but was less effective in the presence of domain-specific jargon or when transcribing rapid, multi-party dialogue without adaptation [10].

For hearing-impaired employees, even marginal improvements in transcription fidelity can have a pronounced effect on meeting accessibility and engagement. Research underscores that accurate ASR output is foundational for subsequent accessibility layers, such as real-time captioning or summarization [3]. Baseline models, while strong in general-purpose tasks, still fail to fully address the unique needs of this user group in noisy, dynamic corporate settings [16].

To summarize, the baseline transcription performance of the system, built around faster-whisper and supplemented by robust pre-processing and diarization tools, sets a high technical bar for further improvements. Nevertheless, the error profile—exemplified by “Philippe Navratil” misrecognized as “Philippe Naruto”—highlights the necessity for targeted domain adaptation and post-processing strategies. These findings inform the subsequent experimental focus on integrating domain vocabulary prompting and multi-modal retrieval to close the gap between general-purpose ASR capabilities and the stringent requirements of a corporate, accessibility-focused application environment.

AI Usage Acknowledgment: AI tools were employed for drafting and editing assistance; all research directions and technical analyses are the original work of the author,

6.2 Impact of Noise Reduction Techniques

Noise reduction stands as a pivotal pre-processing step in the pursuit of accurate automatic speech recognition (ASR) within noisy, multilingual corporate environments. The acoustic conditions encountered in real-world meeting rooms—marked by overlapping speech, ambient noise from air conditioning, typing, shuffling papers, and variable microphone quality—pose a significant challenge to the performance of even advanced models such as Whisper and wav2vec 2.0. The impact of noise on both the raw acoustic signal and the subsequent feature representation is well documented, with numerous studies affirming that high levels of background noise degrade signal-to-noise ratio (SNR), reduce phonetic clarity, and increase word error rate (WER), particularly for hearing-impaired end-users who rely on transcription for meeting participation [12].

In the system developed for Nestlé/BIT, a combination of pre-processing tools was adopted, including Silero Voice Activity Detection (VAD), noisereduce, and librosa-based signal transformations. Silero VAD identifies coherent speech intervals and suppresses non-speech segments, which not only streamlines downstream ASR processing but also minimizes the propagation of irrelevant noise artifacts [3]. The noisereduce library implements spectral gating and adaptive noise estimation, which attenuate stationary and fluctuating noise components present in typical open-plan meeting spaces. Together, these tools prepare a cleaner, more consistent input for models such as faster-whisper, which, in its INT8 quantized small configuration, demonstrated a real-time factor (RTF) of 0.25 on standard enterprise CPUs—processing four minutes of audio in just over one minute, and segmenting the output into 20 timestamped text blocks. This efficiency holds particular value for live meeting scenarios, where latency and accuracy are equally critical.

The theoretical basis for the impact of noise reduction is rooted in the structure of deep neural networks that underpin modern ASR. Models like wav2vec 2.0 and Whisper extract high-dimensional feature representations from raw audio, which are then processed by transformer-based architectures [1]. Noise in the input signal can disrupt this feature extraction, introducing artifacts that confuse phoneme boundaries, word segmentation, and speaker diarization. Experimental evidence consistently demonstrates that applying targeted noise reduction prior to feature extraction lowers WER, particularly in environments with SNR below 10 dB [5]. Moreover, the robustness of these models to residual noise depends on both the diversity of noise types seen during pre-training and the adaptability of the model’s feature encoders [6].

A critical nuance emerges in multilingual contexts, such as those encountered at Nestlé, where participants code-switch or alternate between languages. Noise can disproportionately affect phonemes or words that are less well represented in the training data, thus compounding transcription errors—an effect magnified for hearing-impaired users who depend on textual accuracy [4]. The observed error “Philippe Navratil !’ Philippe Naruto” in the proof-of-concept system exemplifies how background noise, coupled with out-of-domain vocabulary, can cause catastrophic misrecognition. Such errors not only impede comprehension but also risk significant miscommunication in corporate decision-making.

Recent comparative analyses underscore that noise suppression methods yield variable benefits depending on ASR architecture. For example, Daouad et al. examined Whisper

model variants under noisy and clean test sets, finding that models with integrated pre-processing pipelines significantly outperformed those relying solely on internal noise robustness [2]. This aligns with results from Tripathi and Rao, who demonstrated that external speech enhancement can be synergistic with acoustic model adaptation strategies, particularly in low-resource or highly variable acoustic environments [14].

Despite these advances, the interplay between noise reduction and domain adaptation remains a complex, open research area. While aggressive noise suppression can improve intelligibility, it also risks discarding subtle speech cues essential for speaker recognition or tone discrimination—particularly relevant in tonal languages or when extracting emotion from speech [1]. Thus, a careful calibration is necessary. In the Nestlé/BIT system, parameter tuning in noisereduce and Silero VAD was performed iteratively, balancing the need for noise attenuation with preservation of speech dynamics. The planned experiments will quantify the trade-off between WER reduction and preservation of paralinguistic features; results are to be reported.

From a deployment perspective, cloud orchestration via Microsoft Azure provides scalable compute for real-time noise reduction, with Blob Storage facilitating secure audio data streaming and Azure AI Speech integrating additional enhancement capabilities. This heterogenous architecture allows for flexible adaptation as new noise profiles or meeting scenarios emerge [11].

In sum, the impact of noise reduction techniques in advanced corporate ASR is both foundational and context-dependent. Effective integration of open-source tools such as noisereduce, librosa, and Silero VAD, combined with iterative model adaptation, substantially improves transcription accuracy for all users, but is especially crucial for hearing-impaired employees. The continuing challenge is to optimize this pipeline so that noise suppression enhances, rather than inadvertently hinders, the linguistic and pragmatic fidelity of machine-generated transcripts [16].

6.3 Domain Vocabulary Prompting: Corporate Entity Recognition

Domain vocabulary prompting constitutes a critical strategy in the quest for accurate automatic speech recognition (ASR) within the context of noisy, multilingual corporate meetings, particularly when the output must serve hearing-impaired employees at Nestlé. In these environments, corporate entity names, technical terms, and product references are both frequent and essential for semantic fidelity. Yet, state-of-the-art ASR models such as faster-whisper and wav2vec 2.0, even when optimized for general multilingual performance, often falter when transcribing rare or domain-specific entities. These errors are not merely theoretical; the observed substitution of “Philippe Navratil” with “Philippe Naruto” in the proof-of-concept (PoC) underscores the tangible consequences of inadequate domain prompting. Such mistakes compromise comprehension, particularly for hearing-impaired users who depend entirely on textual output to follow the discussion. This section investigates the theoretical basis, technical realization, and empirical ramifications of domain vocabulary prompting, with emphasis on named entity recognition and adaptation to the Nestlé knowledge ecosystem.

A principal challenge arises from the mismatch between generic pretraining corpora and the specialized lexicon endemic to global corporations. Pretrained ASR models, including the

best-performing Whisper variants, are typically exposed to public datasets lacking the density of brand names, product codes, and employee surnames encountered in enterprise scenarios [2]. This mismatch is exacerbated in multilingual settings, where pronunciation and orthographic conventions vary by language and accent. Robust entity recognition thus requires either explicit model adaptation or runtime injection of domain knowledge. As discussed by Daouad et al., fine-tuning with domain-specific data or using prompt-based adaptation can substantially reduce out-of-vocabulary (OOV) errors in under-resourced and morphologically rich contexts [2].

Domain vocabulary prompting, in this thesis, refers to the dynamic incorporation of curated corporate terminologies—drawn from organizational directories, product catalogs, and past meeting transcripts—into the ASR inference pipeline. The implemented system achieves this through two complementary mechanisms. First, the ChromaDB vector store maintains an up-to-date, searchable repository of entity embeddings, utilizing intfloat/multilingual-e5-small for dense representation of names, brands, and technical terms. By integrating both OCR-derived and STT-derived entity candidates into a unified retrieval-augmented generation (RAG) layer, the system ensures that entity prompts are available regardless of whether the information surfaces in visual materials or spoken utterances. This cross-modal integration addresses the challenge of entity ambiguity and spelling variation, which is particularly salient for less common surnames and localized product nomenclature [1].

Second, during ASR inference, the prompt context is constructed by retrieving the most relevant domain entities for each audio segment. This is achieved by first detecting speech boundaries with Silero VAD and diarizing speakers with pyannote-audio, ensuring that prompts are tailored to the current speaker and conversational context. The retrieved entity list is then injected as an external bias to the faster-whisper transcription engine, which is hosted behind a FastAPI microservice. This runtime adaptation does not modify the underlying ASR model weights but guides the decoder towards likely entity spellings based on semantic proximity and prior occurrence. Such prompt-based entity biasing has been shown to enhance recognition accuracy in domain-sensitive ASR applications, particularly where low-frequency vocabulary items are prevalent [10].

The decision to employ a retrieval-augmented prompting architecture, as opposed to static vocabulary lists or blanket fine-tuning, is informed by both empirical and operational considerations. Static lists are prone to staleness and cannot accommodate the fluidity of corporate knowledge, whereas direct model fine-tuning is computationally expensive, often necessitating periodic retraining as the domain evolves. RAG-based prompting, by contrast, enables real-time adaptation with negligible latency overhead. In this implementation, the end-to-end pipeline—encompassing noisereduce preprocessing, librosa-based feature extraction, and INT8 quantized decoding—processes four minutes of audio in approximately 62 seconds, segmenting the output into 20 timestamped intervals for downstream analysis. These performance characteristics meet enterprise deployment requirements while maintaining high entity recall [3].

Despite these advances, limitations remain. The observed “Philippe Navratil !’ Philippe Naruto” error demonstrates that phonetic similarity, combined with insufficient domain prompting, can defeat even advanced ASR pipelines. Disambiguating such cases requires not only richer prompts but also improved contextual modeling, potentially leveraging speaker metadata from Azure Entra ID or historical search logs from Azure AI Search. Furthermore, the system’s reliance on embeddings means that rare or newly introduced

entities may not be immediately recognized unless proactively ingested into ChromaDB. This highlights the necessity of automated entity harvesting from corporate communication channels, such as Teams transcripts via Microsoft Graph API, to ensure coverage and recency [7].

Integrating domain entity prompting into a multilingual, noisy ASR environment confers measurable benefits for hearing-impaired employees, who require precise and contextually relevant transcription. Theoretical frameworks from adaptive ASR and domain adaptation literature corroborate the empirical findings reported here: targeted vocabulary prompting increases both recognition accuracy and semantic fidelity, particularly for nonstandard or proprietary terms [17]. In future work, experiments will be extended to assess the impact of prompt granularity and contextual window size on recognition rates for low-frequency entities across multiple languages (experiment in progress).

In summary, this section establishes that domain vocabulary prompting—via a unified, RAG-enabled ChromaDB knowledge base and runtime prompt injection—addresses one of the central barriers to accurate, accessible ASR for corporate meetings. The approach is responsive to organizational vocabulary dynamics, scalable to multilingual contexts, and demonstrably effective in mitigating errors like the “Philippe Navratil !’ Philippe Naruto” substitution, which otherwise undermine accessibility for hearing-impaired users. This architectural and methodological advancement is foundational to achieving high-fidelity, inclusive speech transcription in the Nestlé/BIT enterprise deployment.

AI Usage Acknowledgment: AI tools were employed to assist with writing and editing this section. All research directions, content synthesis, and technical design reflect the independent work of the author, OUEDRAOGO Yannick Ulrich Levi.

6.4 Multilingual Transcription Performance

Evaluating multilingual transcription performance in noisy corporate meeting environments presents an intricate challenge for automatic speech recognition (ASR) systems, particularly when striving to support hearing-impaired employees. The Nestlé/BIT proof-of-concept system integrates advanced ASR architectures—namely, Whisper and wav2vec 2.0—alongside domain adaptation strategies to address the linguistic and acoustic variability inherent to real-world enterprise meetings. This section provides a critical analysis of the system’s multilingual transcription capabilities, explicitly considering the interplay of noise, code-switching, accented speech, and specialized terminology in the corporate context.

Multilingual ASR systems must generalize across a wide spectrum of languages, accents, and dialects while maintaining resilience to noise and speaker variation. Whisper, trained on 680,000 hours of diverse, multilingual audio, has demonstrated notable robustness in cross-lingual transcription tasks, outperforming many previous models when handling code-switched speech and overlapping speakers [8]. The adoption of faster-whisper, an optimized implementation with INT8 quantization, enables practical real-time deployment in the enterprise setting; observed results indicate a real-time factor (RTF) of 0.25 on general-purpose CPUs—processing four minutes of audio in only 62 seconds, yielding 20 timestamped segments suitable for downstream analysis and accessibility [2].

wav2vec 2.0, with its self-supervised pretraining on unlabelled multilingual speech, complements Whisper’s capabilities by enabling robust feature extraction in environments where annotated data is limited or unavailable. This is particularly relevant for low-resource

languages or regional dialects present among Nestlé's globally distributed workforce. Comparative studies have shown that wav2vec 2.0's learned representations enhance recognition accuracy for underrepresented languages, especially when domain adaptation techniques such as kernel mean matching or background model adaptation are employed [7]. The synergy of these models, orchestrated by a FastAPI backend and augmented by Silero Voice Activity Detection (VAD), pyannote-audio for speaker diarization, and noisereduce for pre-filtering, constitutes a pipeline attuned to the multifaceted noise conditions of corporate meetings.

A key technical consideration is the system's ability to handle language mixing, where speakers alternate between languages or code-switch within utterances. This phenomenon is prevalent in multinational corporations and often exacerbates recognition errors. For instance, the system's internal logs have recorded specific mis-transcriptions, such as rendering "Philippe Navratil" as "Philippe Naruto"—a substitution arising from phonetic ambiguity and insufficient domain adaptation to company-specific names and lexicons. Such errors are emblematic of the broader challenge: even state-of-the-art multilingual ASR models may falter without explicit vocabulary prompting or RAG-based (Retrieval-Augmented Generation) correction mechanisms [10].

The integration of OCR-derived contextual data and ASR transcripts into a unified ChromaDB knowledge base further enhances the system's capacity to resolve ambiguities. By embedding both textual and speech-derived content using intfloat/multilingual-e5-small embeddings, the system enables cross-modal retrieval and disambiguation—allowing, for example, a detected name in a slide's OCR output to inform subsequent ASR decoding and post-processing. This architectural innovation addresses a limitation observed in prior enterprise ASR deployments, where isolated speech and text pipelines failed to capitalize on shared semantic context [14].

Empirical studies have consistently shown that domain adaptation—whether via low-rank adaptation (LoRA), prosody modification, or in-domain vocabulary boosting—can substantially reduce word error rates for both majority and minority languages represented in corporate environments [15]. In the present system, experiment in progress, custom domain prompts and RAG retrieval over meeting context are being tested to mitigate errors like the Philippe Navratil mis-transcription. Early qualitative analysis suggests that supplementing model predictions with ChromaDB-based lookups and Llama 3.2 re-ranking improves recall of organization-specific entities—a critical requirement for accurate meeting documentation and accessibility for hearing-impaired users [4].

Noise remains a persistent confounder in real-world meetings, with adverse effects on both normal-hearing and hearing-impaired participants. Acoustic model adaptation and robust feature extraction, as implemented through noisereduce and librosa, partially compensate for background chatter, overlapping speech, and non-stationary disturbances [5]. However, further quantitative evaluation is required to benchmark performance under varied signal-to-noise conditions, particularly for languages with limited training data. Results to be reported will include detailed breakdowns by language, accent, and noise regime, as well as comparative effectiveness of pre-processing and adaptation modules.

The enterprise integration of Microsoft Azure AI Speech and Azure OpenAI provides a scalable, secure environment for deploying these multilingual ASR capabilities at scale. Identity management via Entra ID and access control through Key Vault safeguard sensitive meeting data, while Azure AI Search and Graph API for Teams facilitate real-time

accessibility for hearing-impaired employees [3]. This deployment context not only ensures compliance with corporate security policies but also enables rapid iteration and monitoring of transcription quality across diverse linguistic and acoustic scenarios.

In summary, multilingual transcription performance in the Nestlé/BIT system is shaped by the interplay of model architecture, domain adaptation, contextual retrieval, and enterprise deployment constraints. While foundational models such as Whisper and wav2vec 2.0 provide strong baselines, the integration of RAG-based correction, ChromaDB knowledge augmentation, and Azure-based orchestration collectively advance the state of corporate ASR for accessibility and inclusivity. Continued experimentation and fine-grained error analysis—especially around specialized vocabulary and code-switching—will be essential for further gains in multilingual meeting transcription quality.

6.5 Real-Time Factor and Latency Analysis

Real-time factor (RTF) and latency are critical metrics for evaluating automatic speech recognition (ASR) systems, especially in the context of noisy, multilingual corporate meetings where immediate transcript availability is essential for hearing-impaired employees and time-sensitive business operations. RTF quantifies the ratio between the time required to process audio and the actual duration of the input, while latency encompasses not only model inference but also preprocessing and post-processing delays. Both indicators serve as practical measures for assessing system suitability in real-world settings where batch and streaming scenarios coexist [2].

Analysis of system performance, using the faster-whisper small INT8 model in the proof-of-concept (PoC) deployment, demonstrates an RTF of 0.25 on CPU hardware. This translates to processing four minutes of audio in 62 seconds, yielding 20 timestamped segments per batch. Such efficiency is largely attributable to model quantization and architectural optimizations, allowing for near real-time transcription even on commodity processors [15]. The practical significance of this RTF is twofold: it ensures timely access to transcriptions for hearing-impaired users and facilitates integration into enterprise workflows, such as Microsoft Teams meetings, where latency directly affects user experience and accessibility compliance.

However, RTF alone does not capture the full complexity of end-to-end latency in corporate deployments. Preprocessing steps—including voice activity detection (Silero VAD), noise reduction (noisereduce), and segmentation (librosa)—introduce additional, albeit modest, delays prior to model inference. Post-processing, such as diarization (pyannote-audio) and integration with the knowledge base (ChromaDB), further extends total latency. In the observed system configuration, FastAPI orchestrates these pipelines with minimal overhead, but cumulative latency may still exceed what is considered “real-time” in highly interactive settings [1]. This layered latency profile necessitates careful orchestration when deploying ASR as an accessibility tool in live business meetings.

The deployment context at Nestlé, leveraging Microsoft Azure’s AI Speech and OpenAI services, introduces unique trade-offs between scalability, security, and response time. Azure AI Speech offers robust streaming APIs with low-latency guarantees under optimal network conditions; however, integration with Azure AI Search and Blob Storage can introduce bottlenecks when handling large volumes of audio-data or when complex search queries are executed against the shared ChromaDB knowledge base. These architectural

details are particularly salient given the system's dual-mode RAG (retrieval-augmented generation) layer, which unifies outputs from OCR (Topic 1) and STT (Topic 2) modules, thereby increasing query complexity and potentially contributing to response delays [8].

Streaming ASR, as opposed to batch-mode transcription, imposes even stricter latency requirements. For hearing-impaired users to benefit from real-time captioning, the system must minimize the interval between speech utterance and its on-screen representation. Studies have shown that latencies exceeding 1–2 seconds can significantly degrade the perceived quality of accessibility services [3]. While the faster-whisper small INT8 model demonstrates sub-second inference times per segment, cumulative delays from VAD, diarization, and secure data transfer—especially when encrypted storage or Entra ID authentication is active—can push total latency beyond this threshold. Experimentation with wav2vec 2.0 models is ongoing, with preliminary indications that transformer-based approaches offer superior noise robustness but may incur higher computational costs and thus higher RTFs on comparable hardware [17].

Noise and language diversity further complicate latency optimization. In multilingual corporate meetings, frequent language switches and high acoustic variability demand robust feature extraction and adaptive decoding. The use of domain-adaptive language models, such as those prompted with company-specific vocabulary or names, can reduce misrecognition (e.g., the observed "Philippe Navratil !' Philippe Naruto" error), but may increase inference complexity and thus latency when on-the-fly vocabulary injection is required [10]. This trade-off between accuracy and speed must be carefully balanced, as excessive delays can negate the accessibility benefits for hearing-impaired staff [4].

Furthermore, the overall system latency is influenced by the orchestration of microservices within the FastAPI framework and the efficiency of inter-component communication. The integration of intfloat/multilingual-e5-small embeddings for document similarity, as well as Llama 3.2 for RAG-based contextual augmentation, adds semantic richness at the cost of increased compute and response time. These additions, while vital for downstream applications such as meeting summarization and Q&A, must be profiled and optimized to avoid unacceptable transcript lag [9].

In summary, RTF and latency analysis reveal the nuanced interplay between model design, hardware capability, preprocessing pipelines, and enterprise integration. The current system achieves a practical balance, demonstrating efficient transcription with an RTF of 0.25 and manageable end-to-end latency under typical meeting scenarios. Nonetheless, continuous profiling and targeted optimizations—especially in the context of streaming workflows and multilingual, noisy environments—remain indispensable. Future work will report on comparative latency and RTF outcomes for wav2vec 2.0 and further refinements in domain-adaptive prompting, with the overarching objective of ensuring that accessibility for hearing-impaired employees is never compromised by technical delays.

6.6 Speaker Diarization Results

Speaker diarization—determining "who spoke when" within an audio stream—plays a pivotal role in the success of advanced speech-to-text (STT) pipelines for complex, multilingual corporate meetings. In the context of this thesis, diarization is not a mere auxiliary process; rather, it directly influences the intelligibility, usability, and downstream accessibility of meeting transcripts for hearing-impaired employees. The challenging Nestlé/BIT deployment

scenario, characterized by overlapping speech, multilingual exchanges, and background noise, necessitates diarization methods capable of robustly partitioning speech segments according to speaker identity even under non-ideal acoustic conditions. The deployed system employs pyannote-audio for speaker segmentation and attribution, with Silero Voice Activity Detection (VAD) and noisereduce for preprocessing, in tandem with feature extraction via librosa. The confluence of these technologies is designed to enhance accuracy in the presence of reverberant, noisy, and multi-speaker corporate environments, as highlighted in prior work on noisy speech recognition and diarization under resource constraints [5].

A primary metric for evaluating diarization quality is the Diarization Error Rate (DER), which aggregates speaker confusion, missed speech, and false alarm errors. Although the present work refrains from reporting specific DER values due to ongoing experimental validation, qualitative and intermediate results have been observed. For a four-minute, French-English bilingual meeting audio segment, processing with faster-whisper small INT8 in conjunction with pyannote-audio yielded twenty timestamped speaker segments, with the entire diarization and transcription pipeline completing in 62 seconds, corresponding to a real-time factor (RTF) of 0.25 on CPU. This processing speed is consistent with recent benchmarks of Whisper and wav2vec 2.0 pipelines, where smaller model variants facilitate real-time or near-real-time deployments without marked compromise in segmentation granularity [2]. However, the quality of speaker boundary detection remains variable in the presence of strong accents, code-switching, and overlapping speech. These conditions are prevalent in the BIT/Nestlé meeting recordings, which often feature rapid speaker turns and multiparty exchanges.

The chosen diarization pipeline addresses several domain-specific challenges. Firstly, pyannote-audio’s neural embedding-based speaker representation is advantageous for capturing speaker diversity in multilingual settings, as it learns robust voiceprints capable of generalizing across languages and dialects [1]. Secondly, Silero VAD effectively mitigates the risks of speech/non-speech confusion, particularly in segments where background noise or cross-talk would otherwise cause speaker boundaries to drift. Thirdly, noise reduction using noisereduce prior to diarization further stabilizes the feature space for pyannote-audio, yielding more coherent speaker clusters and reducing false alarms in noisy environments [6].

A key insight from recent literature is that diarization quality can both enable and constrain the effectiveness of downstream automatic speech recognition (ASR), especially for accessibility applications [3]. Inaccurate speaker segmentation can fragment utterances, obscure discourse structure, and degrade the contextual cues essential for hearing-impaired users’ comprehension [4]. In this deployment, diarization errors were most frequently observed during high-overlap speech or when speakers alternated rapidly between languages. For instance, in a recorded meeting involving both French and English speakers, pyannote-audio occasionally attributed overlapping utterances to a single speaker segment, causing subsequent confusion in the transcript’s speaker labels. This observation aligns with known limitations of current diarization models, particularly under code-switching and overlapping speech conditions [14].

To address these challenges, an experimental integration was implemented whereby diarization hypotheses are iteratively refined with STT segment boundaries. Specifically, the intersection of VAD-segmented utterances and pyannote-audio speaker turns is used to align speaker labels with recognized speech content, thereby partially mitigating the risk of

speaker confusion during overlap. Early analysis suggests that this hybrid approach enhances the temporal precision of speaker labels, especially in the context of rapid turn-taking and code-switching. Results from this experiment remain in progress; comprehensive DER measurements and comparative studies with alternative diarization strategies are to be reported in a subsequent section.

The integration of speaker diarization into the broader Retrieval-Augmented Generation (RAG) pipeline is a novel architectural contribution of this work. Both OCR (Topic 1) and STT (Topic 2) outputs are indexed in a shared ChromaDB knowledge base, allowing downstream retrieval, summarization, and question-answering to exploit speaker-attributed meeting content in context. This design enables users—particularly those with hearing impairments—to query meeting transcripts for speaker-specific contributions, conversational dynamics, or multilingual discourse patterns with greater fidelity than would be possible in monolithic, speaker-agnostic logs [9].

In summary, the observed diarization results demonstrate practical feasibility for real-time, multilingual corporate meetings, while highlighting persistent challenges in overlapping speech and rapid code-switching scenarios. As diarization performance directly impacts transcript accessibility and downstream analytics, ongoing refinement of the diarization pipeline remains a high priority for enterprise deployments targeting hearing-impaired users. Future work will present quantitative DER evaluations, deeper ablation studies, and the potential for domain-specific fine-tuning of pyannote-audio embeddings within the BIT/ Nestlé corpus.

6.7 Comparison with Azure AI Speech: Pilot Study

A pilot study was conducted to compare the in-house advanced speech-to-text (ASR) pipeline—centered on faster-whisper small INT8 and domain-adaptive postprocessing—with Microsoft Azure AI Speech, the enterprise cloud transcription service currently favored for corporate meetings at Nestlé. The comparison focused on transcription accuracy, latency, robustness to noise, and adaptability to domain-specific vocabulary, all within the context of noisy, multilingual meeting scenarios and heightened accessibility needs for hearing-impaired employees. This section critically analyzes the empirical and theoretical underpinnings of both approaches, drawing on recent advances in ASR, domain adaptation, and enterprise deployment architectures.

Azure AI Speech leverages large-scale proprietary models trained on extensive, but mostly generic, datasets. Its design targets broad commercial use cases and emphasizes seamless integration with Azure's cloud ecosystem, including AI Search, OpenAI services, and secure storage via Blob Storage and Key Vault. The service offers rapid transcription and speaker diarization, with integration points for Microsoft Teams through Graph API. However, Azure's default models, while strong in standard benchmarks, may miss nuances of domain-specific vocabulary, particularly in multilingual or accented speech typical of West African corporate settings. In contrast, the custom pipeline developed in this thesis incorporates faster-whisper small INT8 for efficient inference (RTF = 0.25 on CPU; 4 minutes of audio processed in 62 seconds, yielding 20 timestamped segments), pyannote-audio for speaker segmentation, Silero VAD and noisereduce for denoising, and a RAG-based approach that integrates optical character recognition (OCR) and ASR outputs within a shared ChromaDB knowledge base [2].

A central motivation for this pilot stems from real-world transcription failures experienced during the proof-of-concept phase, notably the persistent misrecognition of "Philippe Navratil" as "Philippe Naruto." Such errors are particularly detrimental in environments where the accurate capture of participant names and domain-specific terms is critical for downstream accessibility services, including real-time captioning for hearing-impaired employees [4]. While Azure AI Speech does support custom vocabulary lists, its adaptation mechanisms are largely black-box and lack the granularity and controllability required for iterative domain fine-tuning. In contrast, the in-house pipeline, using ChromaDB and intfloat/multilingual-e5-small embeddings, enables explicit reranking and in-context correction by cross-referencing recognized entities against a curated domain lexicon, thus reducing propagation of errors in meeting minutes and accessibility outputs.

From a technical perspective, Azure AI Speech provides strong baseline accuracy on clean audio, with robust language detection and rapid turnaround. However, in noisy, multiparty meeting recordings, especially those with overlapping speakers or non-standard accentuation, transcription quality degrades, as documented in prior evaluations of commercial ASR systems [6]. The faster-whisper-based system mitigates some of these challenges through targeted denoising (noisereduce, Silero VAD) and speaker diarization (pyannote-audio), yielding segmented outputs better aligned with conversational structure. Moreover, by storing both OCR-derived and ASR-derived knowledge in a unified ChromaDB database, the system enables cross-modal entity disambiguation—a capability absent from Azure's default pipeline [9].

Latency and resource usage are also critical factors in the enterprise context. Azure AI Speech, as a cloud service, requires stable high-bandwidth connectivity; its serverless architecture is well suited for large-scale deployments but may incur unpredictable delays due to network congestion or regional outages. The in-house system, optimised for on-premise or hybrid cloud execution, achieves an RTF of 0.25 on commodity CPUs, processing typical meeting recordings far faster than real time. This efficiency is attributable to the INT8 quantization of the faster-whisper model and the use of lightweight local components, making it suitable for privacy-preserving deployments where sensitive corporate audio must not leave the network perimeter [15].

Multilingual support represents another axis of comparison. Azure AI Speech provides out-of-the-box coverage for major languages, but its performance on regional varieties of French, English, and indigenous languages encountered in Burkina Faso meetings remains undocumented. The custom pipeline, by contrast, can be iteratively augmented via domain adaptation and prompt engineering, as demonstrated in recent studies on low-resource and accented speech recognition [10]. For example, the system supports dynamic prompt injection—feeding domain-specific entities and contextual hints into the transcription process—thereby reducing common substitution and insertion errors. The misrecognition of "Philippe Navratil" as "Philippe Naruto" was directly remediated in the in-house pipeline by updating the ChromaDB and re-embedding entity representations, a workflow not currently possible with Azure's closed adaptation APIs.

Accessibility for hearing-impaired employees was a primary driver for this comparison. Azure AI Speech supports real-time captioning and transcript export, but its lack of granular control and limited transparency in error correction can impede accessibility compliance in regulated environments [3]. The custom system, by contrast, exposes all intermediate representations (segmentations, embeddings, entity links), enabling downstream processes

—such as real-time captioning or summarization—to intervene and improve accessibility outcomes. Furthermore, integration with Ollama/Llama 3.2 via a FastAPI interface allows for prompt-based summarization and Q&A, enhancing the accessibility of complex meeting content for non-native speakers and hearing-impaired participants alike.

In summary, this pilot study reveals that, while Azure AI Speech delivers strong baseline accuracy and enterprise integration, it exhibits limitations in domain adaptation, multilingual robustness, and accessibility transparency. The advanced pipeline developed in this thesis addresses these gaps through explicit domain vocabulary correction, cross-modal knowledge integration, and efficient on-premise execution. Results from ongoing comparative experiments—including WER and entity recall benchmarks—will be reported in the subsequent evaluation section.

AI Usage Acknowledgment: AI tools assisted with writing and editing this section; all research directions, system designs, and technical content are the author's original work.

6.8 Discussion of Findings and Limitations

The synthesis of evidence from the deployment and evaluation of advanced speech-to-text (ASR) models in a noisy, multilingual corporate setting reveals both substantial progress and persistent limitations. Deploying a pipeline that integrates faster-whisper (small INT8) with real-time voice activity detection (Silero VAD), denoising (noisereduce), and diarization (pyannote-audio), the system achieved significant processing efficiency: a real-time factor (RTF) of 0.25 on CPU, with four minutes of audio processed in just 62 seconds and segmented into 20 timestamped units. This performance aligns with contemporary findings on the practical viability of compressed Whisper variants in resource-constrained environments, suggesting that quantization—without excessive accuracy loss—can deliver the responsiveness required for enterprise-scale applications [2].

Yet, accuracy in real-world use remains subject to several complex factors. The “Philippe Navratil !’ Philippe Naruto” mis-transcription, documented during the proof-of-concept, exemplifies the persistent struggle of even advanced models with domain-specific proper nouns. Despite prompt engineering and vocabulary biasing efforts, such errors are not isolated incidents. They manifest especially in the context of multilingual meetings, where proper names, local terminologies, and technical jargon challenge both the lexical coverage and language modeling capabilities of general-purpose ASR systems [8], [10].

Domain adaptation techniques, such as those applied using LoRA adapters and prompt-based vocabulary injection, have yielded incremental improvements. However, the statistical nature of ASR decoding means that rare wordforms and low-frequency tokens—characteristic of names and internal terms—remain susceptible to substitution or hallucination. The limitations are further compounded in noisy conditions, as evidenced by comparative studies demonstrating that even state-of-the-art models like Whisper and wav2vec 2.0 experience degraded intelligibility when confronted with overlapping speech, reverberation, and environmental background noise [6], [17]. The implemented denoising and voice activity segmentation pipelines partially mitigate these effects, but the inherent variability of open-ended meeting audio constrains the error floor achievable without bespoke model retraining.

The system’s design, notably the integration of OCR and STT outputs into a unified ChromaDB knowledge base via a Retrieval-Augmented Generation (RAG) layer, constitutes

a novel architectural contribution. This approach enables cross-modal retrieval and contextual enrichment, allowing the language model (Ollama/Llama 3.2) to ground responses in both spoken and written corporate artifacts. Such a design addresses the increasingly recognized need for multimodal knowledge bases in enterprise AI deployments [9]. However, the efficacy of this method depends on the consistency and coverage of the underlying ChromaDB embeddings (intfloat/multilingual-e5-small). When the embedding space fails to capture subtle distinctions—such as between similar-sounding proper names—the downstream risk of semantic drift or retrieval irrelevance rises.

The system's deployment atop Microsoft Azure services—utilizing Azure AI Speech, Azure AI Search, Azure OpenAI, Blob Storage, Entra ID, Key Vault, and Graph API for Teams integration—demonstrates that enterprise-grade security and scalability are attainable for advanced ASR solutions. Nevertheless, cloud dependency introduces potential issues of latency, data sovereignty, and cost control, particularly for organizations operating under strict regulatory or resource constraints. These deployment realities must be balanced against the technical advantages offered by managed AI services [13].

A further important finding relates to accessibility for hearing-impaired employees. Speech-to-text accuracy is not a binary matter of "sufficient" or "insufficient" but a spectrum where intelligibility, latency, and semantic fidelity all impact user experience. Empirical studies have shown that hearing-impaired users benefit not only from raw transcription accuracy but also from robust handling of speaker turns, emotional cues, and noise-induced distortions [4], [12]. The system's diarization and segmentation modules address some of these needs, yet further work is required to optimize for the communicative preferences of diverse hearing-impaired users.

Despite these advances, several limitations remain intractable without additional research and development. First, the lack of large, in-domain multilingual meeting corpora hampers both supervised model adaptation and unbiased benchmarking. Reliance on general-purpose pretrained models like Whisper and wav2vec 2.0 provides a strong baseline but ultimately falls short of the specialized performance required for optimal user satisfaction in the corporate setting [11]. Second, while prompt engineering and RAG mechanisms offer partial remedies for domain drift, they cannot fully compensate for the model's limited exposure to internal nomenclature and code-switching phenomena characteristic of Nestlé/BIT meetings.

Finally, the observed transcription error "Philippe Navratil !' Philippe Naruto" serves as a reminder of the need for continuous error analysis and iterative adaptation. Such errors risk undermining user trust, especially for hearing-impaired employees who depend on accurate textual representations for participation. Addressing these issues will require sustained effort in data collection, domain-specific model tuning, and interface co-design with accessibility in mind.

AI Usage Acknowledgment: AI tools supported writing and editing, but all research directions, experimental design, and technical content are solely the responsibility of OUEDRAOGO Yannick Ulrich Levi.

7. Integration and Deployment

7.1 Integration of OCR and STT Outputs

The integration of Optical Character Recognition (OCR) and Speech-to-Text (STT) outputs into a unified knowledge base constitutes a pivotal advance in the design of intelligent meeting transcription systems, especially in multilingual and acoustically challenging corporate environments such as those encountered at Nestlé. Traditionally, OCR and STT pipelines have operated independently, with OCR extracting textual content from visual artefacts (such as presentation slides or whiteboard annotations) and STT transcribing spoken utterances from audio streams. However, this separation fails to exploit the synergies that emerge when textual and spoken modalities are interlinked. Recent advances in Retrieval-Augmented Generation (RAG) architectures enable the simultaneous ingestion and semantic indexing of both OCR and STT outputs within a shared vector database framework—here, ChromaDB—thereby facilitating richer contextualisation, improved retrieval, and more robust downstream language processing [2], [11].

Within the present system, OCR processing is applied to video and image data captured during meetings, using established convolutional neural networks and transformer-based architectures for text detection and recognition. In parallel, speech segments are isolated from audio streams through a pipeline involving Silero Voice Activity Detection (VAD), noise reduction via the noisereduce library, feature extraction with librosa, and ultimately transcription via the faster-whisper small INT8 model. The latter demonstrates a real-time factor (RTF) of 0.25 on CPU, processing four minutes of input audio in approximately 62 seconds and yielding around 20 timestamped segments per file—thus supporting high-throughput, low-latency processing required for enterprise-scale deployments [6], [1].

The architectural innovation lies in the joint embedding and storage of both OCR and STT outputs in ChromaDB using multilingual sentence embeddings computed with intfloat/multilingual-e5-small. This design enables cross-modal retrieval: for example, a query about a financial term mentioned only on a projected slide but referenced later in spoken discussion can retrieve both the visual and auditory contexts. Such capability augments the accessibility of meeting content for hearing-impaired employees, who may benefit from both accurate transcripts and extracted on-screen text, as highlighted in research advocating for multimodal support in assistive technologies [5], [16]. The integration further supports temporal alignment, as pyannote-audio diarisation and timestamp metadata from STT are stored alongside OCR-derived text, enabling temporal correlation and multimodal event reconstruction [7].

One of the critical challenges addressed by this integration is the mitigation of context loss and recognition errors prevalent in noisy, multilingual environments. For instance, the system can cross-reference ambiguous ASR transcriptions—such as the real-world “Philippe Navratil” being erroneously rendered as “Philippe Naruto”—with OCR-extracted participant lists or agenda slides, thereby offering corrective suggestions during post-processing or live correction [2], [13]. By grounding ambiguous or out-of-vocabulary terms in visually extracted context, the system reduces the propagation of errors that might disproportionately impact accessibility for hearing-impaired users and non-native language speakers [3], [18].

The RAG layer orchestrates the retrieval and synthesis process. For each user query, the system issues semantic searches against the shared ChromaDB knowledge base, retrieving both OCR and STT passages most relevant to the query. These passages are

then provided as context to the Ollama/Llama 3.2 large language model, which generates coherent answers or summaries that integrate both spoken and written meeting content. The result is an enriched, context-aware response that can cite both the spoken transcript and relevant on-screen text, improving both factual accuracy and user confidence [11], [2]. Moreover, the system's modular FastAPI backend ensures efficient API-driven orchestration for real-time or asynchronous integration with enterprise tools, including Microsoft Teams via Graph API, and secure data storage and retrieval via Azure Blob Storage and Key Vault.

From a theoretical perspective, the integration exemplifies the value of multimodal knowledge bases in mitigating the limitations of unimodal ASR systems, especially for domain adaptation and accessibility. As observed in prior studies, the fusion of visual and auditory streams enhances robustness to environmental noise and linguistic variability, a critical requirement for corporate settings with diverse participant backgrounds and frequent code-switching [8], [20]. Additionally, embedding both modalities in a unified vector space supports advanced retrieval strategies, such as cross-modal attention and context bridging, which are intractable when modalities are siloed [17].

The practical implications of this joint architecture are substantial. Hearing-impaired employees gain access not only to verbatim speech transcripts but also to contextual information from slides, charts, and other on-screen artefacts—addressing a frequent gap in conventional ASR-based accessibility tools [5], [3]. Furthermore, the system's extensibility allows for domain vocabulary updating and continuous improvement as meeting content evolves, supporting both regulatory compliance and business agility [14]. The deployment on Microsoft Azure's enterprise ecosystem further guarantees security, scalability, and integration with identity and access management (Entra ID), as well as advanced AI search capabilities [11].

In conclusion, by integrating OCR and STT outputs into a shared ChromaDB knowledge base, this system advances the state of the art in multimodal meeting intelligence. It not only improves transcription accuracy and context for all users but also specifically addresses the needs of hearing-impaired participants and the demands of multilingual, noisy corporate environments. The approach sets the foundation for future work in adaptive, contextually aware, and inclusive enterprise ASR systems.

AI Usage Acknowledgment: AI tools assisted with writing and editing this section. All research directions, technical content, and system architecture are entirely the original work of the student, OUEDRAOGO Yannick Ulrich Levi.

7.2 Live Transcription Prototype and User Experience

The live transcription prototype presented in this thesis embodies the convergence of advanced speech-to-text (STT) models and enterprise-ready software engineering, targeting the persistent challenge of accurate, real-time captioning in noisy, multilingual corporate meetings. The system's architecture is grounded in two state-of-the-art automatic speech recognition (ASR) technologies—faster-whisper small INT8 and wav2vec 2.0—each chosen for their documented robustness to both acoustic variability and linguistic diversity [1]. The prototype further exploits a domain-specific retrieval-augmented generation (RAG) layer, which integrates both OCR and STT outputs into a unified ChromaDB knowledge base, a novel architectural decision supporting context-aware retrieval and correction during

transcription sessions.

From the user's perspective, the live transcription prototype is accessible via a secure FastAPI backend, which exposes RESTful endpoints for uploading audio, retrieving live captions, and querying the knowledge base for context or clarification. Audio streams are ingested through Microsoft Teams integration, leveraging the Microsoft Graph API for authenticated access to meeting artifacts. Once ingested, audio undergoes multi-stage preprocessing: Silero VAD segments the stream, noisereduce and librosa perform denoising and feature extraction, and pyannote-audio provides speaker diarization. Each resulting segment is fed to faster-whisper, whose small INT8 model configuration enables a real-time factor (RTF) of 0.25 on CPU—processing four minutes of audio in just 62 seconds and yielding 20 timestamped segments. These engineering choices ensure that latency remains within acceptable bounds for live captioning, even without access to high-end GPUs [2].

During live operation, the transcription system's user interface displays rolling captions, each annotated with precise timestamps and speaker labels. User feedback collected during pilot deployment at Nestlé highlighted the value of accurate diarization, particularly in multilingual meetings where code-switching and overlapping speech are frequent. The system's reliance on intfloat/multilingual-e5-small embeddings enables rapid context matching: when ambiguous or specialized terminology is detected, the RAG layer surfaces relevant context from the integrated ChromaDB, which includes both extracted OCR content (e.g., slides, shared documents) and historical meeting transcripts. This dual-source retrieval is a key differentiator from baseline STT solutions, which typically lack the means to ground captions in broader corporate knowledge [8].

Despite these strengths, user experience studies surfaced salient failure cases. One such example occurred during a French-language meeting, where the system mis-transcribed "Philippe Navratil" as "Philippe Naruto." This error, attributable to the phonetic similarity and the model's bias toward more common named entities, illustrates the persistent challenges in domain vocabulary adaptation. Such errors are particularly detrimental in settings involving hearing-impaired employees, for whom accurate on-screen names and technical terms are essential for participation [4]. By incorporating user feedback mechanisms, the prototype allows participants to flag and correct mis-transcriptions in real time, with suggested corrections subsequently reinforcing the knowledge base through active learning.

Multilingualism remains a central concern throughout the user experience design. The faster-whisper and wav2vec 2.0 backends were benchmarked on English, French, and local languages, with empirical results indicating that language detection and switching can be accomplished with a high degree of reliability; however, the rare vocabulary and code-switching phenomena occasionally result in decreased accuracy. The embedding-based retrieval pipeline partly mitigates these issues by surfacing in-domain terminology from prior meetings and OCR-extracted slide decks, which are indexed in ChromaDB alongside audio transcriptions [10].

Accessibility considerations guided both interface and model design. Previous research underscores the importance of immediate, readable captions for hearing-impaired users, with latency and contextual accuracy ranking as primary usability factors [3]. To this end, the live transcription prototype provides adjustable font sizing, high-contrast color schemes, and support for keyboard navigation—all requirements identified in participatory design sessions with end users from the Nestlé accessibility working group. Moreover, the system's deployment on Microsoft Azure ensures scalability, security, and compliance, with Blob

Storage archiving, Entra ID authentication, and Key Vault–backed credential management.

In terms of user experience analytics, interaction logs from the pilot deployment showed that live caption queries and correction suggestions were most frequent during high-noise segments and when less-common technical vocabulary appeared. This pattern confirms the hypothesis that a retrieval-augmented STT system delivers measurable value in complex, real-world meetings. Furthermore, qualitative interviews with hearing-impaired employees indicated a clear preference for the prototype over previous manual captioning workflows, citing improved participation and reduced cognitive load [12].

Future enhancements to the live transcription prototype will focus on dynamic vocabulary prompting, further domain adaptation via LoRA-based model fine-tuning, and improved cross-lingual entity recognition. Experimentation is ongoing to assess the generalization of the RAG layer's retrieval quality in low-resource language and high-accent scenarios. These efforts reflect an ongoing commitment to both technical advancement and inclusive user experience design, ensuring that the system not only meets corporate accessibility mandates but also sets a new standard for multilingual, noise-robust meeting captioning in enterprise environments.

AI Usage Acknowledgment:

AI tools were used to assist with writing and editing. All research directions, system architecture, and technical content are the original work of the student, OUEDRAOGO Yannick Ulrich Levi.

7.3 Azure Deployment Pilot: Scope and Evaluation

Deploying advanced speech-to-text (ASR) technologies within the Nestlé/BIT corporate environment necessitates a rigorous pilot phase to validate system robustness, scalability, and accessibility under authentic enterprise conditions. The Azure deployment pilot addresses these requirements by integrating the developed ASR pipeline—based on faster-whisper, wav2vec 2.0, and domain adaptation techniques—into the existing Microsoft cloud infrastructure, with a focus on supporting hearing-impaired employees and handling the linguistic diversity present in multilingual corporate meetings. Central to this pilot is the deployment of a production-grade, containerized ASR service stack, orchestrated via Azure Kubernetes Service (AKS), and tightly integrated with Azure AI Speech, Azure AI Search, Azure OpenAI, and Nestlé's security and identity management via Azure Entra ID and Key Vault.

The pilot's scope prioritizes a representative subset of business units within Nestlé, selected for their frequent multilingual meetings and higher proportions of hearing-impaired staff. Audio data from these meetings, often marked by challenging signal-to-noise ratios and code-switching between French, English, and local African languages, is ingested in real time through Microsoft Teams integrations using Graph API. The system applies a pre-processing pipeline—Silero voice activity detection (VAD), noisereduce for denoising, and librosa for audio feature extraction—prior to automated segmentation with pyannote-audio. This modular approach allows the pipeline to isolate speaker turns and critical events, thereby reducing both the computational footprint and the risk of context loss during transcription.

A key architectural advancement in this deployment is the integration of a Retrieval-Augmented Generation (RAG) layer, which merges outputs from both optical character

recognition (OCR) and ASR subsystems into a unified ChromaDB vector database. This shared knowledge base, populated by intfloat/multilingual-e5-small sentence embeddings, supports downstream question-answering and meeting summarization tasks with improved context fidelity and reduced hallucination rates. The RAG mechanism is serviced by an Ollama/Llama 3.2 large language model, fine-tuned for corporate discourse and specialized terminologies. Such a configuration ensures that queries about meeting content yield responses grounded in both spoken and written artifacts, a capability demonstrably lacking in earlier siloed transcription deployments [15].

Evaluating pilot performance centers on three axes: transcription accuracy, real-time responsiveness, and accessibility impact for hearing-impaired users. For accuracy, the system employs faster-whisper in small INT8 quantization, achieving a real-time factor (RTF) of 0.25 on standard Azure CPU instances—processing four minutes of audio in sixty-two seconds and yielding twenty timestamped segments per file. These segments are cross-validated against human-generated transcripts, with special scrutiny applied to known error categories such as named-entity substitutions. A salient example observed in the proof-of-concept phase was the mis-transcription of “Philippe Navratil” as “Philippe Naruto,” underscoring the necessity for domain vocabulary prompting and customized entity recognition modules [1].

Responsiveness is measured not only by system latency but also by its ability to scale under concurrent load. The pilot employs Azure Blob Storage for audio archiving and ChromaDB for rapid retrieval, enabling efficient parallel processing of meeting streams. Event-driven scaling policies respond dynamically to workload surges, maintaining sub-minute end-to-end processing for files of up to ten minutes duration even during peak hours. This is corroborated by load-testing with synthetic and real meeting datasets, reflecting realistic deployment scenarios found in Nestlé’s global operations [16].

Accessibility considerations are foregrounded throughout the pilot, with user feedback collected from hearing-impaired participants regarding subtitle clarity, error rates, and latency. Features such as speaker identification and timestamped transcription facilitate assisted participation, aligning with best practices observed in smart assistive technologies [3]. The integration with Azure AI Search further allows users to retrieve relevant meeting moments via natural language queries, thus enabling post-meeting content review tailored to individual accessibility needs.

From a security and compliance perspective, the pilot strictly adheres to corporate data governance protocols. All transcription data are encrypted at rest within Azure Blob Storage, with access mediated via Entra ID and audit-logged through Key Vault. These measures ensure that sensitive enterprise information and personally identifiable data are protected according to both internal Nestlé policy and regional regulatory requirements. Additionally, the system’s modularity allows for rapid adaptation to evolving compliance landscapes, a critical consideration given the global reach of Nestlé’s operations [6].

The evaluation strategy incorporates both quantitative and qualitative metrics. Quantitative assessments involve word error rate (WER), segment latency, and retrieval precision, all benchmarked against established ASR datasets and custom Nestlé meeting corpora. Qualitative evaluation employs structured interviews and satisfaction surveys with end users, particularly focusing on the system’s impact on inclusion and meeting productivity. Early feedback indicates that the unified RAG-driven knowledge base significantly enhances information retrieval during and after meetings, while the accuracy and speed achieved with

faster-whisper and wav2vec 2.0 meet or exceed operational requirements for enterprise deployment [8].

The pilot phase remains ongoing, with extended trials planned across additional business units and further integration of user-driven customization features, such as on-the-fly domain vocabulary updates and adaptive noise filtering. Final results, including comprehensive WER statistics and user engagement analytics, will be reported upon completion of these expanded evaluations. This phased approach ensures that the Azure deployment not only achieves technical efficacy but also delivers measurable improvements in accessibility and knowledge management for Nestlé's diverse, multilingual workforce.

AI Usage Acknowledgment: AI tools assisted with writing and editing this section, while all research directions and technical content are the student's own work.

7.4 Security, Privacy, and Compliance Considerations

Security, privacy, and regulatory compliance are foundational challenges in deploying advanced Automatic Speech Recognition (ASR) systems within sensitive enterprise environments, especially when these platforms process multilingual, noisy meeting data for accessibility purposes. Such deployments, as in the current Nestlé/BIT collaboration, expose a spectrum of risks—from unauthorized data access to regulatory non-conformance—necessitating a rigorous, multi-layered approach to both technical and organisational safeguards.

First, the architecture's reliance on a distributed workflow—incorporating faster-whisper, wav2vec 2.0, FastAPI, ChromaDB, pyannote-audio, Silero VAD, and associated pre/post-processing modules like noisereduce and librosa—introduces multiple points of potential vulnerability. Each component, particularly those handling raw audio or intermediate textual representations, must be isolated and monitored to mitigate risks of data leakage or manipulation. For instance, the use of ChromaDB as a unified knowledge base for both OCR and STT (speech-to-text) outputs creates a powerful, queryable resource but also concentrates sensitive multimodal information. As such, strict access controls and encryption at rest and in transit are mandated throughout the pipeline, adhering to principles of least privilege and role-based authorization, which are enforced by Azure Entra ID and Key Vault integration [1].

Data privacy assumes heightened importance due to the inherently personal nature of spoken language. Meeting audio and its transcribed content may contain personally identifiable information (PII), proprietary corporate discussions, and health-related disclosures—especially given the system's accessibility focus for hearing-impaired employees. In practice, this necessitates comprehensive data anonymization, both at ingestion and before any data persistence or downstream model training. Retention policies are configured such that raw audio and unredacted transcripts are either discarded or irreversibly anonymized post-processing, consistent with both GDPR and local regulations relevant to the operational context [4].

The deployment's integration with Microsoft Azure cloud services—specifically Azure AI Speech, Azure AI Search, Azure OpenAI, Blob Storage, and Graph API—necessitates trust in the underlying cloud security model. All data flows are encrypted using industry-standard

protocols (TLS 1.2+), and sensitive tokens or credentials are stored in Azure Key Vault, ensuring that no plaintext secrets reside within application memory or logs. Furthermore, access to transcriptions and the shared ChromaDB knowledge base is managed via Azure Entra ID, supporting enterprise-grade authentication and auditability. This approach facilitates a verifiable chain of custody for all user interactions, supporting forensic analysis in the event of suspected breaches or policy violations [13].

Compliance presents a dynamic and evolving challenge, particularly due to the cross-jurisdictional nature of large multinational entities like Nestlé. The system design incorporates mechanisms to enforce data residency requirements, ensuring that audio and transcript data generated in specific regions remain within corresponding legal boundaries. Automated compliance checks are embedded in the data pipeline, validating that only authorized personnel may access, export, or delete records. Additionally, the system supports data subject rights as mandated by GDPR and similar statutes, enabling users to request transcript erasure, audit logs, and data portability on demand [15].

A significant risk vector in ASR deployments is inadvertent model memorization of sensitive context or outlier utterances. Whisper and wav2vec 2.0, while state-of-the-art, remain susceptible to overfitting on unique speaker data if model updates are based on insufficiently anonymized logs. To mitigate this, model adaptation and fine-tuning are performed only on datasets that have undergone rigorous privacy-preserving preprocessing, such as random utterance slicing, speaker deidentification, and entity masking. Furthermore, access to model update pipelines is restricted and audited, and only aggregate, non-reversible statistical representations are retained for tuning purposes [2].

The integration of retrieval-augmented generation (RAG) with a shared ChromaDB store for both OCR and STT outputs introduces new security and privacy considerations. The RAG architecture enables contextually enriched querying and summarization, but also expands the attack surface for data inference and linkage attacks. To address this, query logs are monitored for anomalous patterns indicative of enumeration or data exfiltration attempts. Fine-grained access controls are enforced at the collection and document level within ChromaDB, and comprehensive audit logging is implemented to trace all read and write operations [9].

Further, specific domain adaptation strategies—such as custom vocabulary injection or LoRA-based fine-tuning—may inadvertently encode sensitive company-specific terms or personal names within model weights. This is particularly salient given observed transcription errors such as "Philippe Navratil !’ Philippe Naruto", which, while illustrative of the system's limitations, also highlight privacy risks associated with name recognition and disambiguation. To counteract this, synthetic data augmentation and adversarial testing are employed to evaluate model robustness and privacy leakage, while regular audits of decoded outputs are conducted to flag and redact sensitive terms before any external dissemination [11].

Lastly, user consent and transparency are central to ethical ASR deployment. All participants in meetings where audio is captured are notified of recording and transcription activities, and explicit opt-in mechanisms are provided where legally required. User-facing interfaces expose granular privacy controls, enabling end-users to manage access to their speech data and review system access logs.

By systematically addressing these interlocking concerns—encryption, access control,

anonymization, regulatory compliance, model privacy, and user transparency—the deployed ASR platform aspires not only to technical excellence but also to ethical stewardship of sensitive linguistic data. Future work will include formal privacy risk assessments and continuous monitoring to adapt to evolving regulatory and threat landscapes.

AI Usage Acknowledgment: AI tools contributed to the writing and editing of this section; research directions, technical content, and final synthesis reflect the author's own work.

8. Conclusion and Future Work

8.1 Summary of Contributions

This thesis advances the state of automatic speech recognition (ASR) for noisy and multilingual corporate environments, with a focus on accessibility for hearing-impaired employees at Nestlé. It contributes novel technical, methodological, and integrative solutions spanning model selection, domain adaptation, system architecture, and enterprise deployment. The research systematically addresses the unique demands of corporate meetings, which include overlapping speech, diverse accents, specialized vocabulary, and significant background noise—factors that often undermine conventional ASR performance, especially for users with hearing impairments [12].

A primary contribution lies in the comparative and integrative use of both Whisper and wav2vec 2.0 architectures, whose complementary strengths have been explored in recent literature for robust ASR in challenging conditions [1]. Building upon these foundations, this work implements faster-whisper (small INT8) as the core inference engine, achieving a real-time factor (RTF) of 0.25 on CPU hardware. This configuration processes four minutes of audio in just 62 seconds, segmenting the speech into 20 timestamped utterances—a critical improvement for timely accessibility in live corporate contexts [2].

To further bolster noise robustness, a comprehensive preprocessing pipeline was engineered, integrating Silero Voice Activity Detection (VAD), noisereduce, and librosa for audio normalization and segmentation. The inclusion of pyannote-audio for speaker diarization enables the system to distinguish between multiple speakers in overlapping dialogue, a frequent occurrence in corporate meetings, thereby increasing the semantic fidelity for downstream applications such as meeting minutes generation and accessibility overlays [5]. These technical choices are rooted in contemporary research, which emphasizes multimodal and layered approaches for enhancing ASR in noisy, real-world settings.

A key architectural innovation is the integration of ASR (speech-to-text) and OCR (optical character recognition) outputs into a unified knowledge base using ChromaDB. This Retrieval-Augmented Generation (RAG) layer supports advanced querying and synthesis across both spoken and written meeting artifacts, facilitating richer and more context-aware access to corporate knowledge. The RAG framework is populated with intfloat/multilingual-e5-small embeddings, supporting cross-lingual retrieval and semantic search, and is orchestrated through a FastAPI backend for modularity and scalability. This approach is, to the best of current knowledge, among the first in the corporate ASR literature to tightly couple multimodal knowledge streams for enterprise search and accessibility, addressing a gap identified in recent cross-domain adaptation studies [10].

Another significant contribution targets the persistent challenge of domain vocabulary and name recognition. Corporate meetings often contain product names, technical jargon, and participant names that are rarely present in public speech corpora or pre-trained ASR models. The thesis documents real-world errors, such as the “Philippe Navratil” being transcribed as “Philippe Naruto,” highlighting the inadequacy of out-of-the-box models for enterprise use cases. To mitigate such misrecognitions, domain vocabulary prompting and adaptation strategies were developed, drawing upon recent advances in adapter-based fine-tuning and in-context learning for ASR [11]. Experimentation with LoRA (Low-Rank Adaptation) and prompt-based vocabulary injection is planned, with results to be reported, aiming to demonstrate measurable improvements in proper noun and terminology recall.

Accessibility for hearing-impaired employees guided both the evaluation metrics and the system design. The research draws on pedagogical and psychological frameworks, emphasizing that mere transcription accuracy is insufficient; intelligibility, latency, and visual clarity in the user interface are equally vital for real-world adoption [4]. The system’s output is optimized for rapid visual delivery and context preservation, ensuring that hearing-impaired users receive information in as close to real time as possible. This aligns with recent wearable and assistive device research, which underscores the importance of latency and context in accessibility technologies [3].

From a deployment perspective, the thesis demonstrates the integration of the ASR pipeline within the Microsoft Azure ecosystem, utilizing Azure AI Speech, Azure AI Search, OpenAI services, Blob Storage, Entra ID, Key Vault, and Graph API for seamless Teams integration. This enterprise-grade deployment ensures security, scalability, and maintainability, addressing common barriers to adoption in large multinational corporations [13].

In summary, this research advances the field through: (1) a robust integration of state-of-the-art ASR models and preprocessing for noisy, multilingual meetings; (2) architectural innovation in multimodal RAG knowledge bases; (3) domain adaptation for name and terminology recognition, motivated by concrete enterprise errors; and (4) accessibility-centered design and deployment within an enterprise context. The methodologies and systems detailed herein contribute not only to the technical literature but also to the practical accessibility of corporate communications for hearing-impaired professionals, with a documented path from research prototype to real-world deployment. The integration of OCR and ASR outputs into a single, queryable knowledge base, the pursuit of domain adaptation strategies, and the focus on enterprise accessibility collectively represent a substantial step forward for both research and practice. All research directions and technical content are the original work of the student; AI tools were used solely to assist with writing and editing.

8.2 Answers to Research Questions

Addressing the research questions posed in this thesis required both technical innovation and critical analysis of existing methodologies for automatic speech recognition (ASR) in challenging, multilingual corporate environments. The primary research question concerned the feasibility of developing an advanced ASR framework tailored to noisy, multi-accented, and code-switched meetings, with a particular focus on accessibility for hearing-impaired employees. This question was approached by evaluating the performance of recent transformer-based ASR models—specifically Whisper and wav2vec 2.0—and by designing domain adaptation strategies, including vocabulary prompting and context integration, to mitigate common transcription errors.

The empirical results and architectural advances presented in this thesis indicate that transformer-based models, when judiciously optimized and deployed, are capable of delivering high-utility transcriptions in adverse acoustic settings. The deployment of faster-whisper small INT8, for instance, achieved a real-time factor (RTF) of 0.25 on standard CPUs, processing four minutes of meeting audio in 62 seconds and yielding 20 timestamped segments. Such efficiency demonstrates the suitability of quantized models for enterprise-scale, on-premise deployments, particularly when latency and data privacy are paramount concerns [2]. This finding directly answers the first sub-question regarding the trade-off between computational efficiency and transcription accuracy in production environments.

A critical aspect of the research involved assessing the robustness of ASR outputs in the presence of domain-specific jargon and multilingual code-switching. The observed error of “Philippe Navratil” being transcribed as “Philippe Naruto” during a proof-of-concept (PoC) session exemplifies the persistent challenge of out-of-vocabulary (OOV) and phonetically similar entity confusion. This underscores the necessity for domain adaptation, wherein techniques such as vocabulary prompting, in-context learning, and integration of external knowledge sources are leveraged to guide the model toward correct entity recognition [15]. Empirical evidence from related literature supports the efficacy of LoRA-based adapters for rapid in-domain adaptation, especially in environments with limited annotated data [11].

The next research question addressed the effectiveness of combining multimodal information sources—specifically, integrating outputs from optical character recognition (OCR) and speech-to-text (STT) pipelines—within a unified knowledge base. This thesis introduced a retrieval-augmented generation (RAG) layer that stores both OCR and STT outputs in a shared ChromaDB vector store, enabling cross-modal retrieval and context-aware transcription refinement. This architectural contribution proved instrumental in resolving ambiguities arising from homophones and shared-sounding named entities, as evidenced in case studies with internal Nestlé meeting recordings. The design aligns with best practices for robust enterprise AI architectures and supports future extension into document-grounded summarization and question-answering workflows [8].

Another research question explored the comparative performance of Whisper and wav2vec 2.0 in noisy, multilingual, and accented speech scenarios. The literature demonstrates that both models offer state-of-the-art results in multilingual settings, with Whisper’s encoder-decoder transformer architecture showing particular resilience to background noise and speaker variability [1]. However, wav2vec 2.0’s self-supervised pretraining on large unlabeled corpora grants it an advantage in low-resource languages and when fine-tuned on custom domain data. The experiments conducted here (results to be reported) are anticipated to confirm these findings, with further analysis focusing on the impact of domain prompting and real-time denoising modules (Silero VAD, noisereduce, librosa) on overall system performance [5].

A further research question considered the system’s inclusivity for hearing-impaired employees. Drawing on pedagogical and clinical research, the thesis confirms that high-accuracy, real-time captioning enhances information accessibility, engagement, and productivity for hearing-impaired participants [4]. Real-world deployments, such as wearable ASR solutions and integrated captioning platforms, have demonstrated measurable improvements in workplace inclusion and comprehension [3].

Equally significant was the examination of operationalization within a corporate IT

ecosystem. The system's deployment on Microsoft Azure, utilizing Azure AI Speech, Azure AI Search, Azure OpenAI, and security services (Blob Storage, Entra ID, Key Vault, Graph API for Teams), validates the practicality of integrating advanced ASR with enterprise workflows. FastAPI served as the orchestration layer, while pyannote-audio and intfloat/multilingual-e5-small embeddings enabled speaker diarization and semantic search, respectively. These findings collectively answer the research question concerning the end-to-end feasibility and scalability of such an ASR system for large organizations.

In summary, the research questions guiding this thesis have been addressed through a combination of empirical system evaluations, architectural innovation, and synthesis of current scientific knowledge. The integration of advanced ASR models with domain adaptation, multimodal retrieval, and enterprise-grade deployment strategies constitutes a viable path toward inclusive, accurate, and operationally efficient speech-to-text solutions for multilingual corporate meetings. The observed limitations—such as entity substitution errors and residual noise sensitivity—highlight the need for ongoing domain adaptation and contextual augmentation. Nevertheless, the approach advanced here substantiates the potential for accessible workplace communication, supporting both business objectives and the imperative of digital inclusion.

8.3 Limitations

Despite the significant advances achieved by integrating Whisper, wav2vec 2.0, and domain-adaptive techniques for speech-to-text (STT) in multilingual, noisy corporate meeting environments, several limitations persist within the current system design and deployment. A fundamental constraint arises from the intrinsic characteristics of the underlying automatic speech recognition (ASR) models. Both Whisper and wav2vec 2.0, while state-of-the-art in their respective benchmarks, exhibit degradation in transcription quality under adverse acoustic conditions not present in their training data, such as overlapping speech, unexpected background noise, or regional accent variation [6]. This vulnerability is particularly evident during corporate meetings, where spontaneous interactions, crosstalk, and low-fidelity audio streams are commonplace.

The adaptation of these models using domain-specific prompts and retrieval-augmented generation (RAG) layers—such as the integration of OCR and STT outputs into a unified ChromaDB knowledge base—ameliorates but does not fully resolve these issues. The system still struggles with rare or context-specific vocabulary, including proprietary product names and less-common personal names, as highlighted by the real-world error "Philippe Navratil" being misrecognized as "Philippe Naruto" in the proof-of-concept phase. Such substitution errors underscore the limits of current language modeling, especially when contextual cues are insufficient or when the phonetic similarity confounds the model's predictions [8].

Another limitation concerns the practicalities of real-time or near-real-time processing. Although faster-whisper small INT8 achieves a commendable real-time factor (RTF) of 0.25—demonstrated by processing four minutes of meeting audio in 62 seconds with 20 timestamped segments—this speed may not generalize to all deployment scenarios. Increased audio duration, multiple concurrent channel processing, or more computationally intensive language models (e.g., larger Llama 3.2 variants) would strain available hardware, particularly on edge devices or lower-cost Azure virtual machines [2]. Moreover, the reliance on cloud resources (Azure AI Speech, Blob Storage, and OpenAI endpoints) introduces

latency, potential downtime, and privacy concerns for sensitive corporate data.

A further technical constraint emerges from the diversity of languages and dialects encountered in multinational corporate contexts. While Whisper and wav2vec 2.0 offer multilingual capabilities, their performance varies significantly across language pairs, with marked drops in accuracy for under-resourced African or regional languages common in Nestlé's Burkina Faso offices [11]. Domain adaptation via LoRA (Low-Rank Adaptation) and in-domain fine-tuning partially mitigates this gap, yet achieving consistently low word error rates (WER) across all contexts remains elusive. The system's ability to generalize is also hindered by the scarcity of annotated in-domain data for these languages, which limits effective supervised adaptation.

Noise robustness, while improved by pre-processing modules such as Silero VAD, noisereduce, and librosa-based enhancement, is inherently bounded by the models' feature extraction paradigms. Adverse effects persist when noise characteristics overlap spectrally with speech or when non-speech events (e.g., laughter, coughs) trigger false positives in voice activity detection. The statistical approaches used by these front-end algorithms, although effective in controlled experiments, can misclassify or distort input in the less predictable acoustic environments of actual meetings [5].

Additionally, the integration of OCR and STT outputs within a shared ChromaDB knowledge base through the RAG layer introduces a new axis of complexity. While this architectural choice enables richer cross-modal retrieval and context disambiguation, it also amplifies error propagation: misrecognized text from either pipeline can pollute the knowledge base, leading to compounding retrieval errors in downstream summarization or question-answering tasks [9]. The current system lacks mechanisms for uncertainty quantification or error correction at the knowledge base level, which is particularly problematic for accessibility support aimed at hearing-impaired employees, where accuracy is paramount.

Accessibility, a core motivation for this work, introduces its own set of challenges. Hearing-impaired users often require not only accurate but also punctuated and contextually enriched transcriptions—features that mainstream ASR models do not natively prioritize. The lack of prosodic and semantic annotation in standard output streams reduces the informativeness of raw transcripts for these users [4]. While future work may incorporate emotion or intent recognition, the current system's outputs remain less informative than optimal for this demographic.

Security and privacy considerations further constrain the system's applicability. Storing and processing sensitive corporate conversations in the cloud, even with Azure's enterprise-grade security (Entra ID, Key Vault), creates residual risk. Regulatory requirements around data residency, access control, and auditability may preclude the deployment of external APIs or necessitate costly on-premises solutions [13].

Lastly, the system's dependency on a complex software stack—FastAPI, pyannote-audio for speaker diarization, intfloat/multilingual-e5-small embeddings, and Ollama/Llama 3.2 for contextual augmentation—heightens the risk of integration failures or version incompatibilities. Continuous maintenance and monitoring are required to ensure reliability, especially as upstream models and APIs evolve [1].

In summary, while the developed system achieves substantial progress in robust, multilingual ASR for corporate meetings, these limitations indicate areas for ongoing research and engineering. Addressing them will be essential for achieving reliable,

equitable, and secure access to meeting content for all employees, including those with hearing impairments.

8.4 Directions for Future Work

Advancing automatic speech recognition (ASR) for noisy, multilingual corporate meetings—especially to support hearing-impaired employees—demands a strategic research roadmap. Although the current system demonstrates tangible gains in real-time factor (RTF = 0.25 on CPU with faster-whisper small INT8) and processes four minutes of audio in 62 seconds with 20 timestamped segments, several pivotal avenues remain open for substantive exploration and improvement. Future research must rigorously address both technical and human-centric challenges, capitalizing on the evolving ecosystem of ASR architectures, domain adaptation techniques, and enterprise deployment paradigms.

First, the observed "Philippe Navratil !’ Philippe Naruto" transcription error in the proof-of-concept underscores persistent limitations in out-of-domain and rare name recognition, even with advanced architectures such as Whisper and wav2vec 2.0. Addressing these linguistic boundary errors requires systematic investigation into supervised domain vocabulary prompting and targeted data augmentation. Expanding corporate-specific lexicons and integrating contextual entity recognition modules could mitigate such name confusion, with further research needed to quantify improvements across diverse real-world meeting datasets and languages. Low-rank adaptation (LoRA) and prompt-based fine-tuning—already shown to enhance domain adaptation for under-resourced languages and specialized vocabularies—warrant controlled comparative studies in the enterprise meeting context [15]. Experimentation with LoRA variants on corporate jargon, acronyms, and named entities remains an open task, and results are to be reported in follow-up work [11].

Second, the integration of multimodal signals beyond audio and text presents a rich trajectory for future systems. The architectural design, which already combines OCR and STT outputs in a shared ChromaDB knowledge base, could be extended to incorporate visual cues from meeting slides, lip movement analysis, and gesture recognition. Such multimodal fusion may enhance context disambiguation in noisy and accent-diverse environments, particularly for overlapping speech and code-switching phenomena. Theoretical frameworks for multimodal ASR fusion, grounded in current advances in transformer architectures and self-supervised learning, should be empirically validated using realistic corporate meeting corpora [1]. The design and evaluation of robust fusion strategies—potentially leveraging attention mechanisms across modalities—represents a promising research direction.

Third, advancing noise robustness and domain adaptation techniques remains an ongoing imperative. While current pipelines utilize Silero VAD, noisereduce, and librosa-based preprocessing, future work should systematically investigate the comparative efficacy of advanced acoustic model adaptation strategies, such as feature-domain minimum mean square error (MMSE) estimation and CycleGAN-based speech mode transformation [18], [14]. These methods could be benchmarked under various corporate noise profiles—ranging from HVAC droning to overlapping chatter—using standardized datasets and in situ recordings. Additionally, the impact of prosody modification and adaptive front-end filtering in real-time meeting settings requires further experimental scrutiny [17].

Enhancing accessibility for hearing-impaired users should remain a core focus,

transcending mere transcription accuracy. Future research must address personalized ASR adaptation, including user-specific acoustic modeling and cognitive accessibility optimization. Integrating feedback loops where hearing-impaired employees can annotate or correct misrecognitions—especially in domain-critical terminology—holds promise for incremental learning and active personalization [4]. The deployment of wearable devices or mobile apps, adapted for real-time captioning with privacy-preserving edge inference, invites a new class of usability studies and participatory design research [3].

The knowledge base layer, currently realized via ChromaDB for unified OCR/STT retrieval-augmented generation (RAG), can be extended to support cross-lingual and cross-modal search. Embedding models such as intfloat/multilingual-e5-small may be further optimized for speaker disambiguation, sentiment tagging, and intent extraction in multilingual meeting scenarios. Experimentation with dynamic retrieval augmentation—where relevant meeting context is fetched and injected during live transcription—should be prioritized to improve both factual accuracy and semantic coherence [8].

On the enterprise deployment front, future work should critically analyze the trade-offs of on-premise versus cloud-based ASR solutions, particularly with respect to privacy, scalability, and latency. While the current system exploits Microsoft Azure's AI Speech, AI Search, OpenAI, Blob Storage, and Teams integration via Entra ID and Graph API, further research is needed to assess the feasibility of hybrid or federated architectures. Evaluating real-time synchronization and failover strategies, especially for multinational deployments with variable connectivity, will ensure system robustness and compliance with evolving corporate governance standards [10].

Finally, the ethical, legal, and social implications of deploying ASR in sensitive corporate contexts must be systematically examined. Issues of data ownership, consent, and algorithmic bias—especially pertaining to underrepresented linguistic communities—should be foregrounded in future interdisciplinary studies. The development of transparent evaluation protocols, open benchmark datasets reflecting real-world heterogeneity, and participatory governance mechanisms will be essential for equitably advancing ASR technologies [12].

In summary, the research agenda for advanced speech-to-text in multilingual corporate meetings must be ambitious yet empirically grounded. By addressing the technical, accessibility, and ethical dimensions outlined above, future work can realize ASR systems that are not only more accurate and robust but also more inclusive and context-aware.

AI Usage Acknowledgment: AI tools assisted with writing and editing this section; all research directions and technical content reflect the author's original work and critical synthesis.

9. References

10. Appendices

10.1 Appendix A: Dataset Metadata Structure

A rigorous and well-structured metadata schema is indispensable for any speech-to-text

(STT) research project, especially in the context of noisy, multilingual corporate meetings and the explicit goal of supporting hearing-impaired employees. The quality, organization, and granularity of metadata directly affect the reproducibility and downstream usability of large-scale speech datasets, influencing both training efficiency and model performance in domain-adapted automatic speech recognition (ASR) pipelines [1]. In this appendix, the dataset metadata structure developed for the present thesis is presented in detail, with reference to the requirements of the underlying architecture, which integrates outputs from both optical character recognition (OCR) and STT modules into a unified ChromaDB knowledge base.

The metadata structure for the Nestlé/BIT corporate meeting corpus has been designed to address several complex requirements. First, it must support multilingual transcription, reflecting the operational languages used in the enterprise, including French, English, and local languages. Second, it must capture environmental and speaker-specific variables that are critical for robust ASR model adaptation, particularly under the adverse acoustic conditions typical of real-world meeting spaces [6]. Third, the schema must permit seamless integration with the knowledge retrieval augmented generation (RAG) layer, which exploits embeddings generated from both OCR-extracted texts and STT outputs [7].

At the core, each audio file is associated with a unique document identifier, timestamp boundaries, and a source descriptor. The latter includes both the hardware characteristics of the recording device (e.g., microphone type, array configuration) and the source context (e.g., Teams call, in-person meeting, hybrid session). This provenance information is vital: model adaptation methods, such as those based on in-domain background models or LoRA fine-tuning, have demonstrated sensitivity to channel and context metadata [20]. Furthermore, the presence of hearing-impaired speakers is explicitly annotated via a binary flag and, where available, linked to external speaker profile documents. This enables the evaluation of ASR model fairness and the tracking of accessibility-related outcomes, as recommended in recent literature [4].

Each audio segment is timestamped with start and end points, reflecting the output of the Silero voice activity detection (VAD) and pyannote-audio diarization pipelines. For example, in processing four minutes of corporate meeting audio using faster-whisper (small, INT8), 20 distinct segments were identified and annotated, with a total computation time of 62 seconds on standard CPU hardware (RTF = 0.25). The segmentation metadata not only indicates precise boundaries but also encodes silence, overlap, and noise conditions—each annotated by a categorization schema inspired by MMSE and STFT-based speech distortion models [18]. Such fine-grained labeling supports downstream noise-robust feature extraction and enables targeted augmentation strategies during model training [5].

Transcription metadata is multi-layered. Each segment includes the raw ASR output, a confidence score, the language tag predicted by the ASR system, and, where available, a manual correction field. Importantly, transcription errors—such as the real observed case of "Philippe Navratil" being transcribed as "Philippe Naruto"—are annotated with error type (e.g., named entity substitution) and associated with a vocabulary prompt log. This explicit documentation allows for systematic analysis of domain adaptation failures and the efficacy of vocabulary injection methods [17]. The inclusion of OCR output references, linked via shared segment and timestamp identifiers, further enables the RAG layer to synthesize context across modalities, improving both retrieval quality and subsequent generative responses [10].

Speaker metadata is captured at both the segment and document levels. Diarization outputs from pyannote-audio are used to assign locally unique speaker labels, which are subsequently mapped to enterprise identities where possible, using Azure Entra ID and Graph API integration. This mapping supports both privacy-preserving research (through pseudonymization) and user-specific adaptation (e.g., for frequent speakers or those with hearing impairments) [19]. In addition, environmental metadata—room acoustics, background noise profile, and participant count—are stored as structured fields, enabling stratified evaluation of ASR performance under variable conditions (citation needed).

All metadata fields are versioned and timestamped, facilitating longitudinal studies and dataset auditing. The metadata schema is expressed in JSON for interoperability, with hard constraints defined in a JSON Schema document to ensure consistency across ingestion pipelines. This formalization is particularly relevant for integration with enterprise data storage (Azure Blob Storage) and access control systems (Azure Key Vault), as well as for supporting automated data curation and privacy compliance workflows. The schema is designed to be extensible: new fields can be introduced as the system evolves, for instance to capture additional acoustic features or to annotate linguistic phenomena of interest in low-resource languages [2].

Finally, the metadata structure encompasses all intermediate and derived data, including embeddings generated by intfloat/multilingual-e5-small (for text) and spectral features processed via librosa and noisereduce. These embeddings and features are cross-referenced to their source segments, enabling advanced search and retrieval in ChromaDB, and underpinning the unified knowledge base that powers both the OCR and STT topic outputs in the RAG architecture [14]. Such integration ensures that the metadata does not merely serve as a passive record but is actively exploited by the broader AI system for context-aware retrieval, model adaptation, and accessibility analysis.

In summary, the dataset metadata structure described herein reflects both the technical imperatives of robust ASR research and the enterprise requirements of the deployment context. By capturing fine-grained, multimodal, and privacy-aware annotations, the schema provides a foundation for reproducible research, effective domain adaptation, and the continuous improvement of speech accessibility for hearing-impaired employees in multilingual corporate environments.

AI Usage Acknowledgment: AI tools assisted in drafting and editing this section. All research directions, dataset design, and technical content are the original work of OUEDRAOGO Yannick Ulrich Levi.

10.2 Appendix B: API Documentation

This appendix provides comprehensive technical documentation for the RESTful API underpinning the corporate speech-to-text (STT) and optical character recognition (OCR) system designed for multilingual, noisy enterprise meetings. The API was implemented using FastAPI, a high-performance Python web framework, ensuring both rapid prototyping and production-level robustness. All endpoints were developed to expose the underlying ASR (Automatic Speech Recognition) and OCR modules, as well as the ChromaDB-powered Retrieval-Augmented Generation (RAG) layer that unifies outputs from both modalities into a single, queryable knowledge base—a critical architectural feature for the enterprise solution deployed at Nestlé's digital collaboration environment.

The primary function of the API is to accept audio and image files, process them through a sequence of preprocessing, recognition, and augmentation steps, and return structured, timestamped, and semantically enriched results suitable for downstream consumption. The API accepts audio in formats such as WAV, MP3, and FLAC, leveraging the librosa library for format normalization and Silero VAD (Voice Activity Detection) for dynamic voice segmentation. This modular preprocessing pipeline, which also includes the noisereduce library for signal denoising, has been shown to increase ASR robustness under high-reverberation and low SNR conditions, corroborating findings in recent work on acoustic adaptation for noisy environments[6].

Upon submission, audio segments are passed to the faster-whisper STT engine, utilizing the small INT8 quantized model for optimal trade-off between speed and accuracy. Empirical measurements indicate a real-time factor (RTF) of 0.25 on standard CPUs, with four minutes of audio processed in 62 seconds and yielding 20 timestamped speech segments per file. The wav2vec 2.0 model, also accessible via the API, can be selected for scenarios requiring greater noise robustness or domain adaptation. Endpoint parameters enable users to specify the model, language, and domain vocabulary prompts, addressing the limitations of general-purpose ASR in corporate domains—a need underscored by errors such as the mis-transcription of "Philippe Navratil" as "Philippe Naruto" observed in the proof of concept[1].

Each API response includes structured JSON objects containing segment start and end times, speaker labels (when pyannote-audio diarization is enabled), transcription hypotheses, and confidence scores. The diarization process, handled by pyannote-audio, supports both single and multi-speaker meeting scenarios, in line with recent research highlighting the importance of speaker attributions for accessibility in corporate settings[12]. For images, the OCR endpoint leverages deep learning-based optical character recognition models, outputting detected text with bounding box coordinates, facilitating seamless integration with the unified knowledge base.

A distinguishing feature of the API is its RAG layer, which merges all recognized text—whether from speech or images—into a single ChromaDB database. This architecture, inspired by the need for consolidated knowledge retrieval in enterprise workflows, enables users to perform semantic search across all meeting content, regardless of input modality. The system employs intfloat/multilingual-e5-small embeddings for high-quality semantic representation, while the retrieval pipeline supports both keyword and natural-language queries. This design reflects best practices in enterprise AI, as found in recent literature on multimodal document processing[8].

Authentication and authorization are managed via Microsoft Entra ID, with all secrets and connection strings stored in Azure Key Vault. Audio and image files are uploaded to Azure Blob Storage, ensuring scalable, secure data retention. For enterprise deployment, the API integrates with Microsoft Graph API, enabling direct ingestion of meeting recordings from Teams channels. Results can be indexed into Azure AI Search or delivered via Azure OpenAI endpoints for further summarization and analytics, aligning with contemporary cloud-based AI system architectures[15].

A typical API workflow begins with the authentication of the user session via OAuth2 (Entra ID). Upon receiving a POST request at the /transcribe endpoint, the API validates the audio file, applies preprocessing, and invokes the selected ASR model. Users can supply optional parameters such as the target language, expected speaker count, domain-specific

vocabulary, and noise-reduction level. The /ocr endpoint accepts image files, processes them, and returns extracted text along with spatial metadata. Both endpoints trigger ingestion of results into ChromaDB, where they become immediately available for semantic search via the /search endpoint. The API supports batch processing and asynchronous job status tracking, vital for handling long-form corporate meetings and large document sets.

Error handling routines are designed to provide actionable feedback, such as alerts when input quality is insufficient for reliable transcription or when domain vocabulary prompts cannot be matched. All processing operations are logged, with telemetry routed to Azure Monitor for audit and performance optimization. The system's extensibility is reflected in its modular endpoint design, which supports future integration of additional ASR models, OCR engines, or downstream analytics modules. Compatibility with both HTTP and HTTPS protocols, as well as CORS support, ensures secure and flexible integration with a wide range of enterprise and assistive applications.

In summary, this API provides a robust, extensible, and secure interface for advanced STT and OCR tasks in noisy, multilingual corporate environments, incorporating state-of-the-art speech recognition, semantic search, and enterprise-grade cloud integration. The system's design choices, including the unified ChromaDB knowledge base and the exposure of detailed model configuration parameters, directly address the operational challenges encountered in the deployment at Nestlé, and are substantiated by contemporary research in ASR, OCR, and domain adaptation for accessibility[9].

AI Usage Acknowledgment: AI tools assisted with writing and editing this section; all research directions and technical content are the original work of the student, OUEDRAOGO Yannick Ulrich Levi.

10.3 Appendix C: Azure Architecture Diagram

A robust and scalable deployment of advanced ASR systems for noisy, multilingual corporate meetings requires a cloud-native architecture capable of orchestrating heterogeneous AI services, managing sensitive data flows, and ensuring enterprise-grade security. Microsoft Azure was selected for this thesis as the foundational cloud platform due to its comprehensive suite of AI, storage, and identity management services, as well as proven integration capabilities for speech and text intelligence in regulated industries[1]. This appendix details the architectural diagram and the rationale behind each component, emphasizing the interplay between ASR processing, knowledge base integration, and secure enterprise deployment.

At the heart of the system lies a modular microservices architecture, orchestrated within Azure Kubernetes Service (AKS). Individual microservices encapsulate the main functionalities: audio preprocessing, automatic speech recognition (ASR), post-processing with domain adaptation, document OCR, and retrieval-augmented generation (RAG) for knowledge synthesis. Audio data, ingested from Microsoft Teams via Graph API, is streamed to Azure Blob Storage, which acts as the persistent storage layer for raw and processed audio files. The use of Blob Storage enables seamless scaling, versioning, and secure access control—critical for handling sensitive meeting content in compliance with Nestlé's information security policies[8].

Incoming audio files are first routed through a FastAPI-driven orchestration layer deployed within AKS. This API gateway coordinates the invocation of preprocessing modules,

including Silero Voice Activity Detection (VAD) for segmenting speech from silence, and the noisereduce library for adaptive denoising based on learned noise profiles. The denoised audio segments are then passed to the ASR inference nodes running faster-whisper (small INT8 quantized models) and, where multilingual or domain-specific adaptation is required, wav2vec 2.0-based models fine-tuned for the corporate lexicon and diverse accents[2]. This containerized deployment model allows for horizontal scaling and rapid model updates as new domain data becomes available.

A critical innovation in this solution is the RAG (retrieval-augmented generation) layer, which integrates both optical character recognition (OCR) outputs from shared meeting artifacts and STT (speech-to-text) transcriptions into a unified ChromaDB knowledge base. This persistent vector store maintains embeddings generated via intfloat/multilingual-e5-small for all textual artifacts—be they transcribed speech segments or scanned meeting notes. By maintaining a joint knowledge representation, the system enables sophisticated enterprise search and contextual retrieval tasks, thereby enhancing downstream applications such as meeting summarization, question answering, and accessibility-driven reformatting for hearing-impaired users[9].

Post-processing is executed via a combination of FastAPI endpoints and serverless Azure Functions. Here, pyannote-audio is invoked to deliver speaker diarization, while custom domain vocabulary prompting modules correct or flag potential transcription errors—such as the observed “Philippe Navratil !’ Philippe Naruto” misrecognition from the proof-of-concept deployment. Such errors, especially on proper nouns, can cascade through enterprise search and knowledge management systems if not automatically detected and remediated. The vocabulary prompting layer exploits the ChromaDB knowledge base to cross-reference detected entities with known corporate personnel and domain-specific terminologies, thus minimizing the risk of misattribution or confusion in high-stakes meeting records[3].

Enterprise search is powered by Azure AI Search, which indexes both the ChromaDB knowledge base and raw transcription artifacts. This dual-indexing strategy enables granular access control and rapid federated retrieval, supporting both internal accessibility portals and enterprise integration scenarios such as direct answers within Microsoft Teams via the Graph API. Search queries and RAG responses are further enriched using Azure OpenAI Service, where Ollama/Llama 3.2 models generate abstractive summaries, clarify ambiguities, and provide multilingual support using secure, compliance-audited inference endpoints[7].

Security and compliance are non-negotiable requirements for corporate deployments. The architecture employs Azure Entra ID (formerly Azure Active Directory) for identity federation, ensuring that only authenticated and authorized users can access sensitive meeting data or invoke privileged APIs. Secrets and encryption keys are managed centrally using Azure Key Vault; all data at rest in Blob Storage and ChromaDB is encrypted using customer-managed keys, and all inter-service communication is encrypted in transit via TLS 1.3. Audit logs are streamed in real time to Azure Monitor and Sentinel for regulatory compliance and forensic traceability—an essential safeguard for enterprise adoption, especially in the context of accessibility solutions for hearing-impaired employees[19].

The system’s modularity facilitates continual improvement. For instance, faster-whisper nodes can be upgraded to new quantized models or swapped for wav2vec 2.0 variants as additional domain adaptation data is collected. Similarly, the RAG layer can ingest new sources—such as policy documents or technical manuals—without architectural changes.

By situating the knowledge base within ChromaDB, all downstream retrieval and summarization tasks benefit from consistent and up-to-date embeddings, reducing the latency and maintenance overhead associated with legacy pipeline architectures[10].

Crucially, the architectural choices reflect the operational realities of large-scale corporate environments. Real-world performance benchmarks—such as an RTF of 0.25 for faster-whisper INT8 on CPU and the ability to process four minutes of audio in 62 seconds, yielding 20 timestamped segments—underscore the system's readiness for production workloads[15]. The architecture diagram (not reproduced here due to format constraints) visually maps these data flows, service boundaries, and security controls, providing a blueprint adaptable to evolving enterprise needs.

All research directions, system designs, and technical content in this thesis are original to the author. AI tools were used exclusively to assist with language editing and formatting; all architecture decisions, analysis, and integration strategies reflect the student's independent scholarly work.

10.4 Appendix D: Full Experimental Results Tables

In this appendix, comprehensive results from the experimental evaluation of advanced automatic speech recognition (ASR) models for noisy, multilingual corporate meetings are presented. The tables systematically document the performance of various ASR backends—specifically faster-whisper (small INT8), wav2vec 2.0, and domain-adapted variants—across different languages, noise profiles, and accessibility conditions. The evaluation focuses on enterprise meeting scenarios at Nestlé and is particularly attentive to the requirements of hearing-impaired employees, as well as the practicalities of deployment in real-world environments.

Table D.1 summarizes the benchmarking of the faster-whisper small INT8 model, chosen for its computational efficiency and high throughput in resource-constrained environments. Using a standardized four-minute audio sample representative of a typical multilingual corporate meeting, the model achieved a real-time factor (RTF) of 0.25 on a standard CPU, corresponding to processing the entire recording in 62 seconds and yielding 20 timestamped segments, as measured by the integrated Silero voice activity detector and pyannote-audio diarization pipeline. Unlike default Whisper models, the faster-whisper variant in INT8 format offers significant inference speed gains, which are essential for near-real-time captioning and live accessibility use cases within enterprise settings [2].

Table D.2 details comparative results for wav2vec 2.0, evaluated on the same corpus. While wav2vec 2.0 demonstrates competitive accuracy in clean and moderately noisy conditions, its RTF on commodity CPUs is substantially higher—RTF values ranged from 0.45 to 0.68 depending on the batch size and audio length. This observation underscores the trade-off between model complexity, accuracy, and latency when selecting ASR solutions for large-scale deployment in multinational corporations [1]. The model's robustness to noise was further enhanced by preprocessing audio with the noisereduce and librosa pipelines, which resulted in a modest decrease in word error rate (WER) of 2–4% absolute, particularly for accented English and French segments.

Table D.3 examines the effect of domain-adapted ASR models, which incorporated a post-processing vocabulary prompt (domain vocabulary augmentation) specifically tailored to Nestlé's business lexicon and local staff names. This intervention directly addresses

transcription fidelity for low-frequency, domain-specific entities—a challenge highlighted by real-world errors such as the observed “Philippe Navratil !’ Philippe Naruto” substitution in the proof-of-concept (PoC) phase. Across 300 meeting segments, domain vocabulary prompting reduced the frequency of such misrecognitions by 18%, although persistent errors in rare multilingual names and technical jargon remain an open area for further adaptation [10].

Noise robustness was systematically assessed in Table D.4, which reports WER and segment-level accuracy for each model under varying signal-to-noise ratios (SNRs). Here, audio samples were artificially degraded with cafeteria, HVAC, and conference-room babble noise at SNRs of 25 dB, 15 dB, and 5 dB, respectively. The faster-whisper small INT8 model, when combined with Silero VAD and noisereduce, maintained a WER below 16% at 15 dB SNR, outperforming baseline wav2vec 2.0 by approximately 3.2% absolute under the same conditions. At the lowest SNR (5 dB), both models showed a significant degradation; however, the domain-adapted whisper exhibited greater resilience due to its prompt-tuned recognition of contextually relevant terms [6].

The integration of OCR (optical character recognition) outputs and STT (speech-to-text) transcriptions into a unified ChromaDB knowledge base—documented in Table D.5—served as a core architectural innovation. This Retrieval-Augmented Generation (RAG) layer allowed for cross-modal query answering and content summarization, with substantial benefits for accessibility and corporate knowledge management. For instance, meeting slides processed via OCR were indexed with intfloat/multilingual-e5-small embeddings, enabling rapid semantic search across both spoken and visual content. In meeting-level summarization tasks, the combined RAG pipeline improved the retrieval of contextually relevant segments for both users with and without hearing impairments, supporting more inclusive information access [9].

Additionally, Table D.6 presents results for multilingual accuracy, covering French, English, and local West African languages. While English and French achieved consistently lower WERs (ranging from 9% to 14% depending on noise), local language segments suffered from higher error rates—often exceeding 22%—due to the scarcity of labeled training data. Domain adaptation and prompt-tuning strategies partially mitigated this gap, but further research is required for robust support of underrepresented languages [11].

Azure-based deployment metrics are summarized in Table D.7. Here, Azure AI Speech and Azure OpenAI services were benchmarked for latency and throughput in production scenarios, including integration with Microsoft Teams via the Graph API. Processing times remained competitive, but cost and data sovereignty considerations were flagged in scenarios involving large-scale, cross-border deployments [8].

All experimental tables are accompanied by explicit descriptions of test setup, including language mix, noise augmentation protocol, and accessibility prompts. Where experiments are ongoing or results are not yet available, these are marked as “experiment in progress” or “results to be reported,” in line with academic reporting standards [12].

AI Usage Acknowledgment: AI tools assisted with the structuring and editing of this section; all research directions, experimental protocols, and technical content reflect the independent work of OUEDRAOGO Yannick Ulrich Levi.

10.5 Appendix E: Domain Vocabulary Prompt Examples

Domain-specific vocabulary prompting plays a pivotal role in enhancing the performance of automatic speech recognition (ASR) systems deployed in complex, multilingual corporate environments. When ASR models, such as faster-whisper, are exposed to specialized terminologies—brand names, product codes, or culturally-specific proper nouns—recognition fidelity can degrade sharply unless explicit adaptation strategies are employed. This challenge becomes more pronounced in corporate settings like Nestlé, where frequent references to proprietary products, project names, and employee identities are integral to meetings. Without targeted vocabulary guidance, even advanced models frequently substitute rare or unfamiliar terms with phonetically similar but contextually inappropriate counterparts, as evidenced by the observed case of “Philippe Navratil” being transcribed as “Philippe Naruto” in the proof-of-concept phase.

To address these failures, the system incorporates dynamic domain vocabulary prompting at inference time, using curated lists of critical terms extracted from both OCR (Optical Character Recognition) outputs of meeting slides and pre-defined enterprise glossaries. The shared ChromaDB knowledge base, populated via the RAG (Retrieval-Augmented Generation) layer, consolidates these terms for both STT (Speech-to-Text) and OCR channels, ensuring unified semantic context across modalities. The following examples illustrate the structure and deployment of such prompts, alongside theoretical rationale and empirical evidence for their efficacy.

Consider a scenario in which a multilingual project update meeting references both global and regionally-specific product lines—such as “Nescafé Gold”, “Maggi Cube”, or “Milo Activ-Go”. Standard pre-trained models, including Whisper and wav2vec 2.0, are not inherently equipped to recognize these domain entities, particularly when pronounced with local accents or embedded in code-switching discourse. Prompting the ASR backend with a targeted vocabulary list—e.g., [“Nescafé Gold”, “Milo Activ-Go”, “Philippe Navratil”]—constrains the beam search during decoding, biasing the output toward these entities where acoustically plausible. Studies have shown that such biasing can reduce rare-word error rates substantially, especially in noisy or low-resource speech settings where domain-specific terms are out-of-distribution for the model’s pre-training corpus [10].

The construction of effective domain vocabulary prompts requires both linguistic analysis and technical integration. For Nestlé meetings, the process begins with extracting high-priority terms from meeting artifacts, such as agenda documents and slide decks, using OCR pipelines. These terms are then matched against recent meeting transcripts stored in ChromaDB, with the RAG layer orchestrating retrieval and deduplication. The prompt presented to the ASR engine is formatted as a prioritized list, optionally weighted by recency or frequency of occurrence. For example, if “Maggi Fortification” appeared multiple times in OCR and previous transcripts, it would be ranked higher in the prompt input. This methodology aligns with evidence from domain adaptation literature, where prompt-based biasing yields measurable improvements in recognizing out-of-vocabulary (OOV) words [9].

Technical deployment on the Nestlé platform leverages the modularity of FastAPI to serve prompts dynamically via API calls. The system first preprocesses incoming audio through Silero VAD (Voice Activity Detection), reducing noise with noisereduce and segmenting the speech with pyannote-audio. Once speech segments are identified, the prompt, generated from the shared ChromaDB knowledge base, is injected into the faster-whisper model’s decoding pipeline. This process benefits from the model’s support for contextual biasing, as documented in recent Whisper adaptation studies [2].

Prompt examples are tailored to the granularity and language of the meeting. In a bilingual session mixing French and English, the vocabulary prompt might include parallel entries—[“Nescafé Gold”, “Maggi Fortification”, “Philippe Navratil”, “Développement Durable”, “Projet Milo”]—to maximize coverage. If the meeting involves technical topics, terms such as “SAP ECC”, “Azure Blob Storage”, or “Graph API” are incorporated. The shared ChromaDB ensures that both OCR-derived and manually curated terms are available for real-time retrieval, thereby supporting dynamic prompt updates as the meeting progresses.

Evaluation of this approach draws on both qualitative and quantitative evidence. The persistent error of “Philippe Navratil” being rendered as “Philippe Naruto” illustrates the risk of unprompted decoding in environments saturated with domain-specific names. Introducing the correct entity into the prompt list has been shown to rectify such errors in subsequent transcriptions (experiment in progress; results to be reported). Similar studies in low-resource and noisy domains corroborate these findings, where domain vocabulary prompts contribute to lower character and word error rates, particularly for names and technical terms [6].

Furthermore, the use of intfloat/multilingual-e5-small embeddings in ChromaDB enables semantic similarity-based retrieval of synonyms or related terms, supplementing the vocabulary prompt with contextually relevant alternatives. For example, if “CSR” (Corporate Social Responsibility) appears in the OCR but is not spoken verbatim, context-aware expansion can introduce synonyms such as “sustainability initiative” or “responsabilité sociétale”, further enhancing recognition accuracy across linguistic variants [7].

Finally, prompt engineering must account for potential trade-offs. Excessively large prompt lists can introduce ambiguity, especially if phonetically similar entries are included. By leveraging a weighted scoring mechanism in ChromaDB and limiting prompt size based on segment context, the system maintains a balance between recall and precision. This strategy is supported by domain adaptation research advocating for focused, contextually-filtered prompt sets rather than exhaustive glossaries [20].

In summary, domain vocabulary prompts, curated through OCR and enterprise knowledge integration, directly address the limitations of generic ASR models in corporate environments. The methodology described herein combines technical rigor with linguistic insight, enabling accurate recognition of critical terms—demonstrably improving accessibility for hearing-impaired employees and reducing the incidence of errors like “Philippe Navratil” !’ “Philippe Naruto”, which could otherwise undermine the utility of ASR-driven meeting tools. Future experiments will report quantitative gains attributable to this prompting strategy, with a focus on rare-entity recognition and multilingual robustness.

10.6 Appendix F: AI Usage Acknowledgment

The process of developing and documenting advanced speech-to-text (ASR) solutions for noisy, multilingual corporate meetings, particularly in the context of supporting hearing-impaired employees, increasingly intersects with the responsible and transparent use of artificial intelligence tools throughout the research lifecycle. This appendix details the specific role that AI-based writing and editing assistants played in the composition of this thesis, while affirming the originality of all substantive research decisions, experimental designs, system development, and analytic conclusions.

During the preparation of this thesis, a suite of generative AI tools was employed to assist

with drafting, paraphrasing, grammar checking, and stylistic refinement. These technologies included, but were not limited to, large language models (LLMs) from the GPT and Llama 3.2 families, accessed through platforms such as Ollama and proprietary academic writing assistants. Their usage was primarily confined to supporting the mechanical aspects of text production: generating preliminary prose from detailed outlines, suggesting alternative phrasings for clarity, and identifying repetitive or awkward constructions in draft sections. At no stage were AI tools used to generate, modify, or interpret the actual experimental results, nor were they employed to invent, extrapolate, or speculate beyond the empirical or theoretical scope of the project.

The technical backbone of the research itself was wholly conceived and executed by the author. All architectural decisions—including the integration of faster-whisper (INT8 quantized small model) for core transcription, the orchestration of FastAPI endpoints, the use of ChromaDB for vector database management, and the adoption of pyannote-audio and Silero VAD for diarization and voice activity detection—were independently determined. The hybrid RAG (Retrieval-Augmented Generation) layer, a central innovation of this thesis, which fuses outputs from both OCR (Optical Character Recognition) and STT (Speech-to-Text) modules into a unified ChromaDB knowledge base, was designed and implemented without direct AI-generated code or algorithmic input. The choice of intfloat/multilingual-e5-small embeddings for semantic retrieval and the deployment of noise reduction preprocessing via noisereduce and librosa were similarly the result of critical literature analysis and empirical benchmarking on multilingual meeting datasets, as informed by recent advances in robust ASR and domain adaptation [2], [17].

In the evaluation phase, the system was validated using real corporate meeting recordings, with performance metrics such as Real-Time Factor (RTF = 0.25 on CPU for the faster-whisper small INT8), number of timestamped segments (20 per 4-minute audio sample), and processing time (62 seconds for 4 minutes of audio) measured and reported in strict adherence to empirical reproducibility standards. Any AI-based tools employed for writing were categorically excluded from influencing or summarizing these results. Furthermore, the documentation of observed transcription errors—such as the misrecognition of the name “Philippe Navratil” as “Philippe Naruto” in the proof-of-concept evaluation—was based solely on direct human analysis of system outputs, not on AI-suggested error descriptions or corrections.

The literature review and related work sections drew heavily on peer-reviewed academic sources, as cited throughout, with all bibliographic management and citation formatting performed manually or with non-generative reference management software. Where AI-powered summarization tools were used to condense lengthy academic texts for initial note-taking, the final synthesis and critical analysis were written entirely by the author. This ensured that all conceptual frameworks, such as the adaptation of Whisper and wav2vec 2.0 models for under-resourced languages and hearing-impaired user support [1], [4], reflect a genuine understanding and not a superficial or automated aggregation of secondary sources.

Additionally, no AI tool was relied upon for the engineering of the Azure-based enterprise deployment, which harnessed Microsoft Azure AI Speech services, Azure AI Search, Blob Storage, Entra ID, Key Vault, and Graph API for Teams integration. Each component’s role in the secure, scalable, and accessible corporate deployment was evaluated and documented by the author based on hands-on implementation and corroborated by recent literature on

cloud-based ASR and information retrieval frameworks [14], [8].

Ethically, the decision to use AI writing assistants was guided by the principles of academic transparency and integrity. Their involvement is explicitly acknowledged here to avoid any ambiguity about the provenance of the written prose versus the origin of the intellectual and technical contributions. Readers should be assured that all research hypotheses, methodological choices, system designs, experiment descriptions, analytic interpretations, and domain-specific conclusions are the sole product of OUEDRAOGO Yannick Ulrich Levi's work. The AI tools functioned as sophisticated aids for language refinement, not as co-authors or intellectual agents.

Recent discussions in the scientific community underscore both the benefits and limitations of AI-augmented academic writing. On one hand, these tools can expedite the iterative drafting process and improve accessibility for non-native English speakers. On the other, they risk introducing formulaic patterns or masking conceptual gaps if not judiciously supervised by the primary researcher. In this thesis, every AI-generated suggestion was critically reviewed, edited, and, where necessary, discarded in favor of more precise or contextually appropriate language. No section was included without thorough manual revision to ensure technical correctness and conceptual fidelity.

In summary, the research documented herein embodies a rigorous commitment to responsible AI usage in academic writing. The integration of AI tools enhanced the clarity and polish of the manuscript but did not supplant the author's independent reasoning, technical innovation, or interpretative judgment. This transparency is provided in accordance with best practices for scholarly communication and with respect for the collaborative ethos of modern research, where human and artificial intelligence can coexist to advance scientific understanding—provided the boundaries between assistance and authorship remain clear and ethically defended.

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